LECTURE 8

PARALLEL TEMPERING

Saifuddin Syed

PROBLEMS WITH SERIAL MCMC

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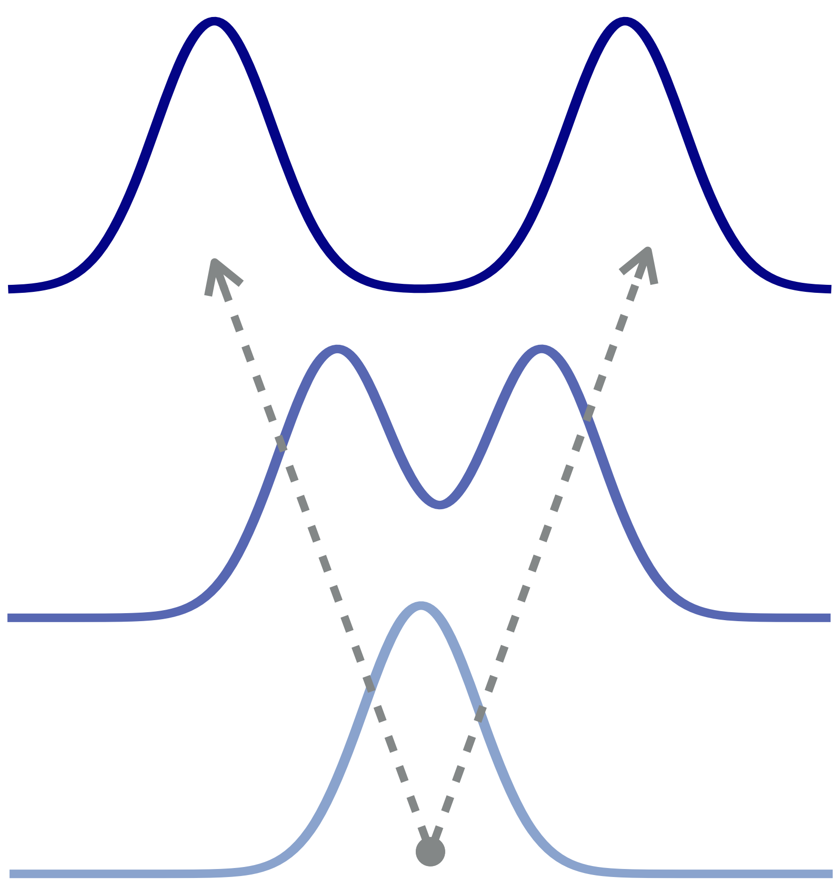
- **Algorithmic:** Single chain tries to simultaneously efficiently explore & be accurate
- **Solution:** Build an interacting particle system
  - Delegate exploration to tractable **reference** chain
  - Communicate to **target** through the **interpolating distributions**



Intractable Target

Interpolating Distributions

Tractable Reference

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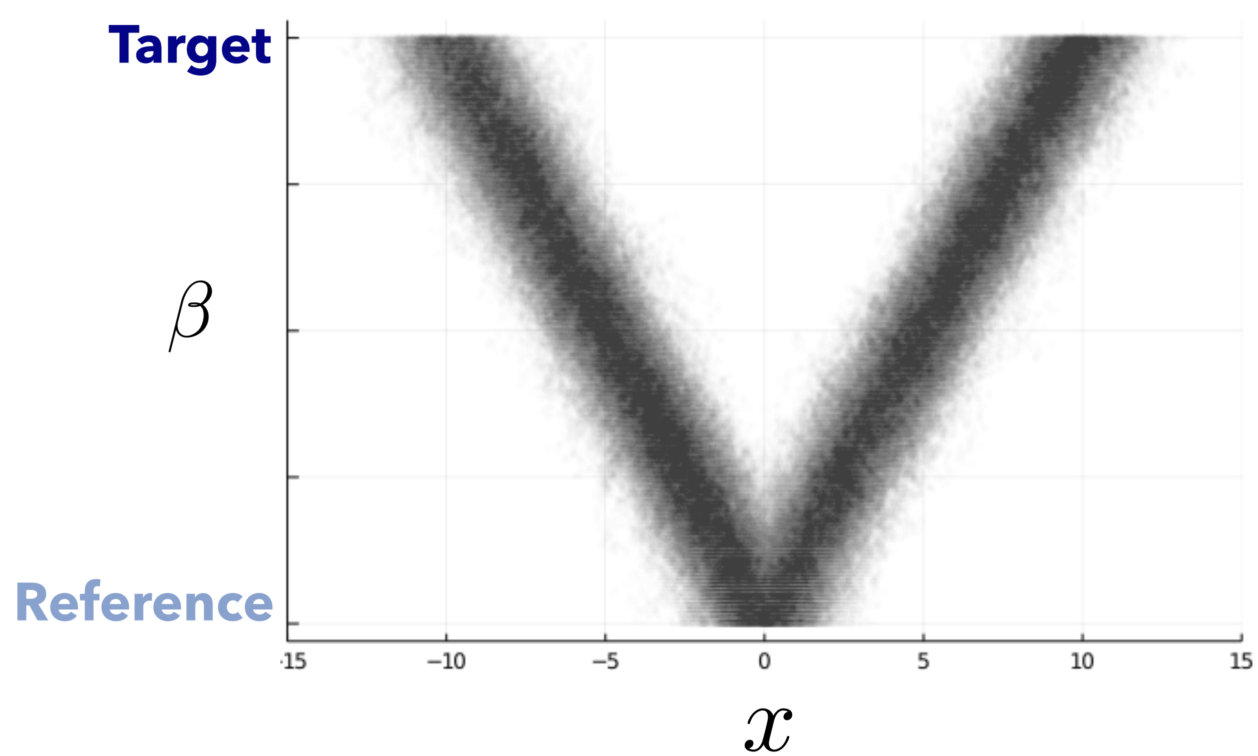

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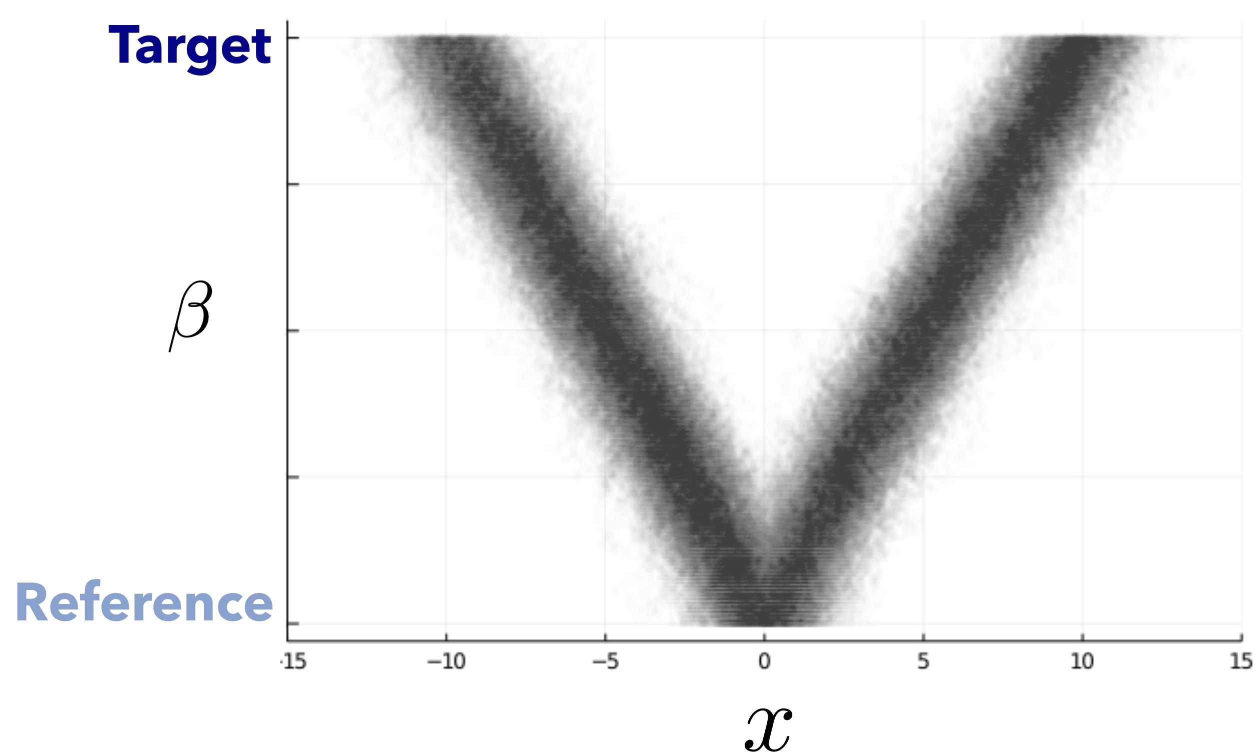

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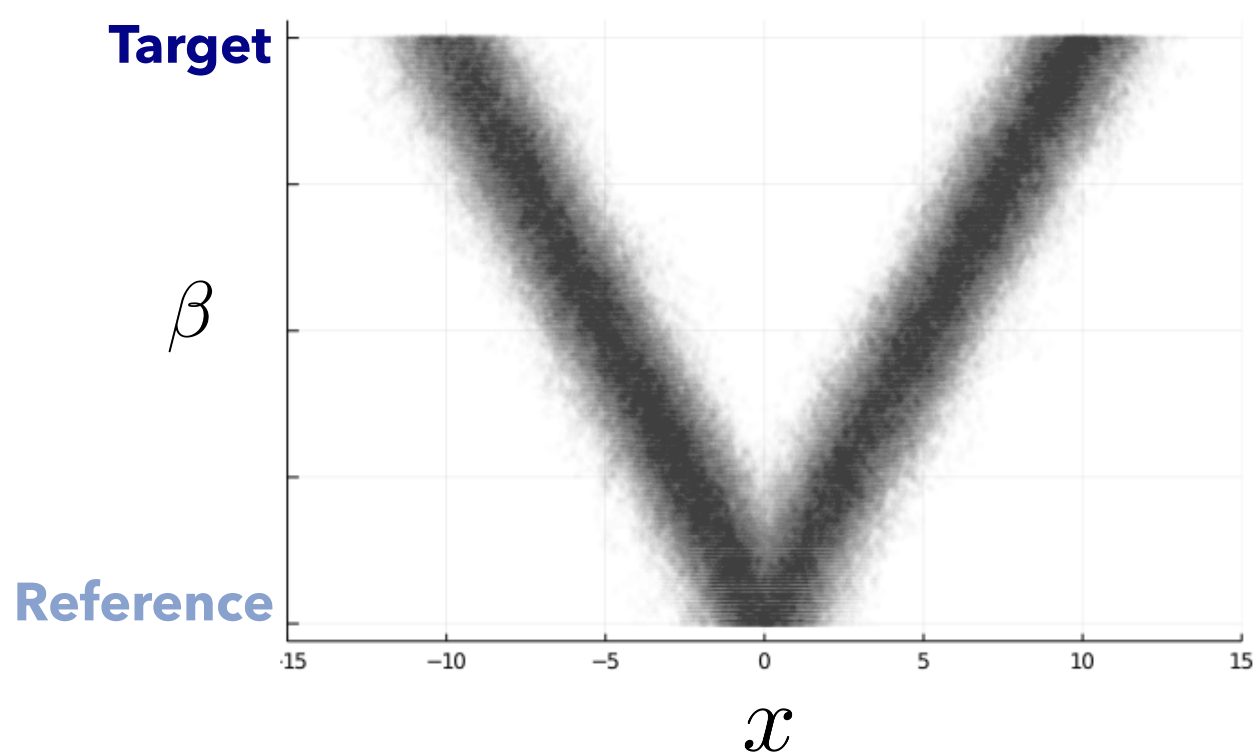

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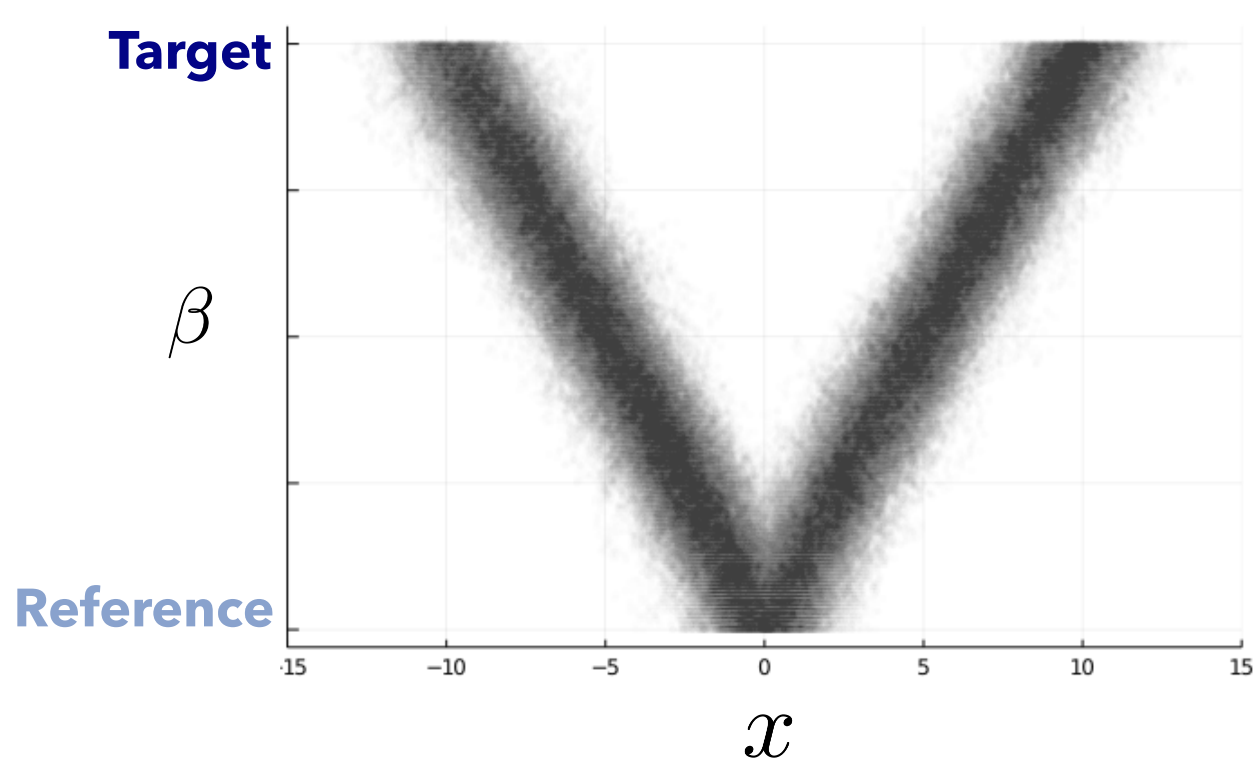

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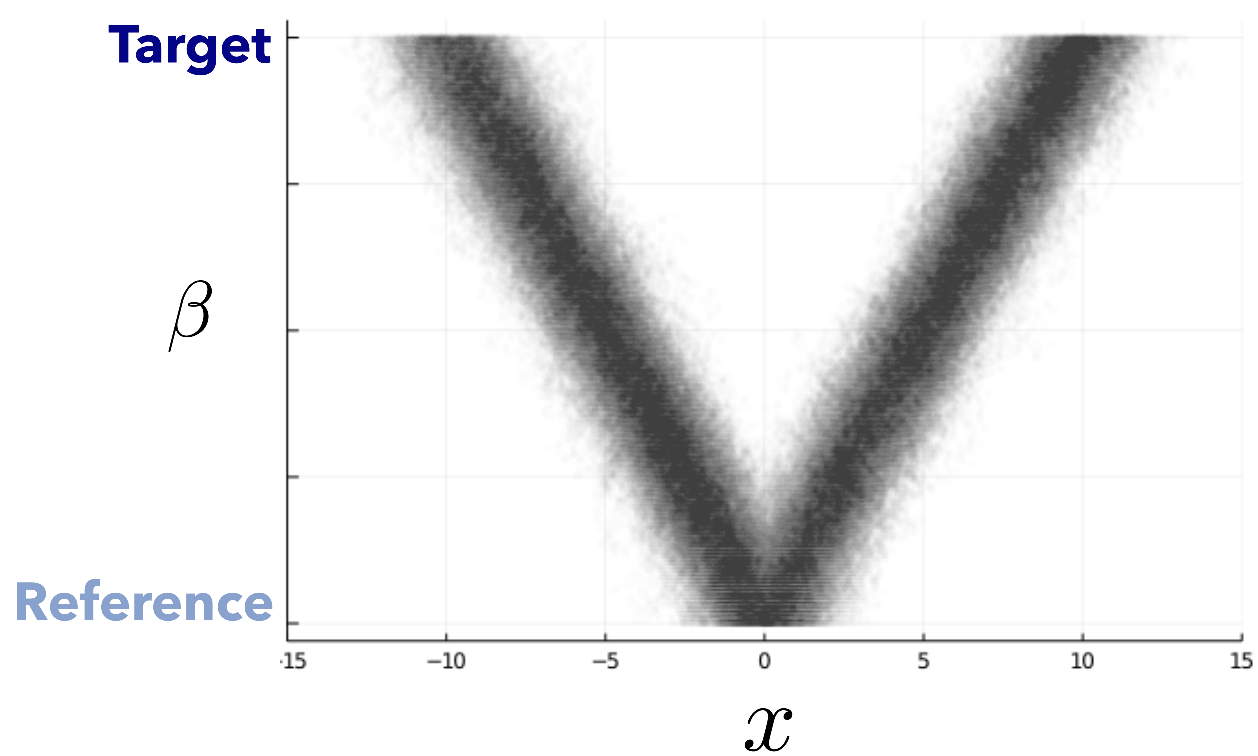

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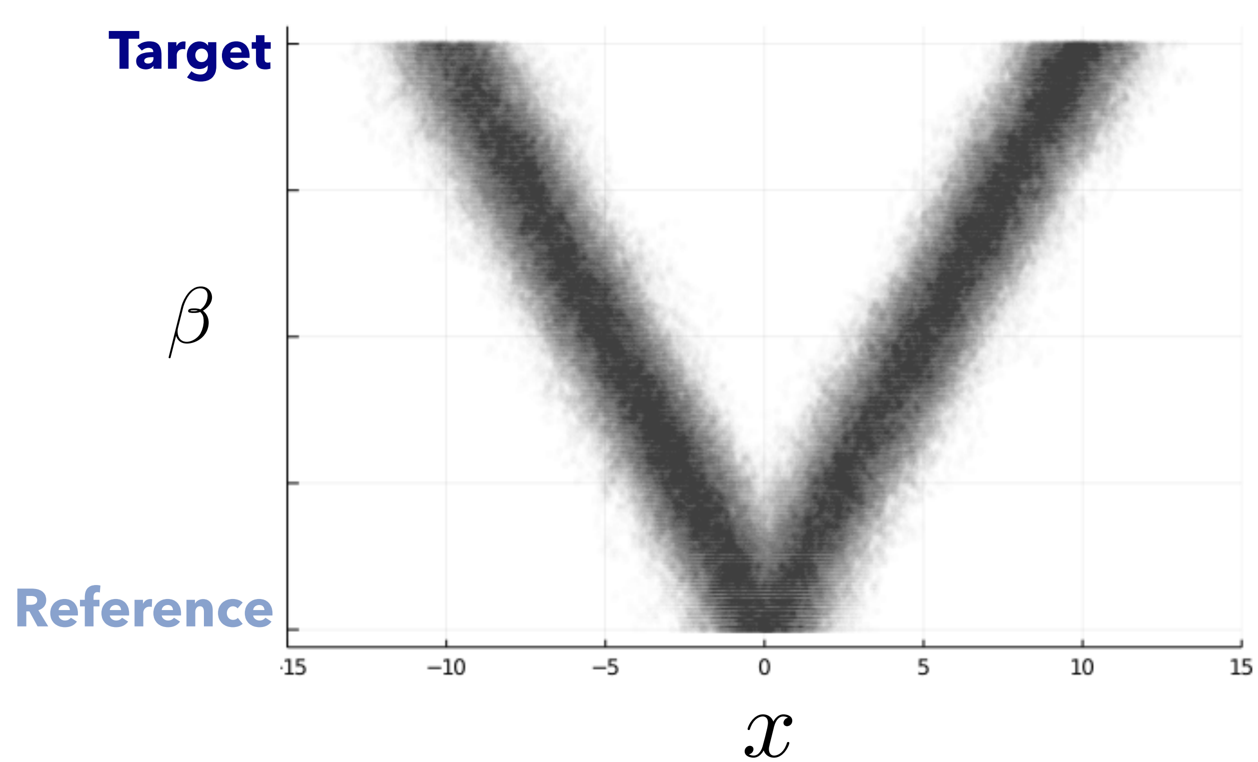

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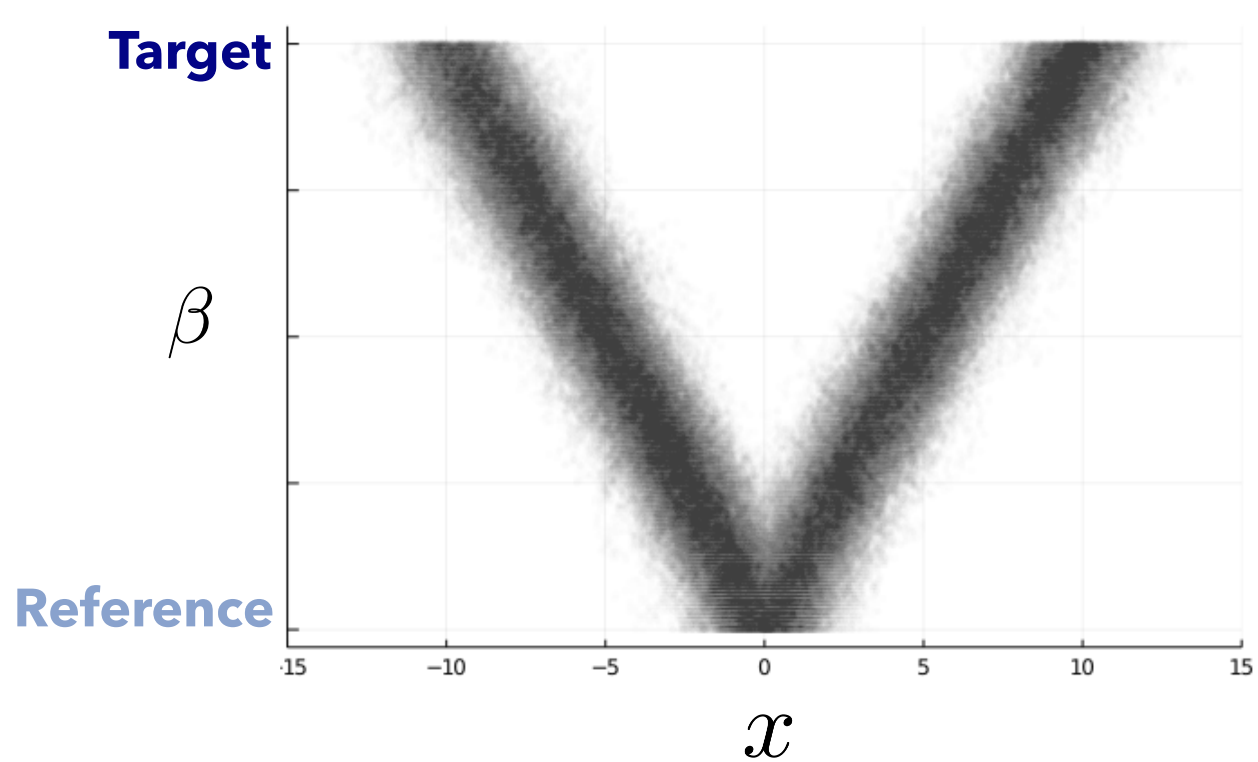
  - Can use the reference to assess the quality and ensure robustness



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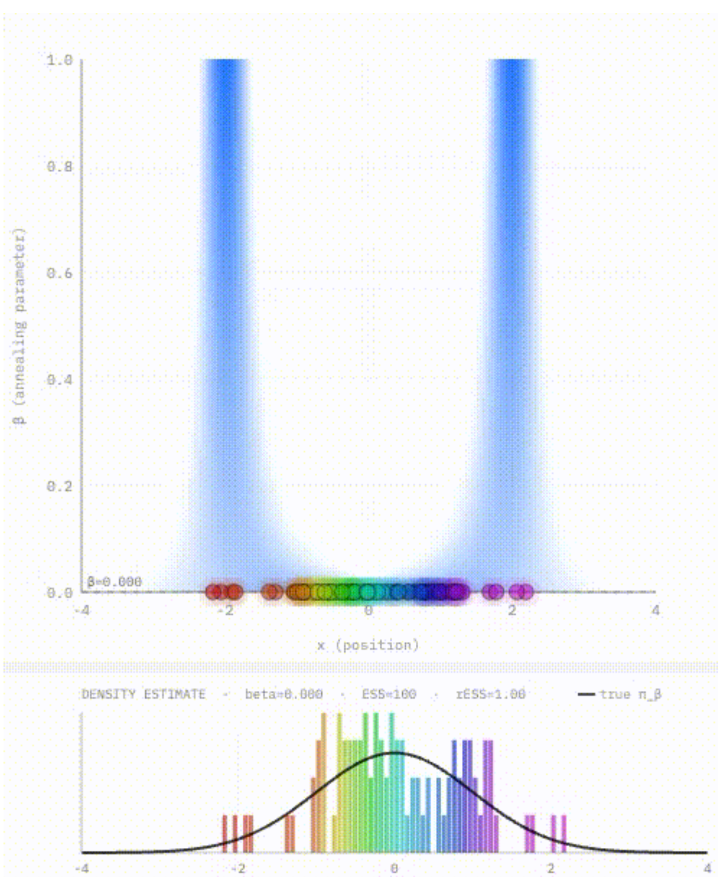

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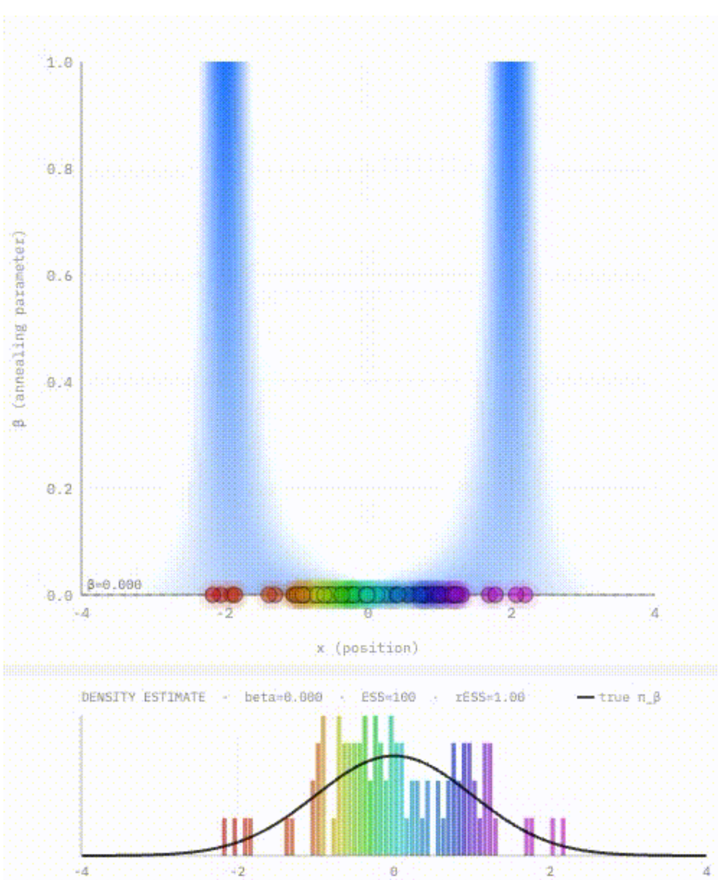

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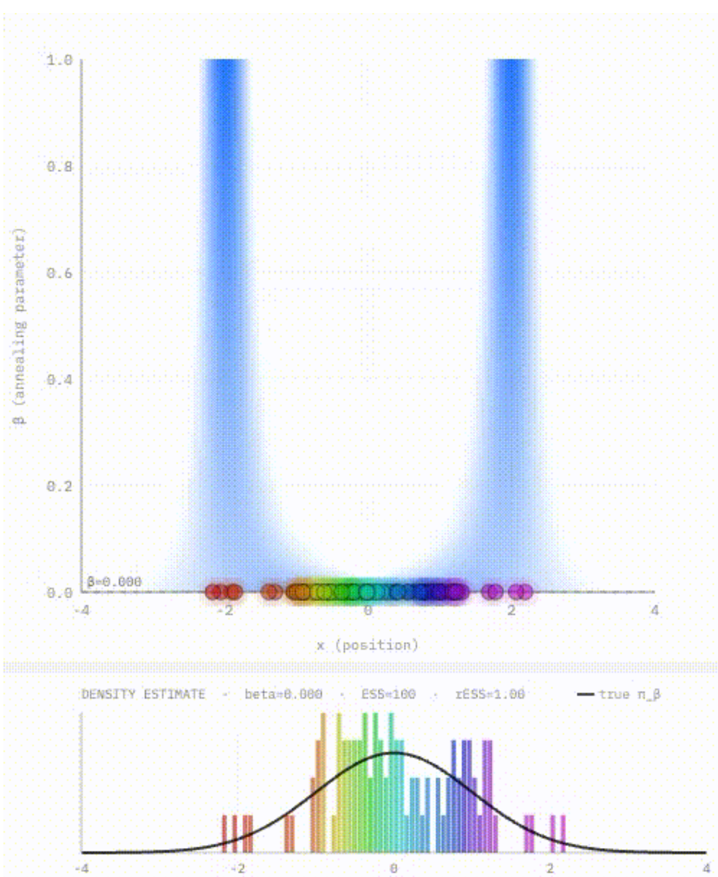

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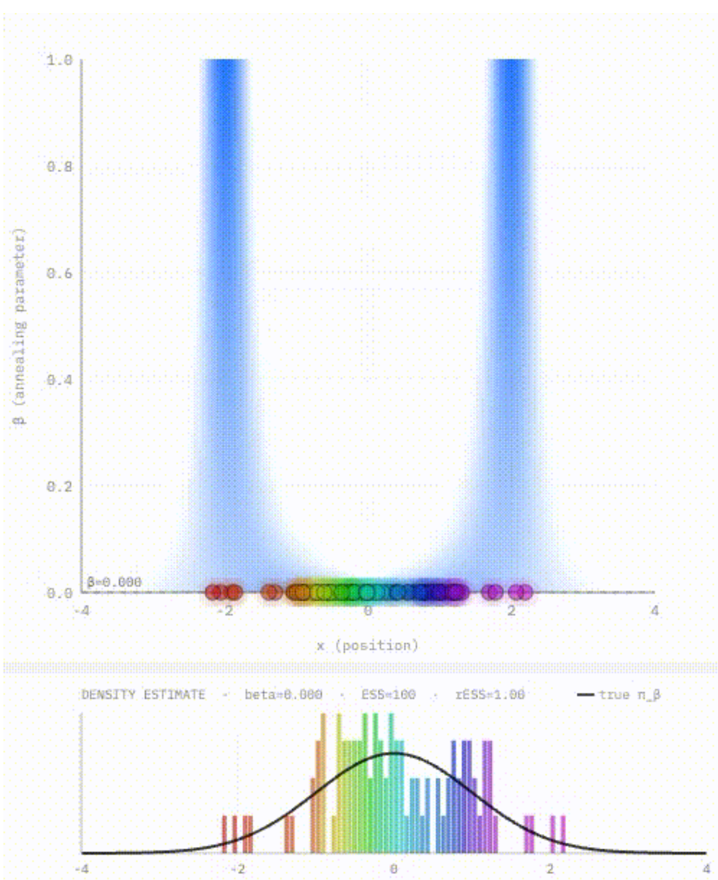

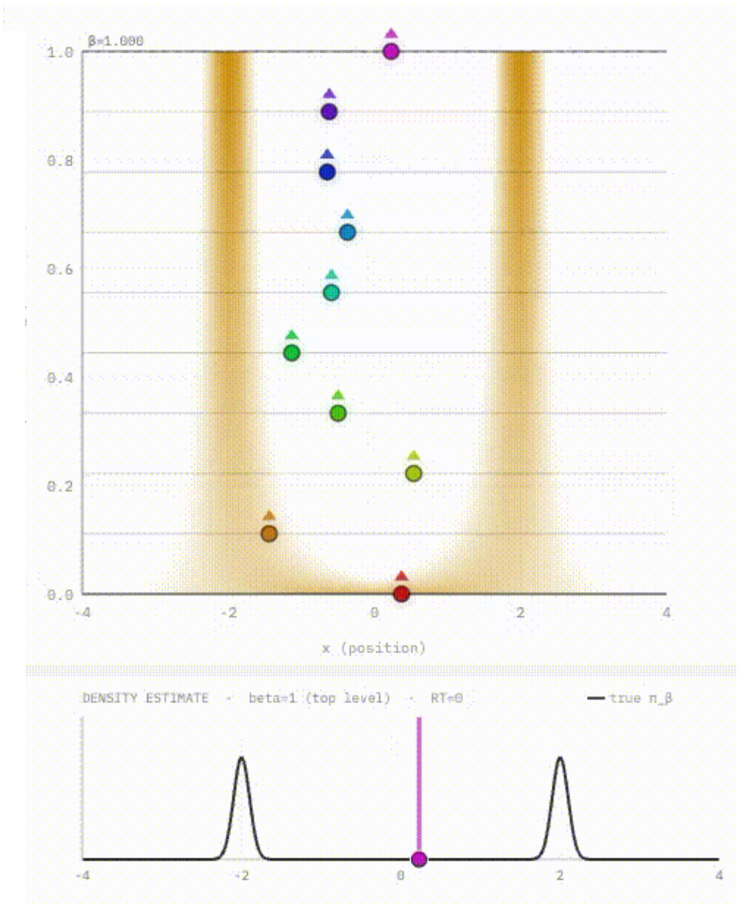

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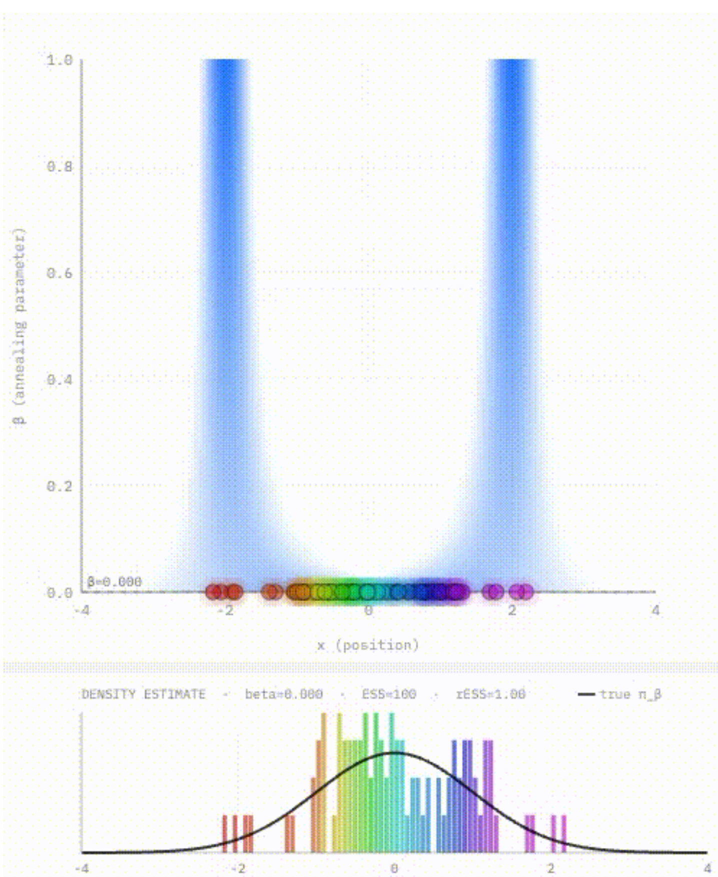

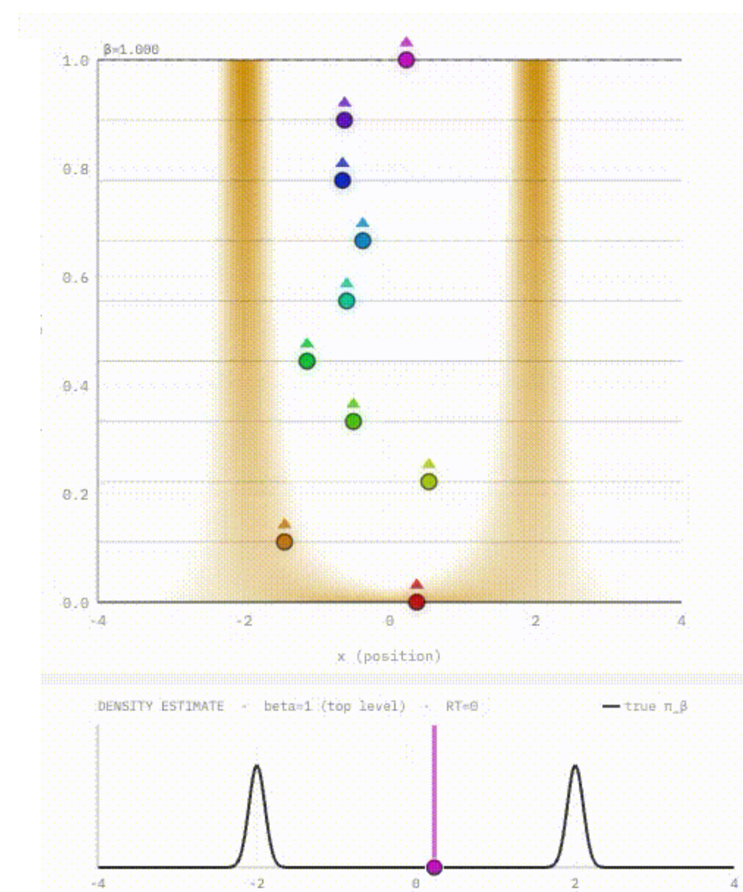

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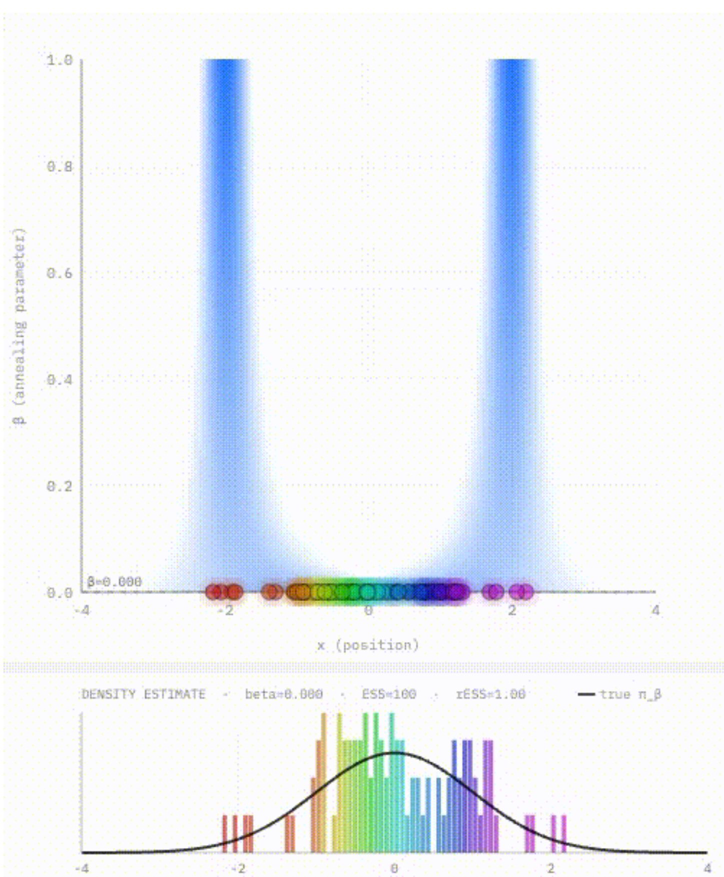

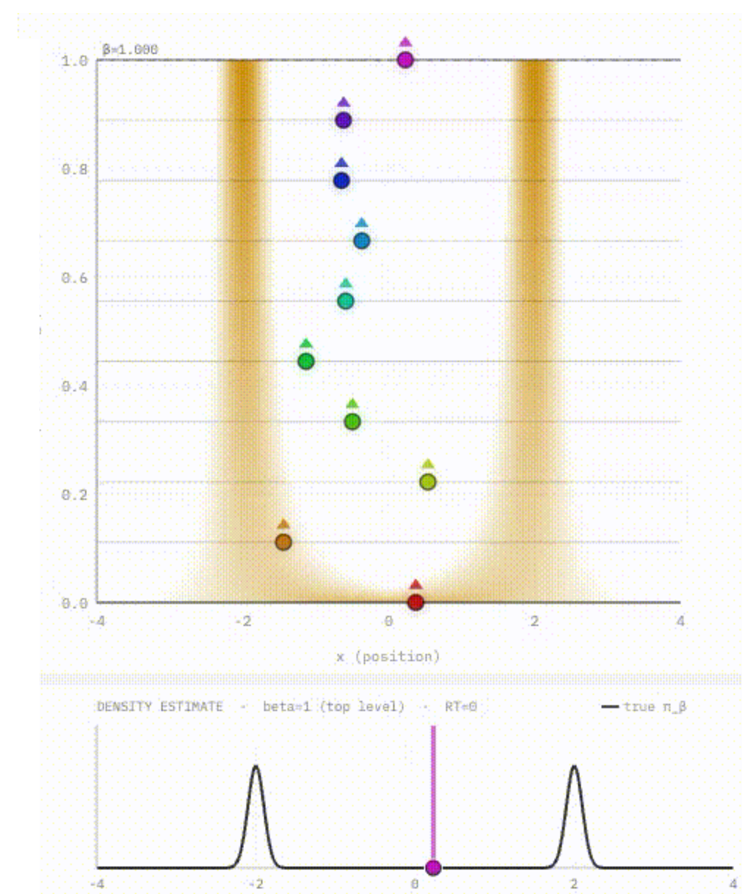

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- Built using importance sampling

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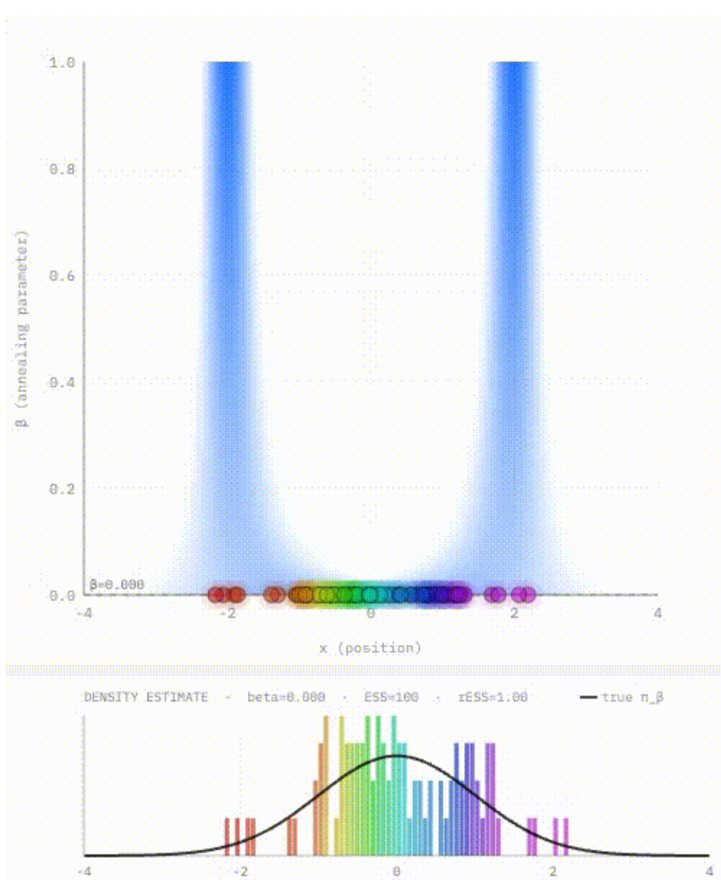

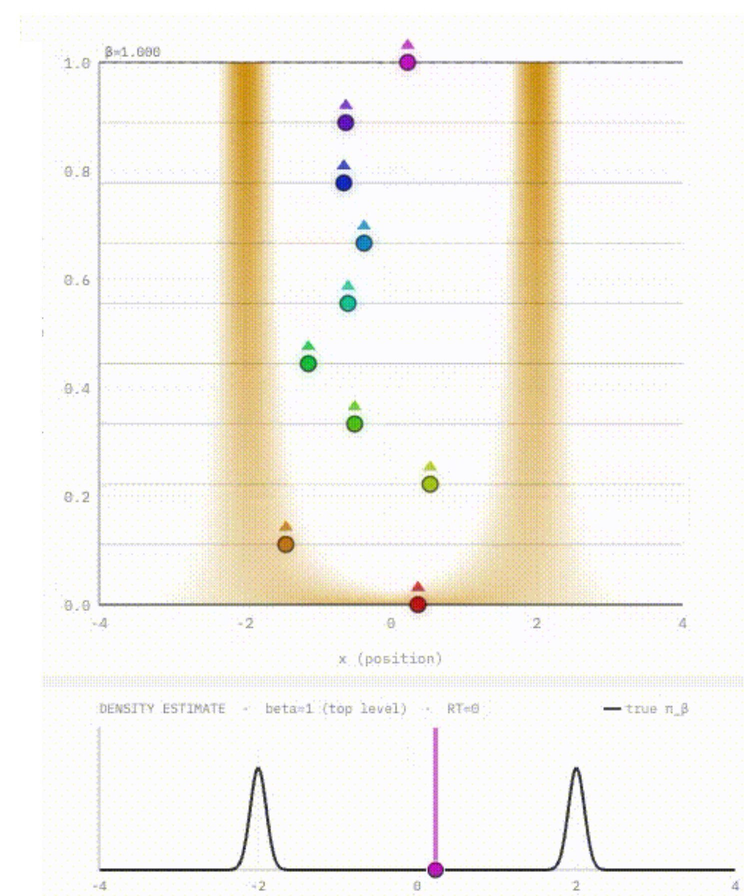

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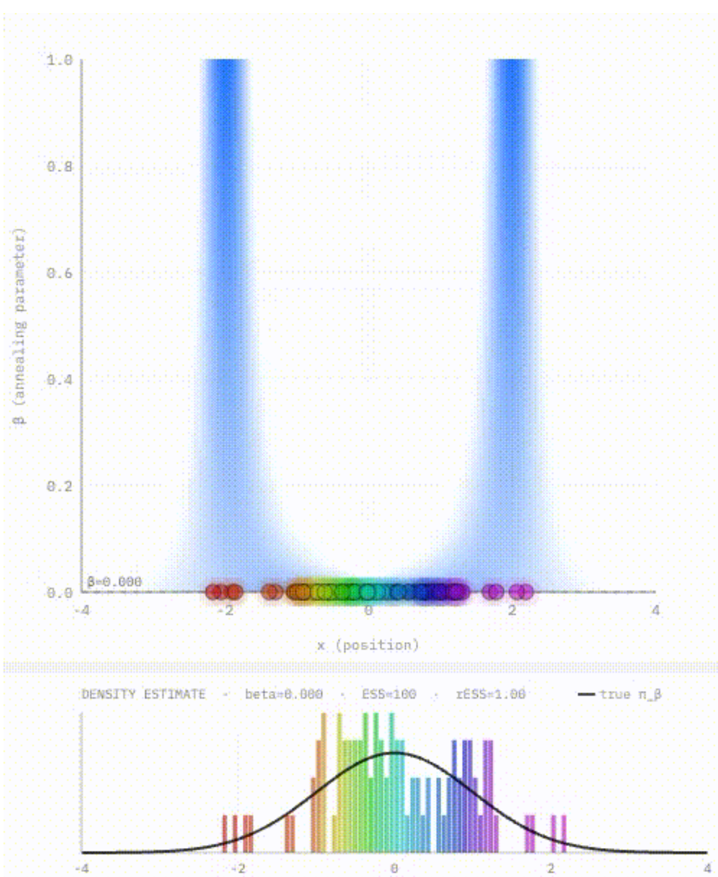

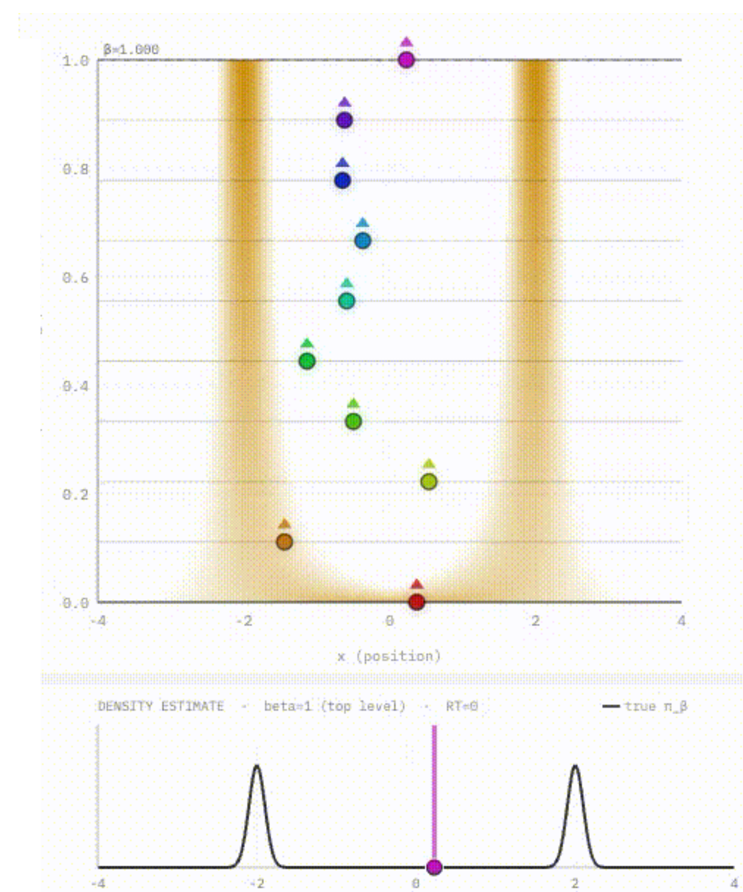

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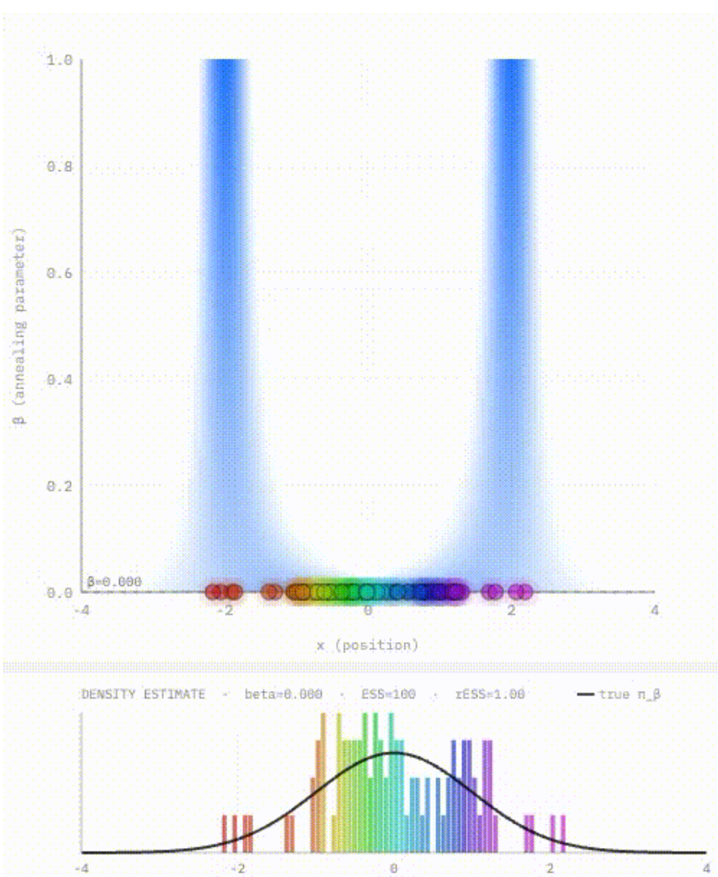

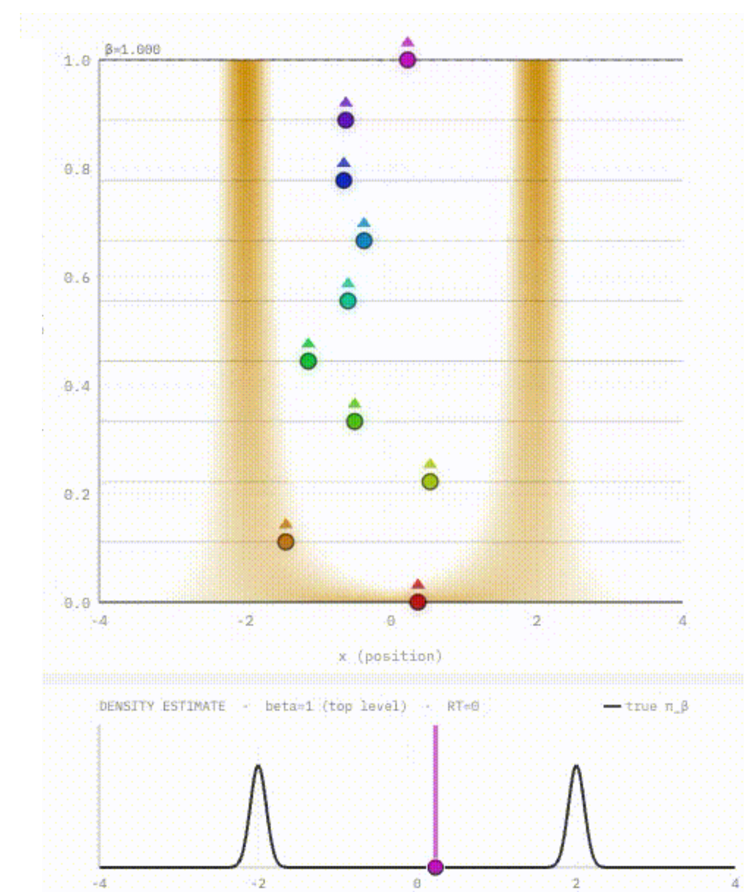

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Parallel Annealing

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- Samples generated in sequentially
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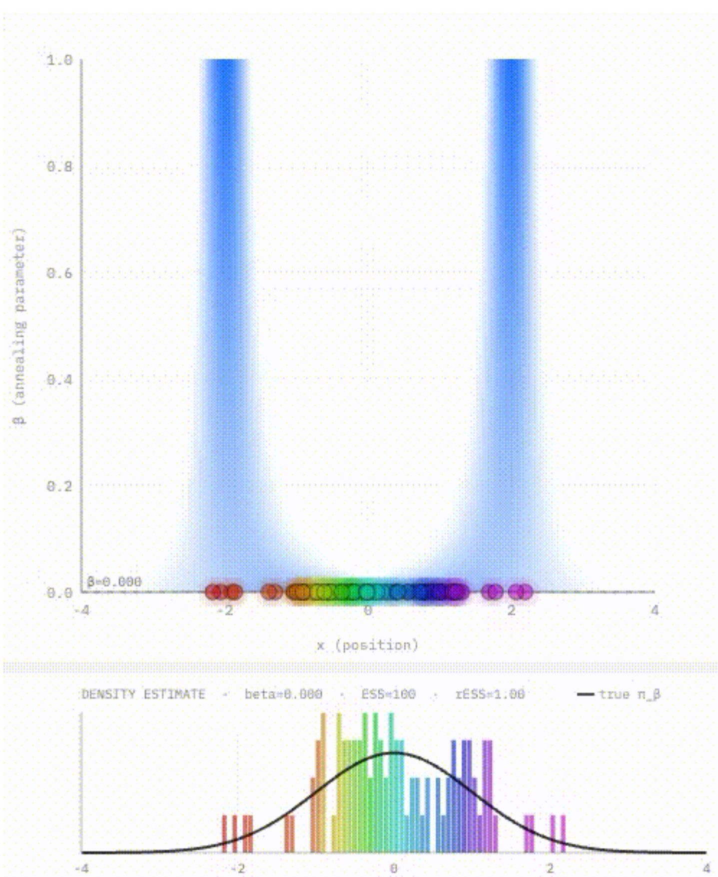
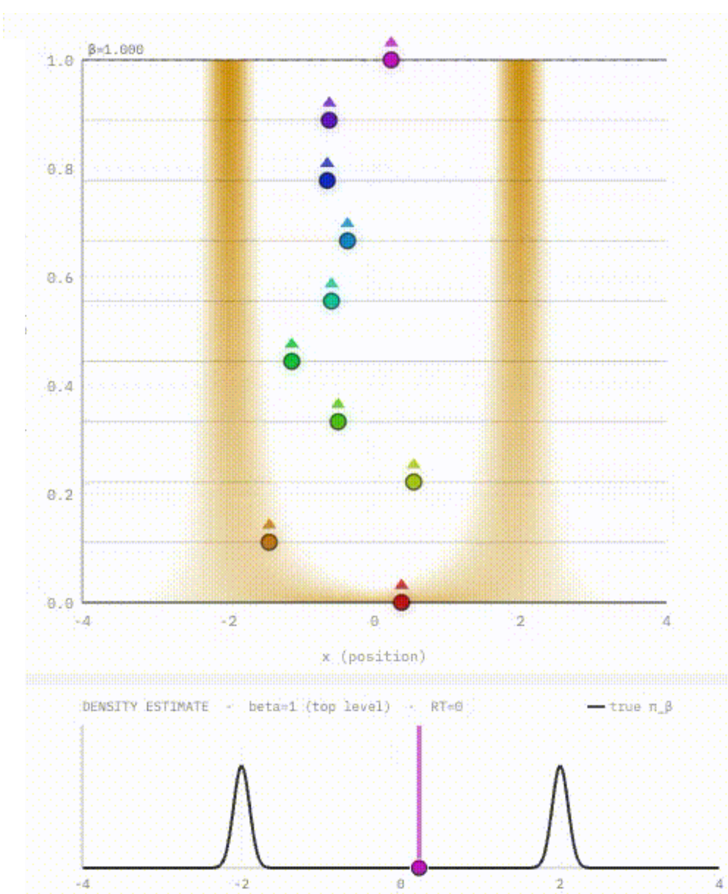
- Interacting → Parallel Tempering





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- Distributed → Simulated sampling

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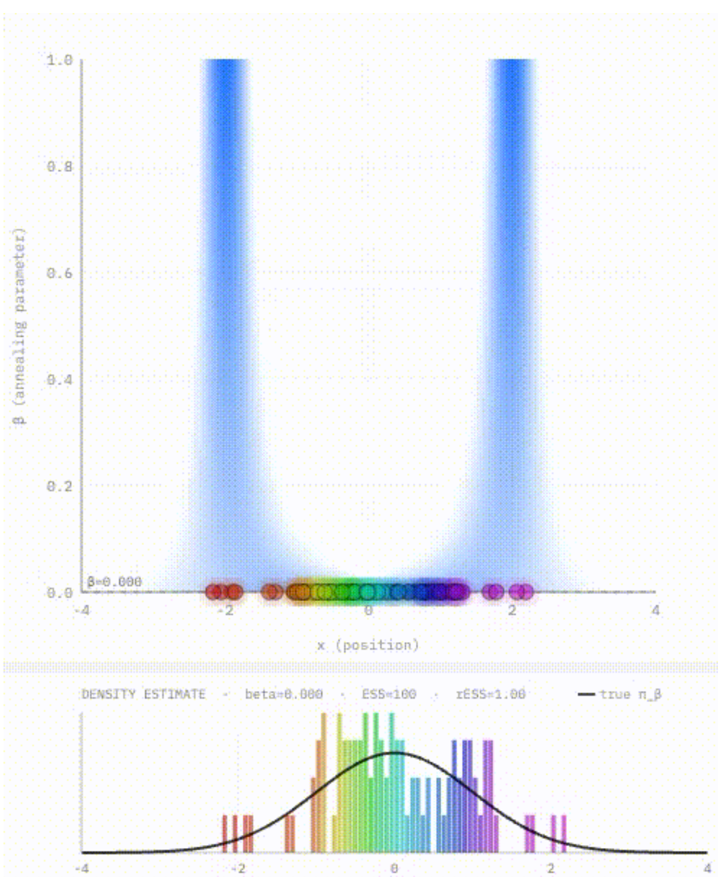

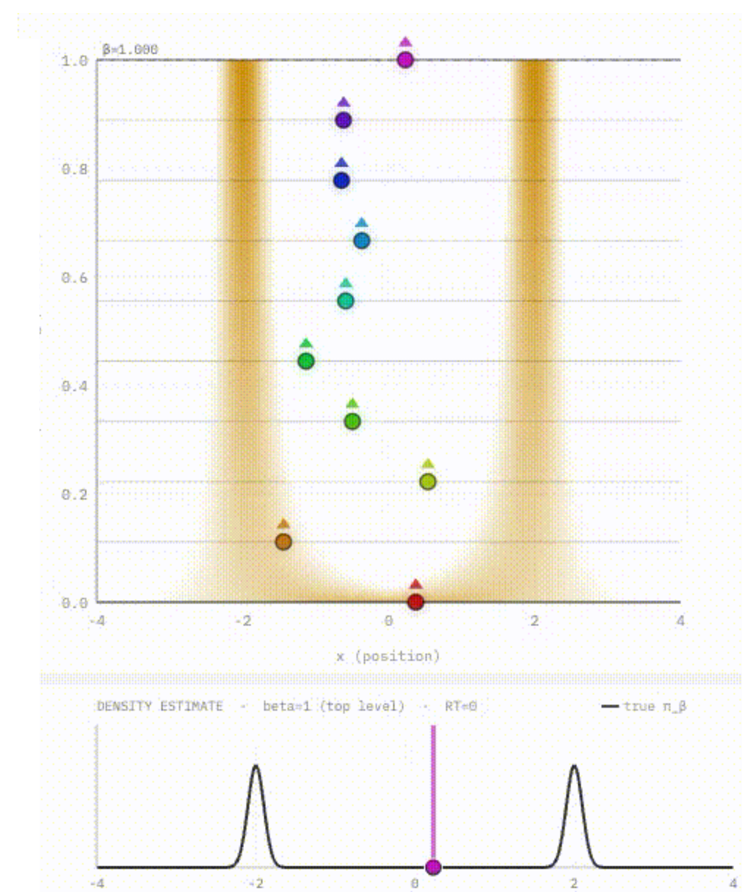

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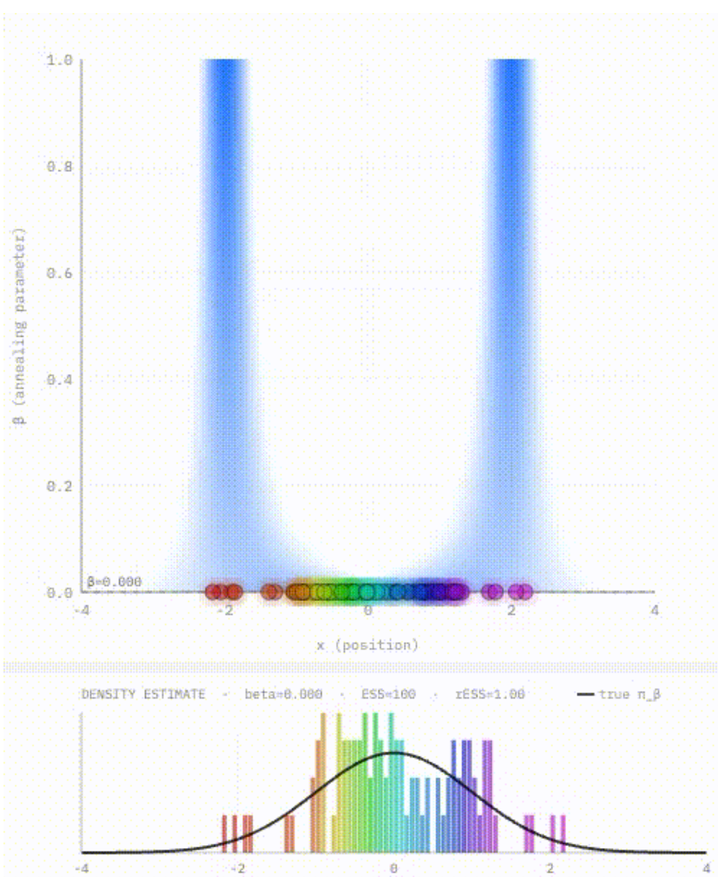
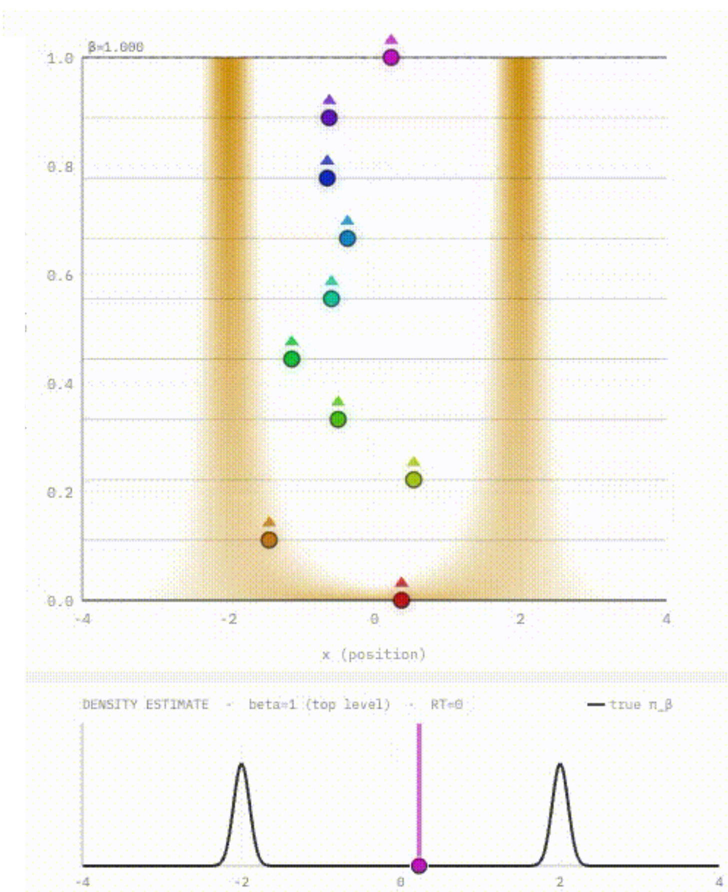
- Distributed → Annealed Importance Sampling





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    - Annealed importance sampling (AIS)

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  - **Parallel Annealing:**
    - Parallel Tempering (PT)
    - Simualted Tempering (ST)
  - **Sequential annealing:**
    - Annealed importance sampling (AIS)
    - (Annealed) Sequential Monte Carlo samplers (ASMC)

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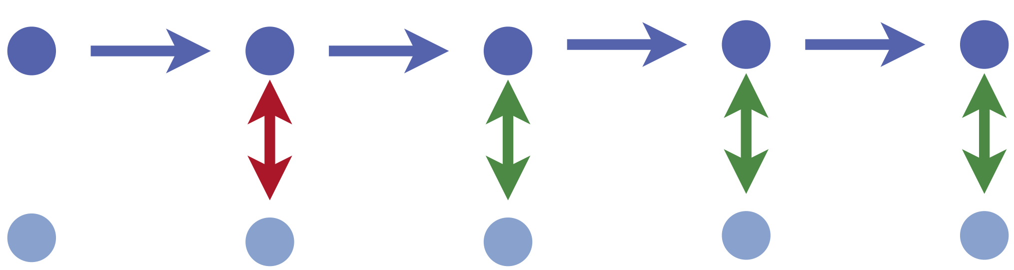
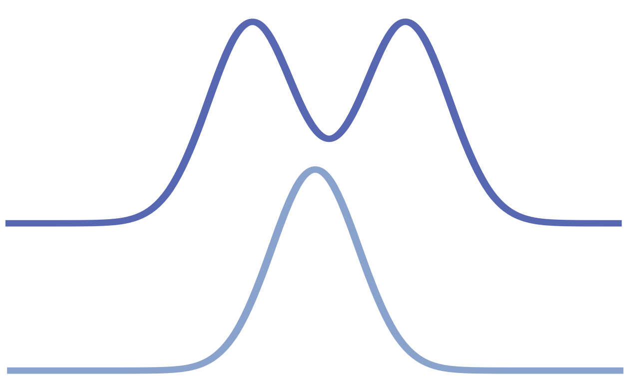
  - **Tuning annealing algorithms**
  - **Normalising constant estimation**

RECALL STABILISED MCMC

- Referenced stabilised MCMC is **stable** when reference is close to target
    - Acceptance rate is stable





RECALL STABILISED MCMC

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- Referenced stabilised MCMC is **fragile** when the reference share little support with target
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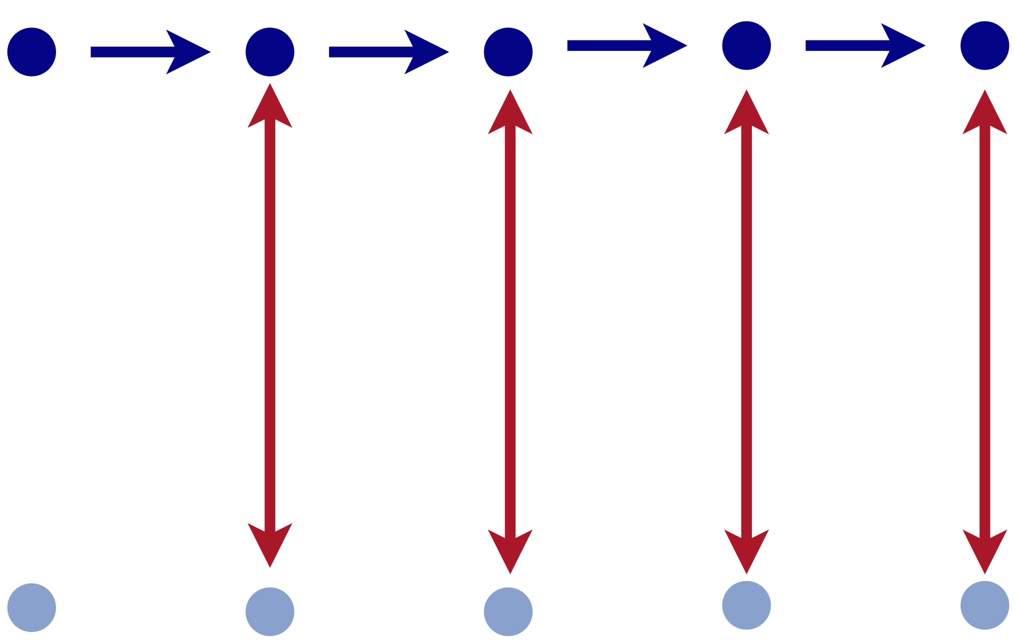

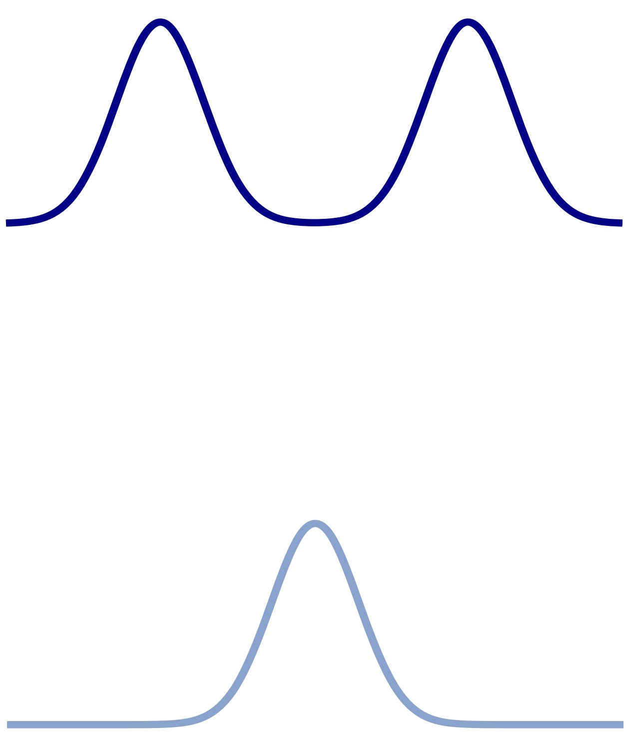

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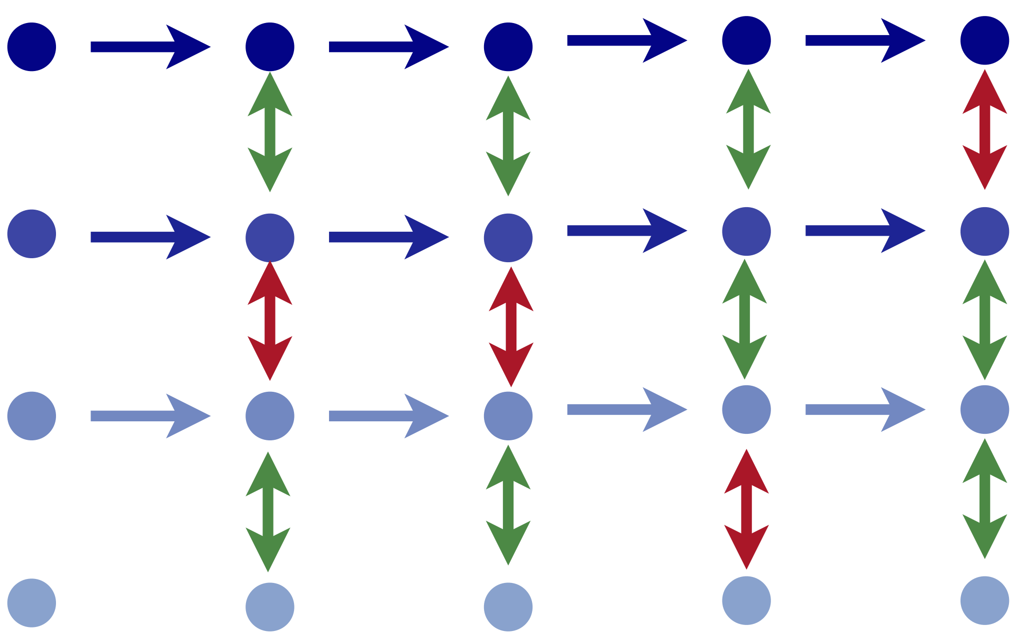
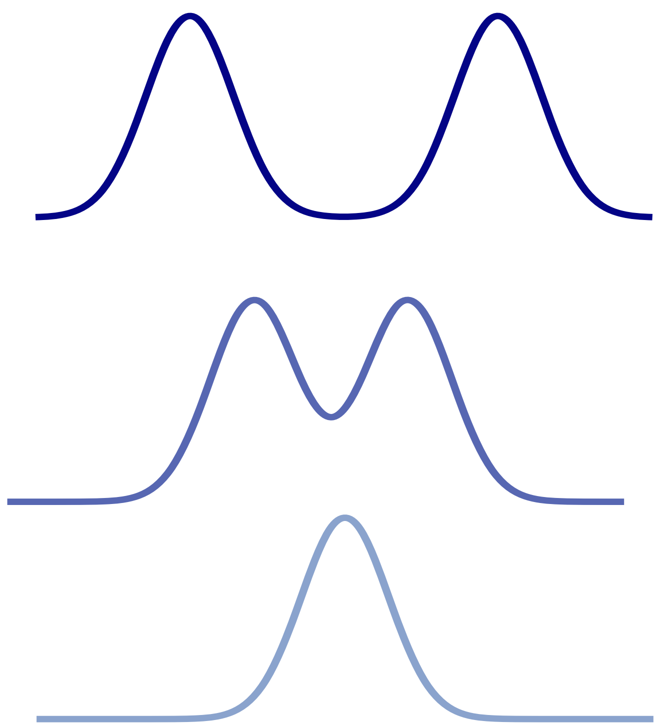
- Introduce multiple interpolating distributions with enough overlap to maintain stability





PARALLEL TEMPERING

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- Uses stabilised MCMC to target annealing distributions in **parallel**

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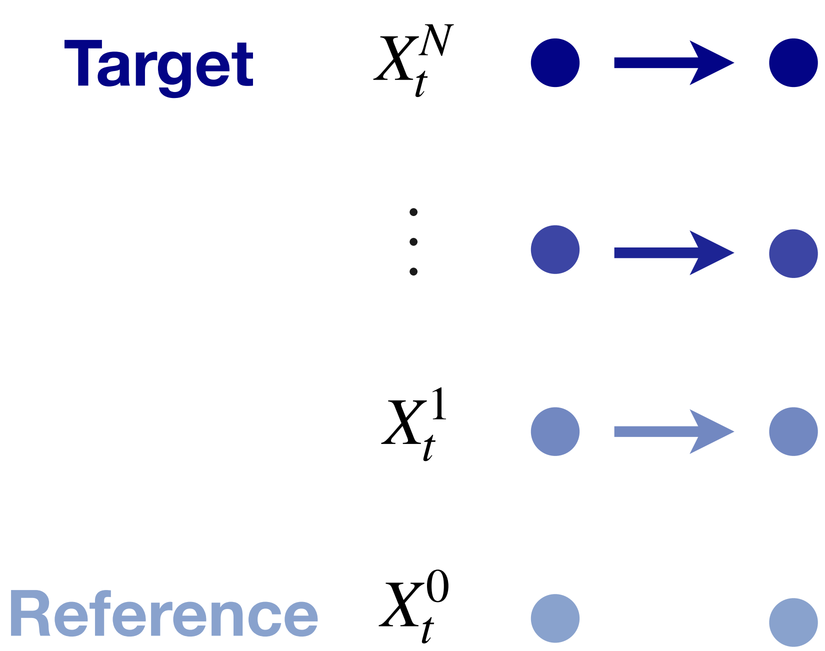

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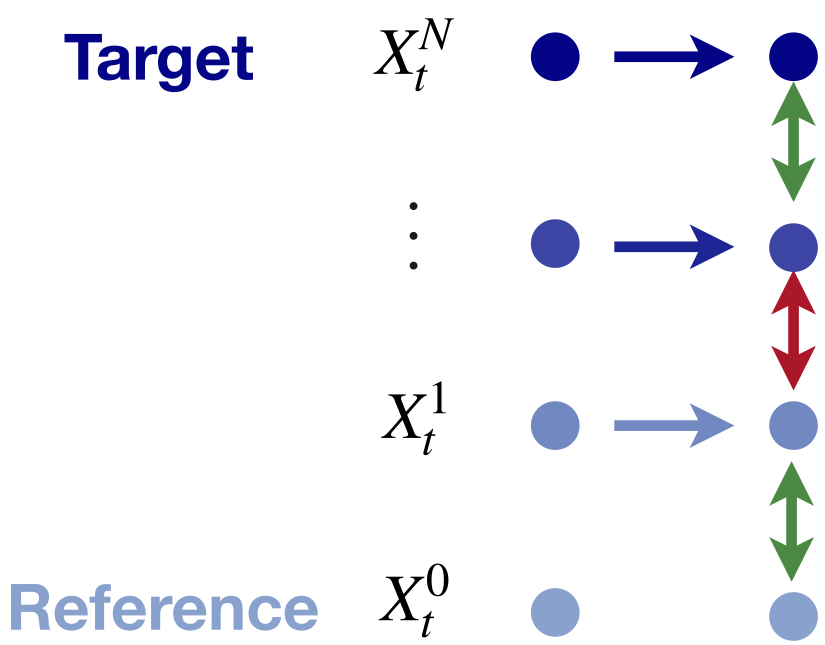

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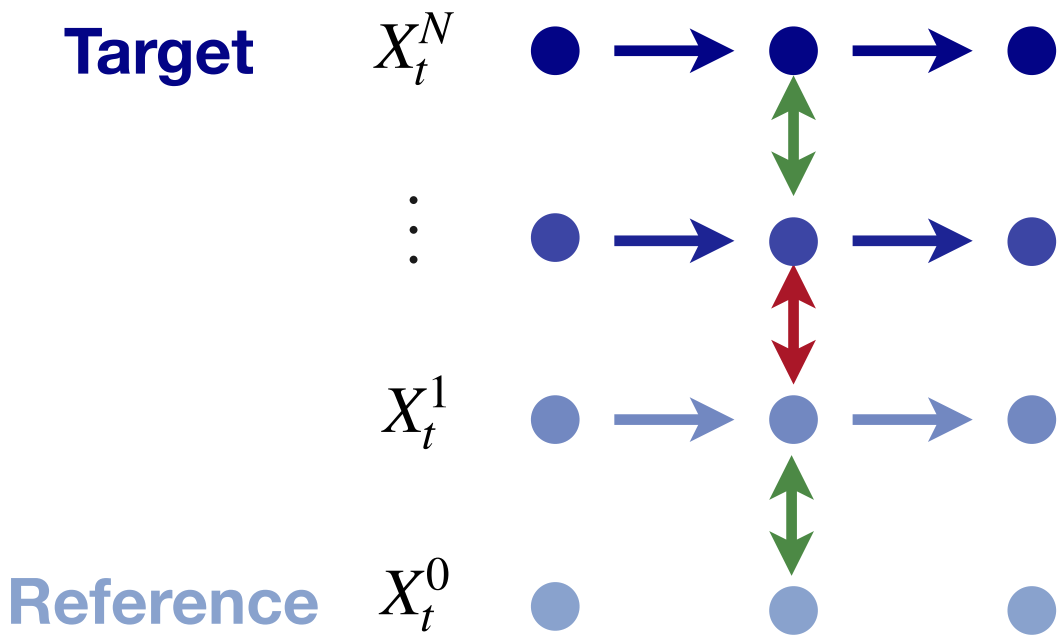

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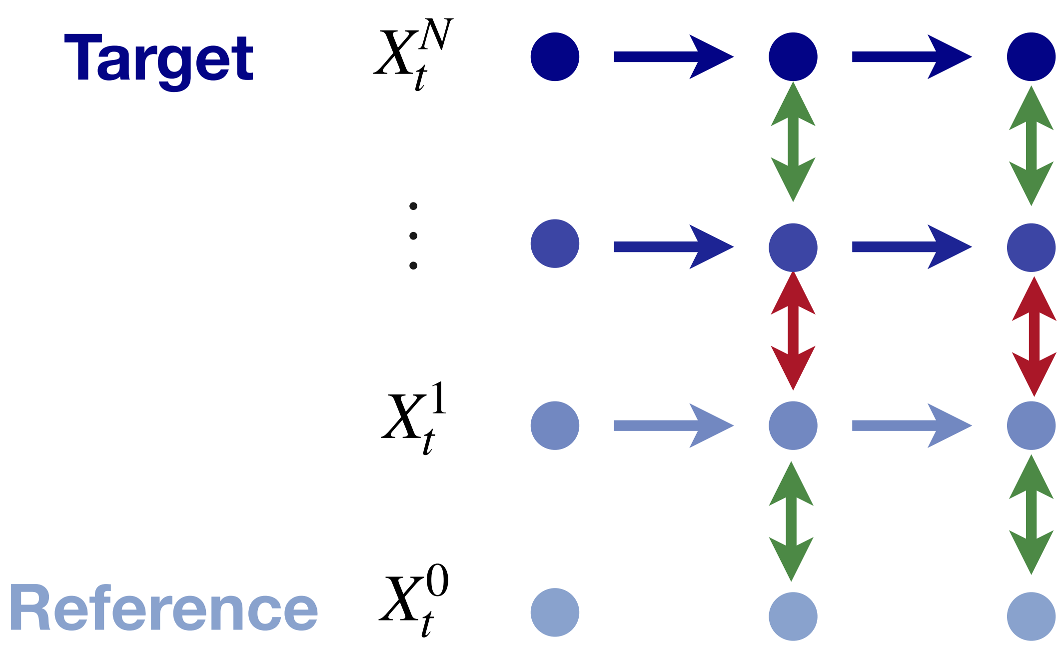

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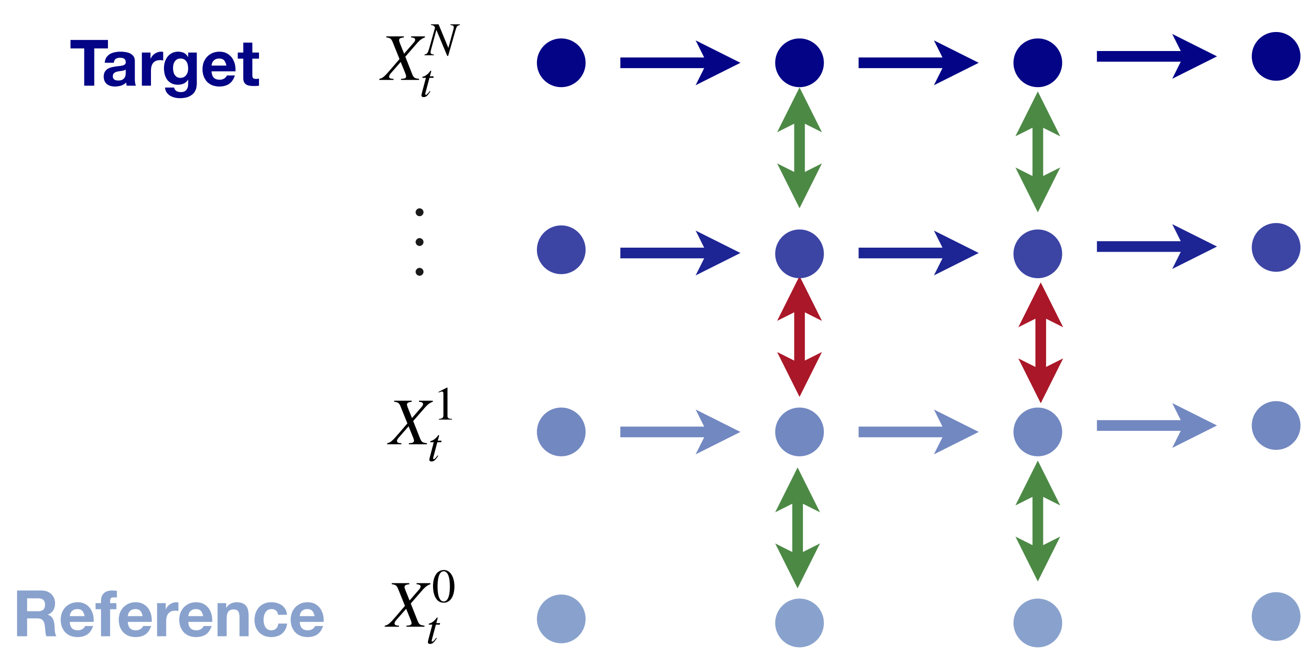

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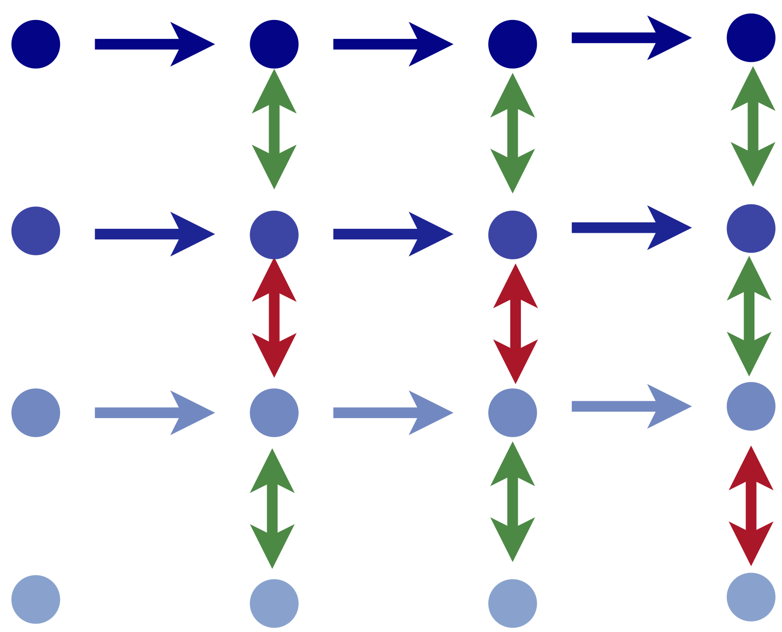

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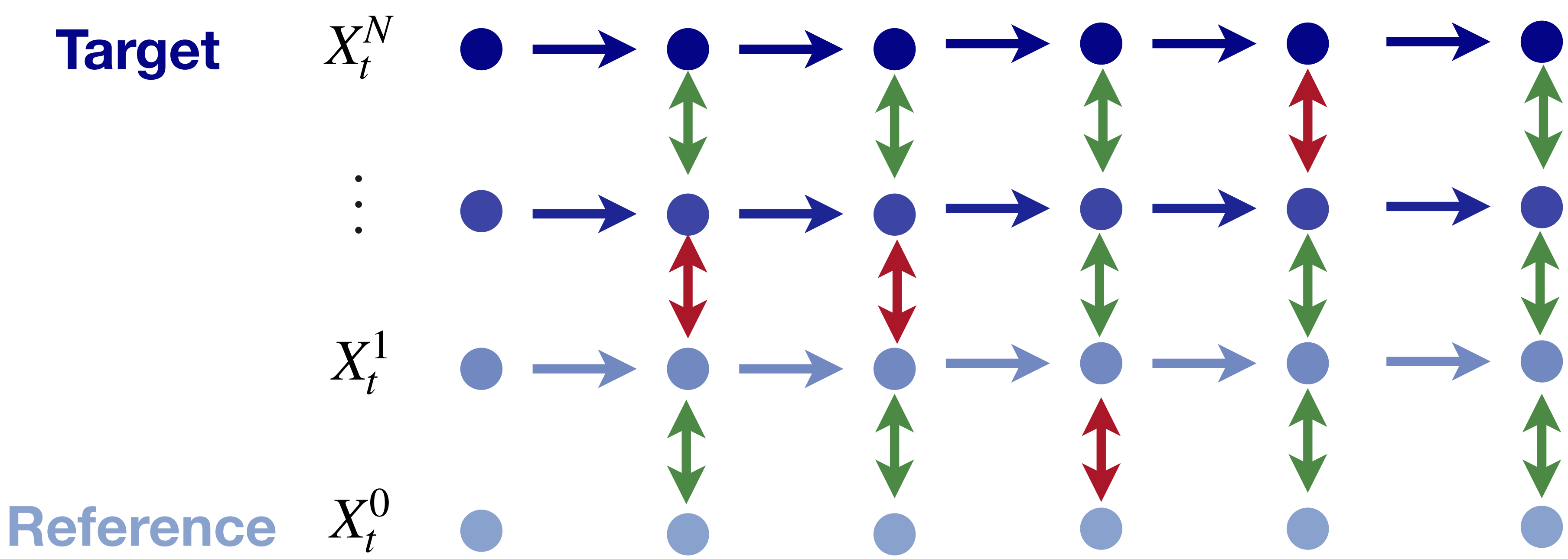

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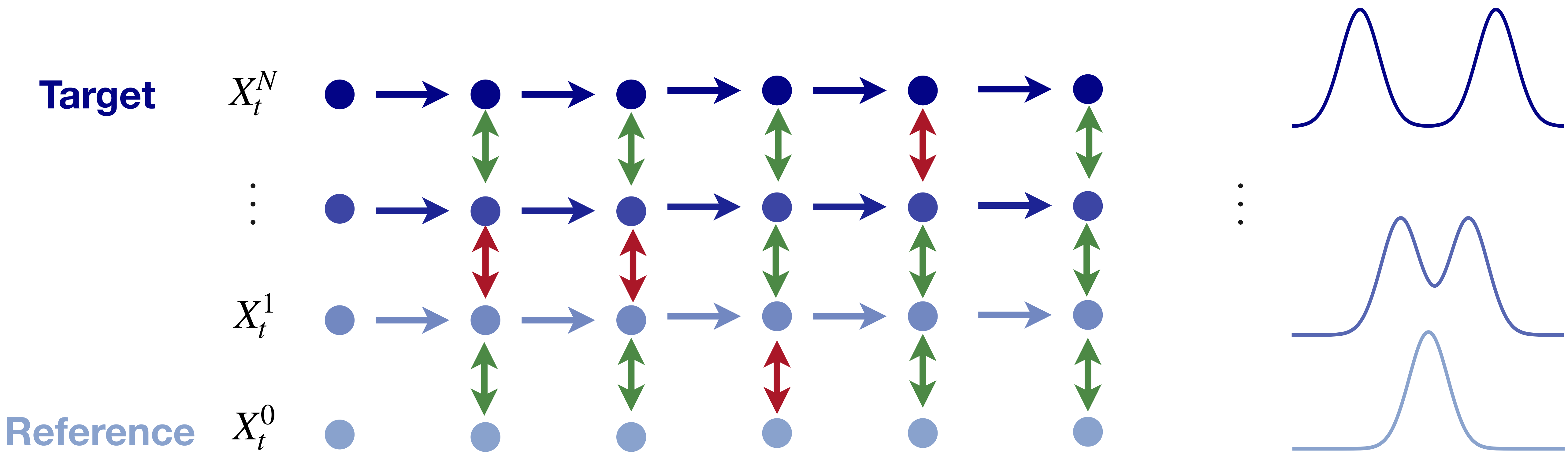

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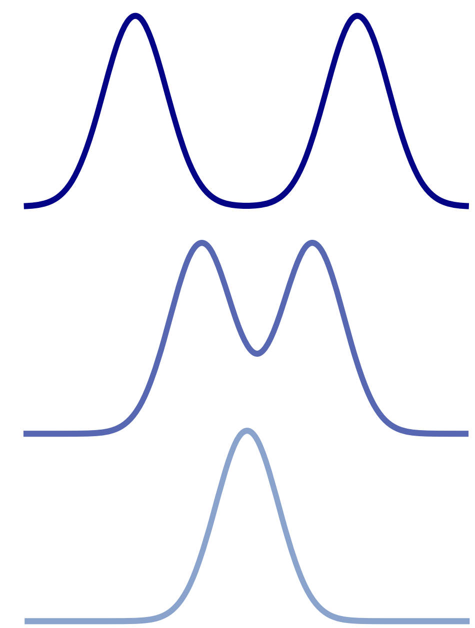
- Parallel tempering constructs MCMC chain $\mathbf{X}_t$ on $\mathbb{X}^{N+1}$ targetting $\pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$

$$\mathbf{X}_t = \begin{pmatrix} X_t^N \\ \vdots \\ X_t^n \\ \vdots \\ X_t^0 \end{pmatrix} \begin{array}{l} \sim \pi_{\beta_N} \\ \\ \sim \pi_{\beta_n} \\ \\ \sim \pi_{\beta_0} \end{array}$$

Target States

Interpolating States

Reference States



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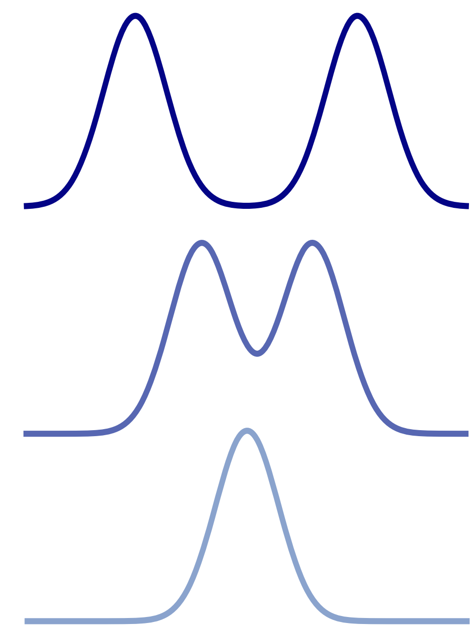

- Generate $\mathbf{X}_t$ from $\mathbf{X}_{t-1}$ using by applying the local exploration and communication moves

$$\mathbf{X}_t \sim \mathbf{K}_{\mathrm{PT}}(\mathbf{X}_{t-1}, \mathrm{d}\mathbf{x}') \qquad \mathbf{K}_{\mathrm{PT}} = \mathbf{K}_{\mathrm{expl}} \mathbf{K}_{\mathrm{comm}}$$

LOCAL EXPLORATION

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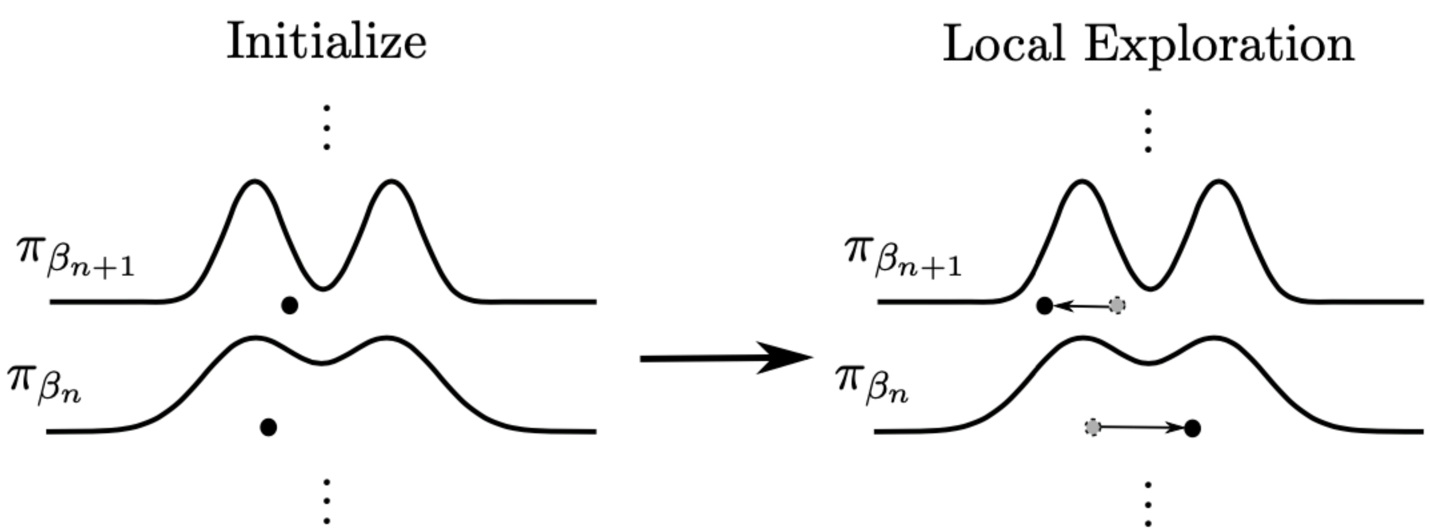

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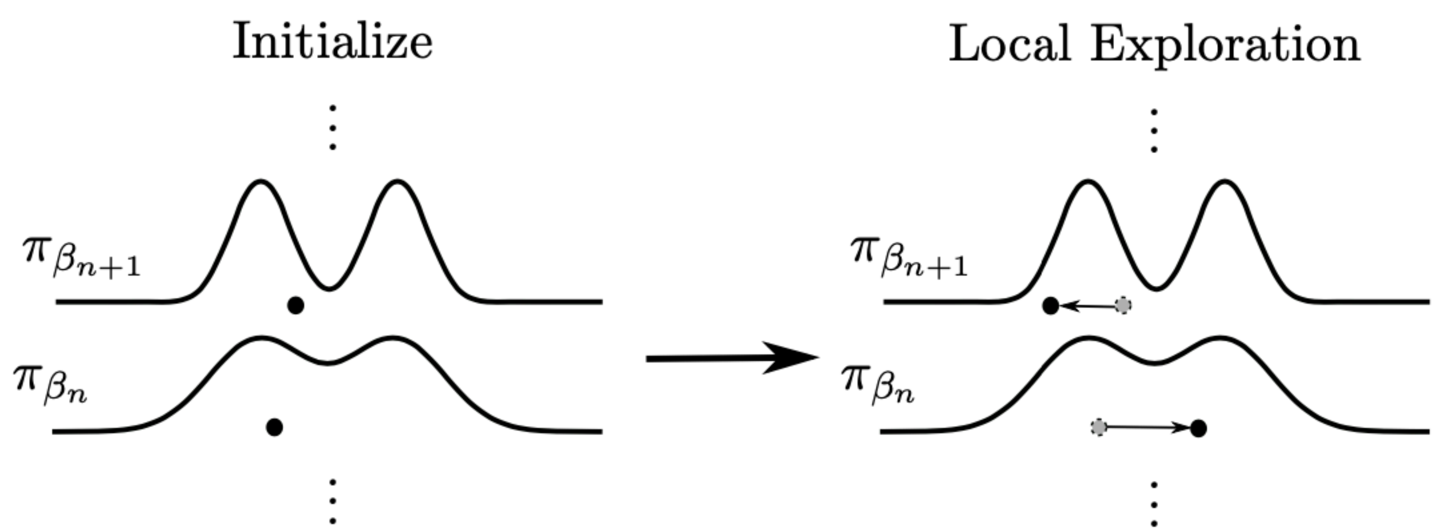

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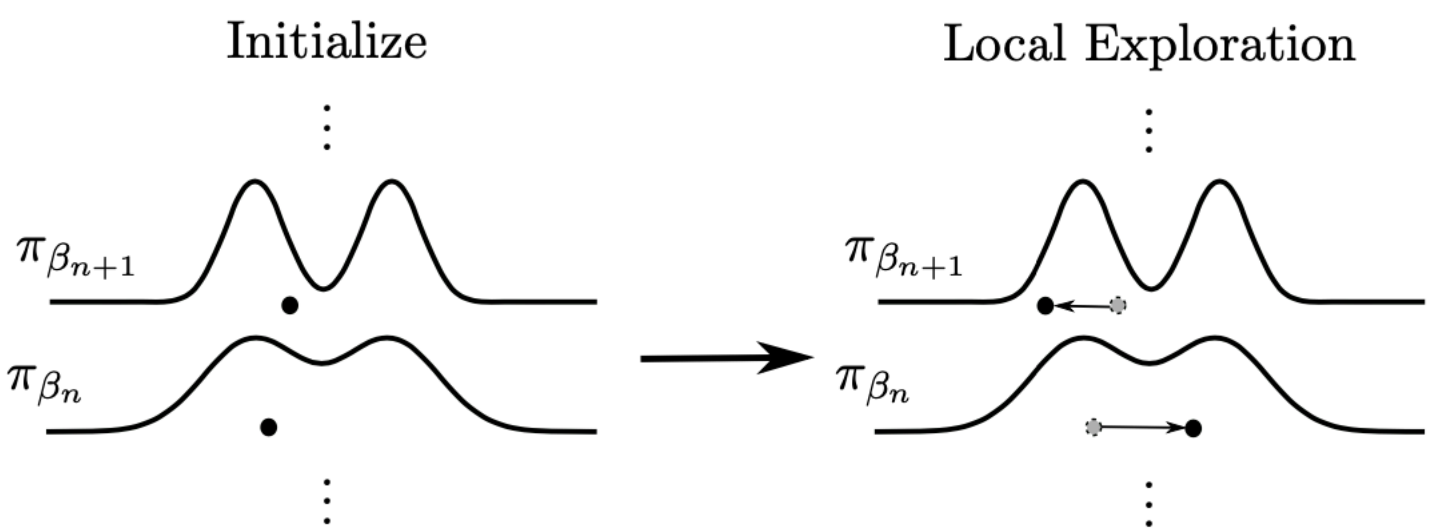

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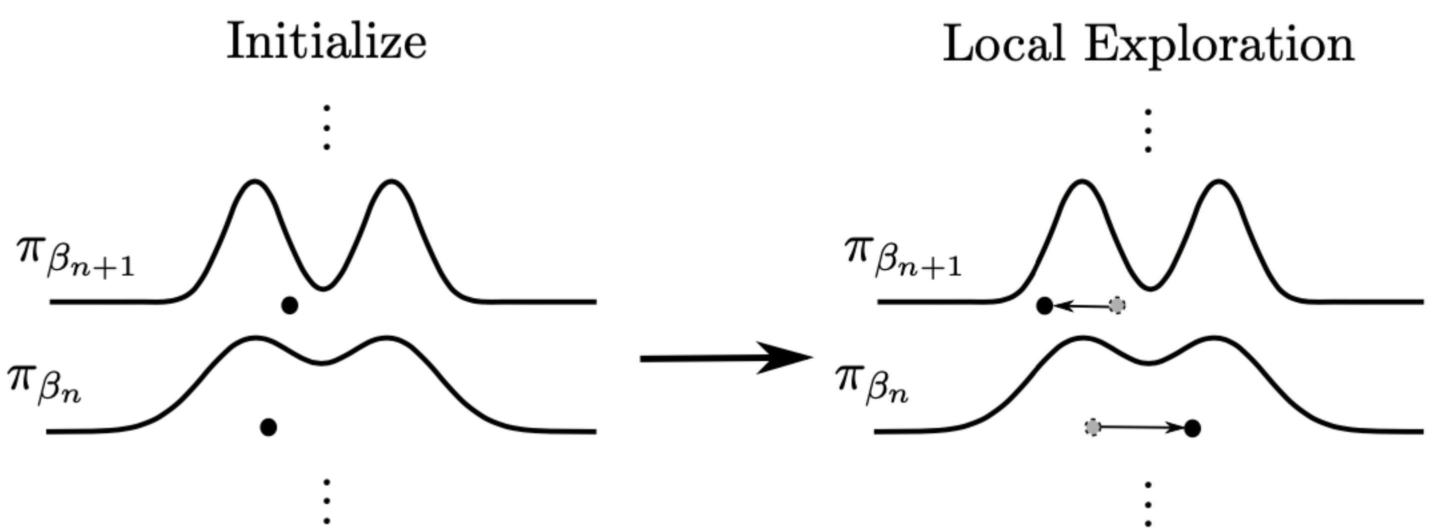

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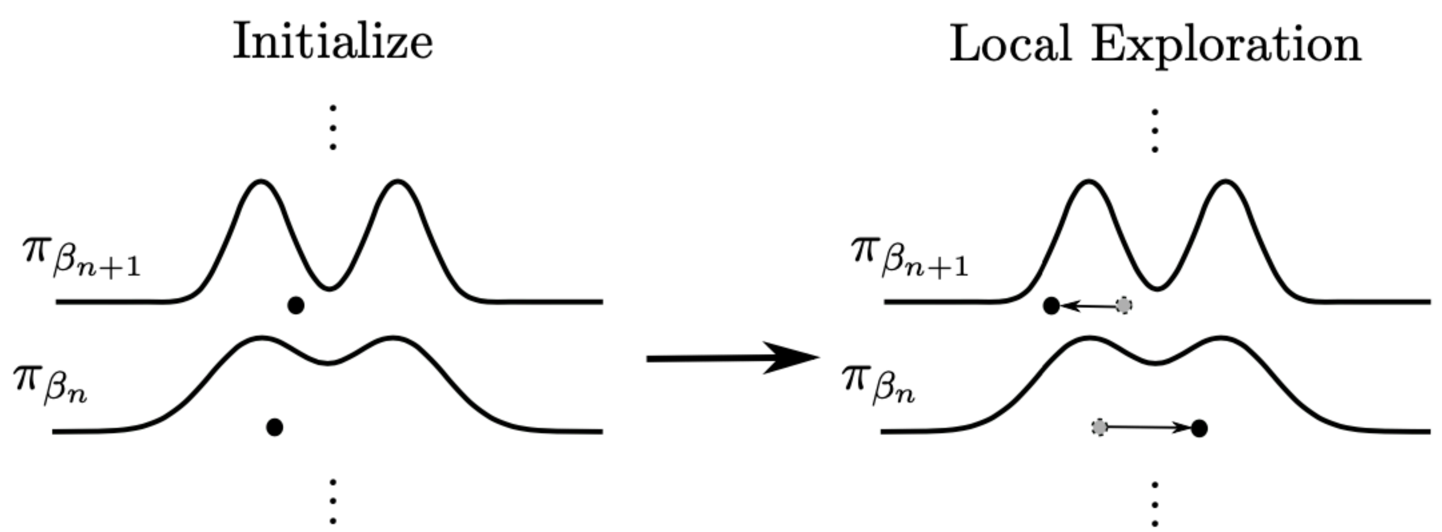

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    - Assume that an independent sample from reference

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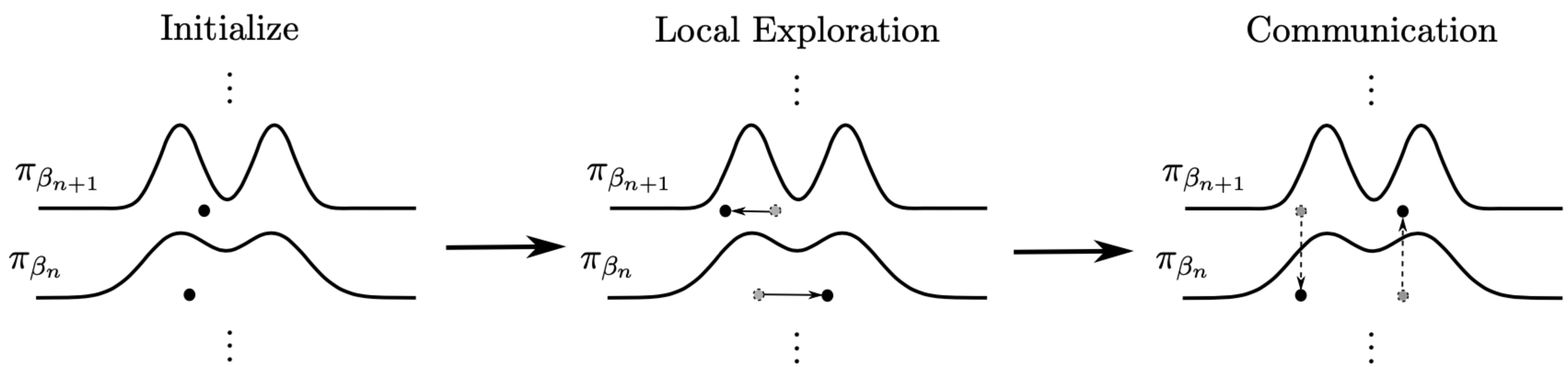
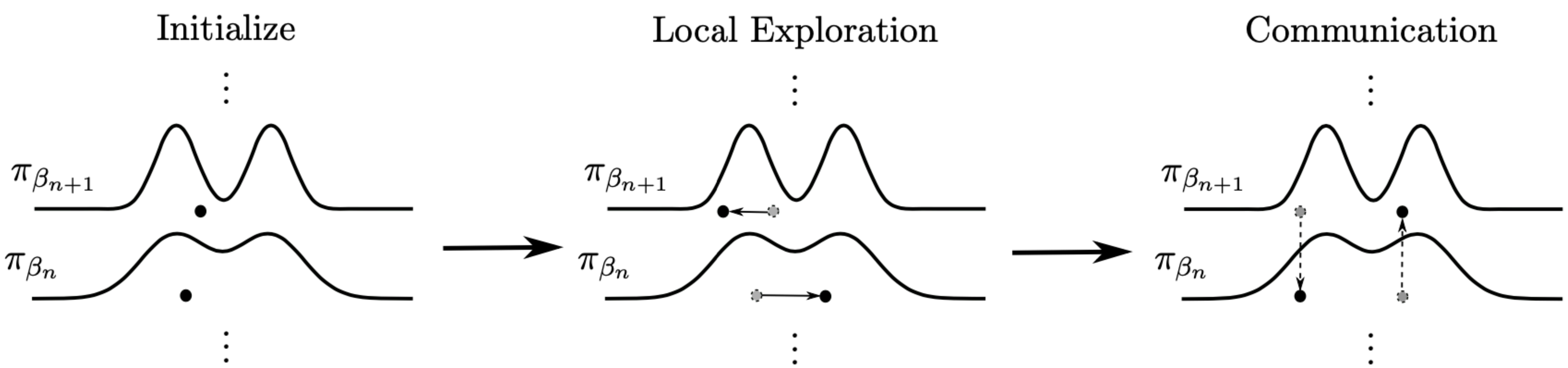
$$= \eta(\mathrm{d}x'^0) \prod_{n=1}^{N} K_{\beta_n}(x^n, \mathrm{d}x'^n)$$

SWAP MOVE


Initialize
Local Exploration
Communication
$\pi_{\beta_{n+1}}$
π_{β_n}

SWAP MOVE



- The n-th swap move takes the state $\mathbf{x}$ proposes $\mathbf{x}^{(n,n+1)}$ swapping components n and $n+1$

$$\mathbf{x}^{(n,n+1)} = (x^0, \ldots, x^{n-1}, x^{n+1}, x^n, x^{n+2}, \ldots, x^N)$$

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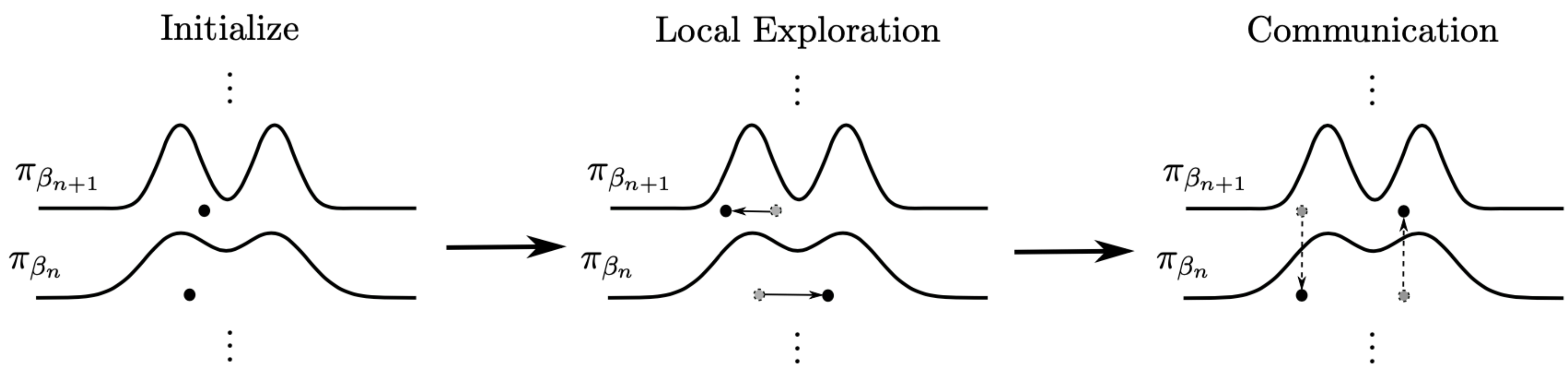

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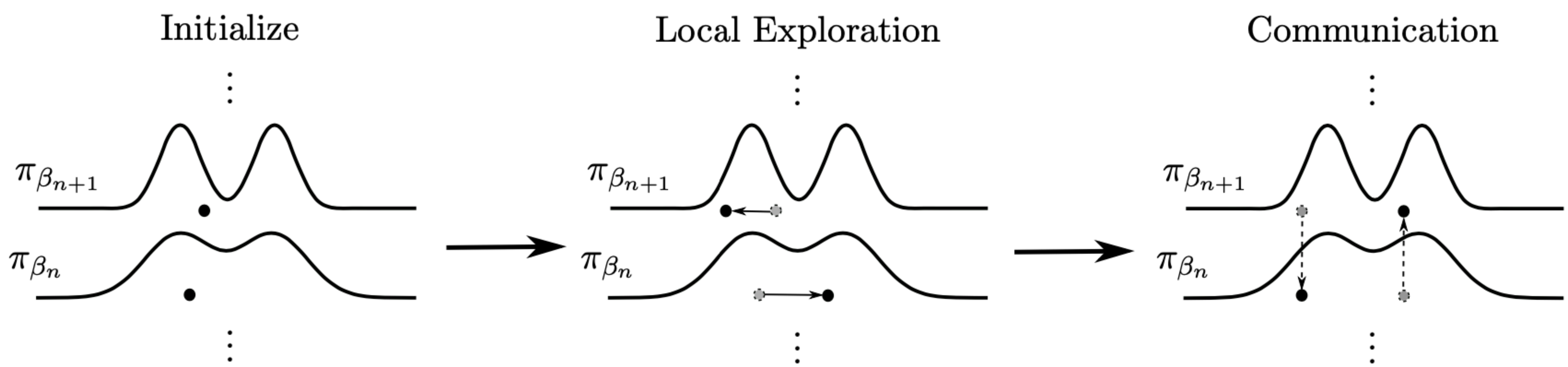

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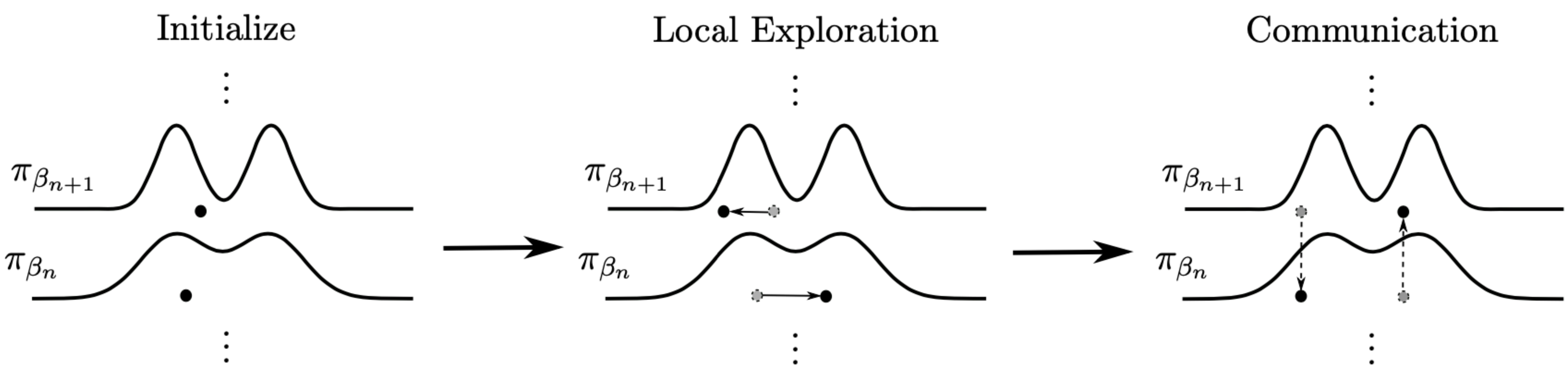

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COMMUNICATION

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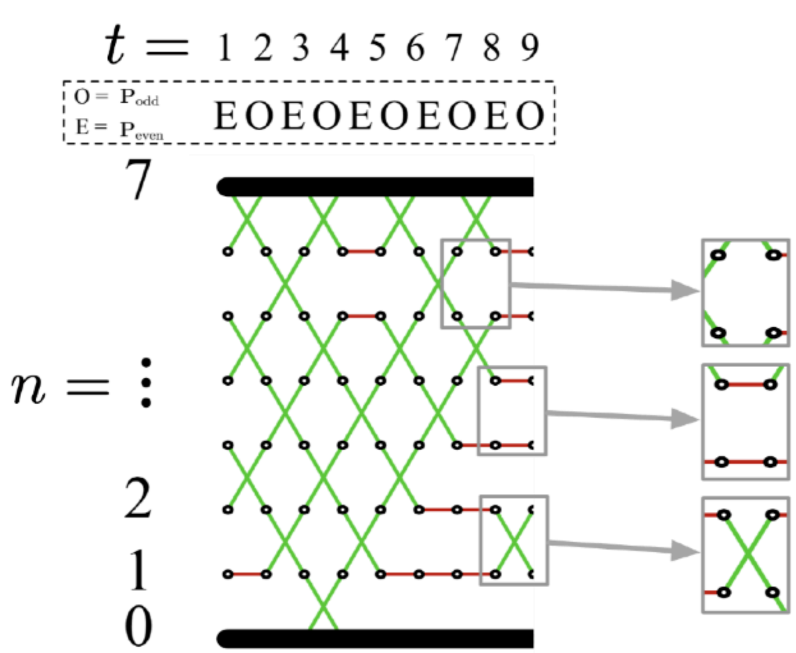

Proposal indicators

$P_7^{(5,6)} = 0$

$P_8^{(3,4)} = 1$

$P_8^{(1,2)} = 1$

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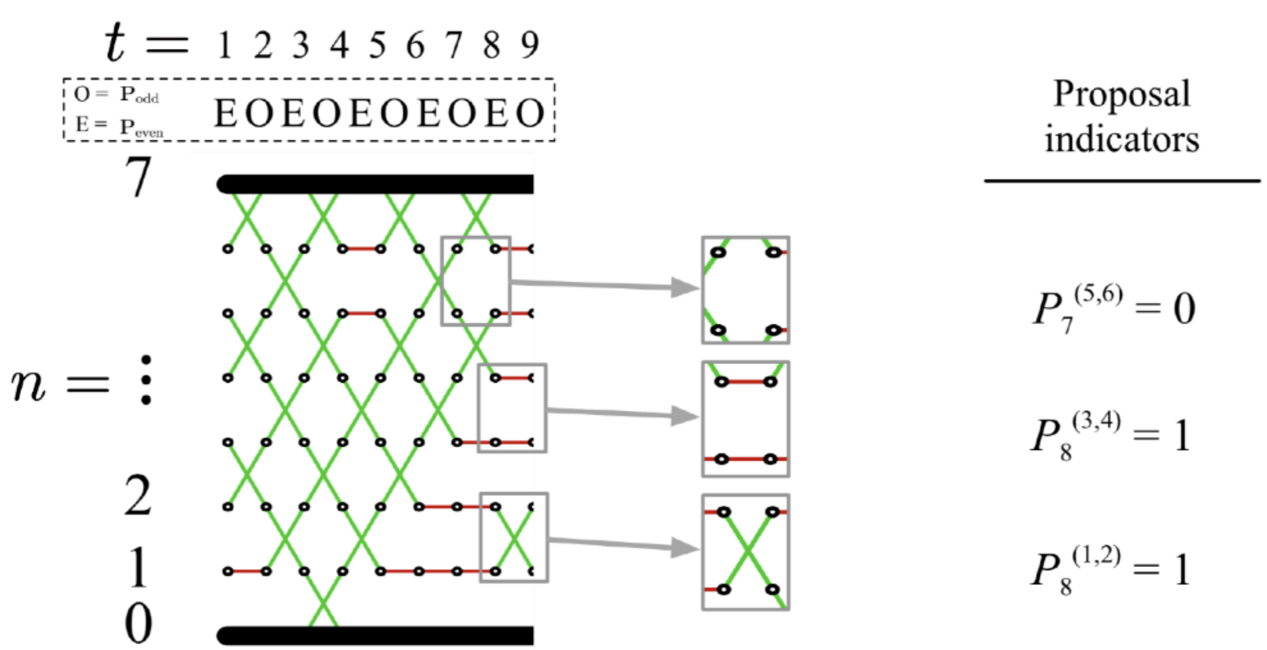

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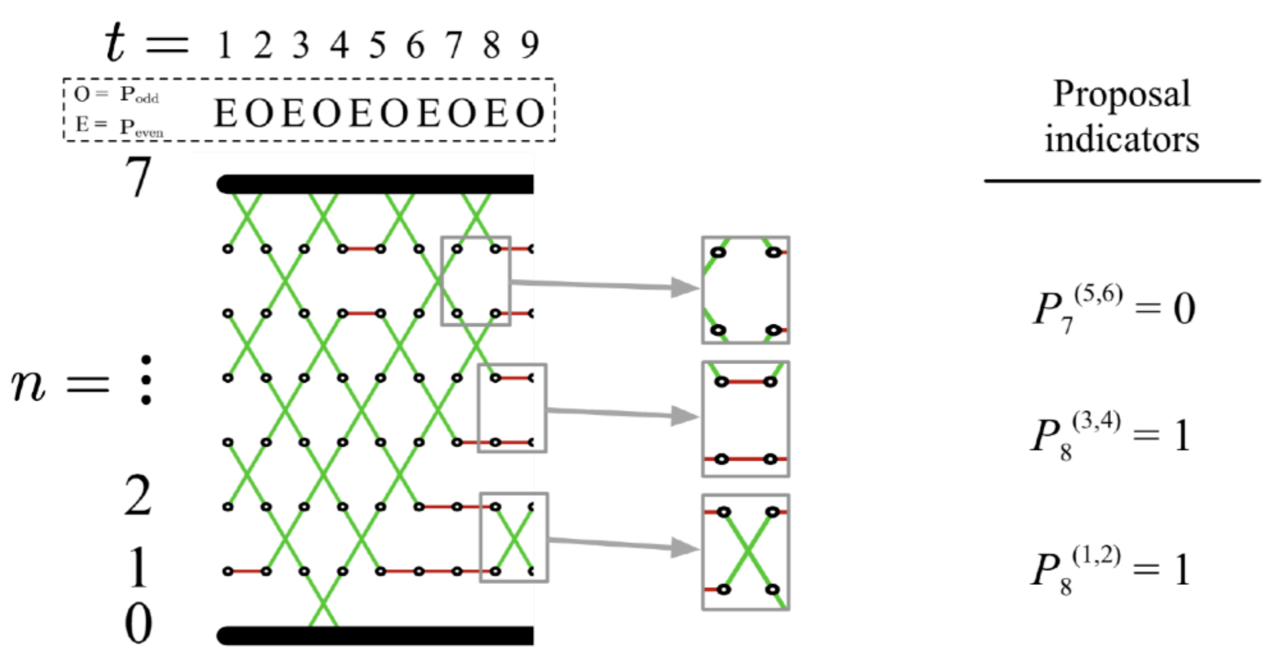

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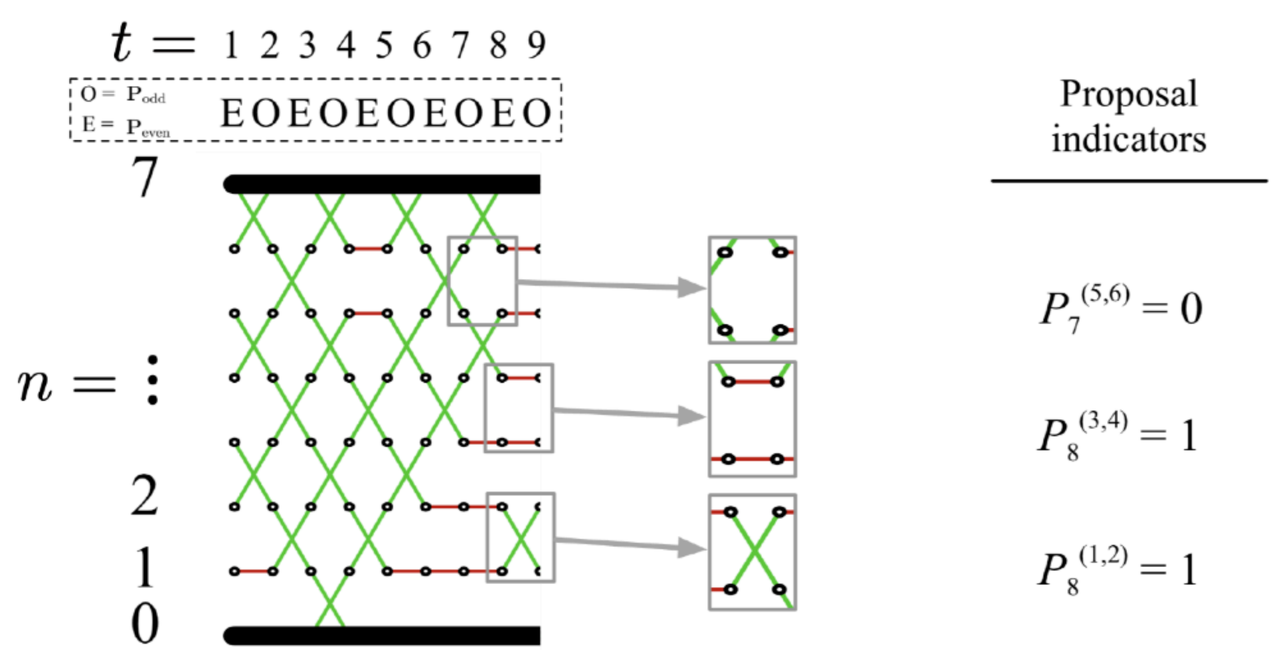

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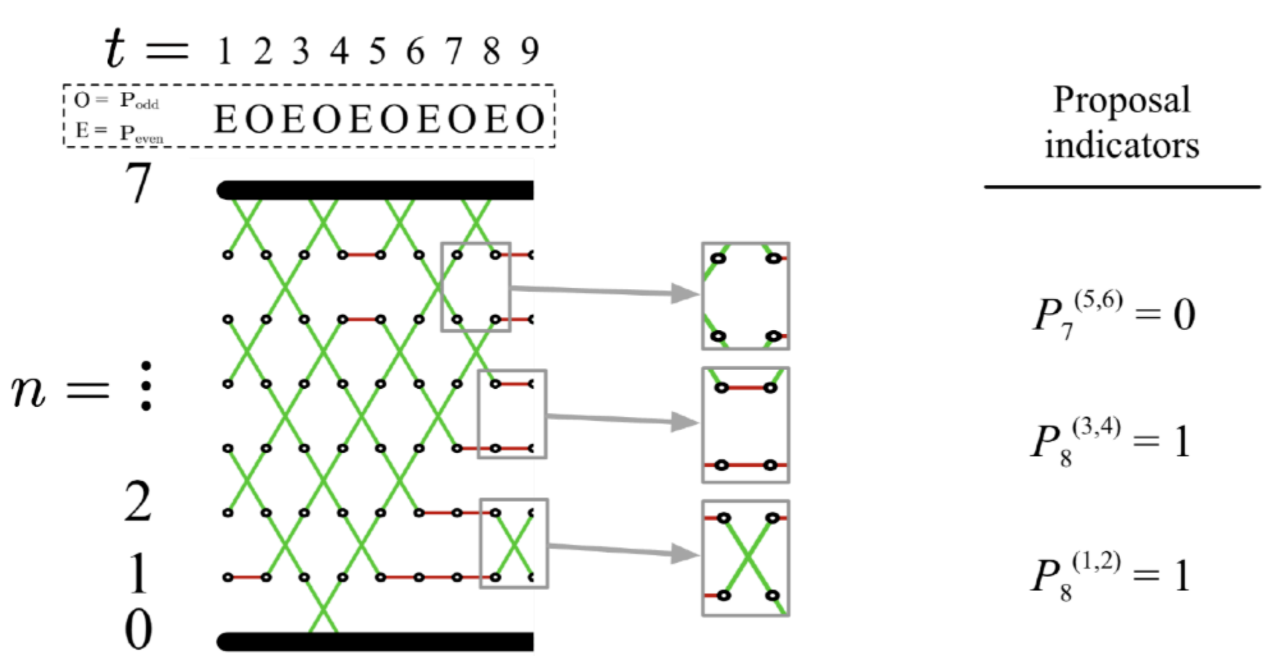

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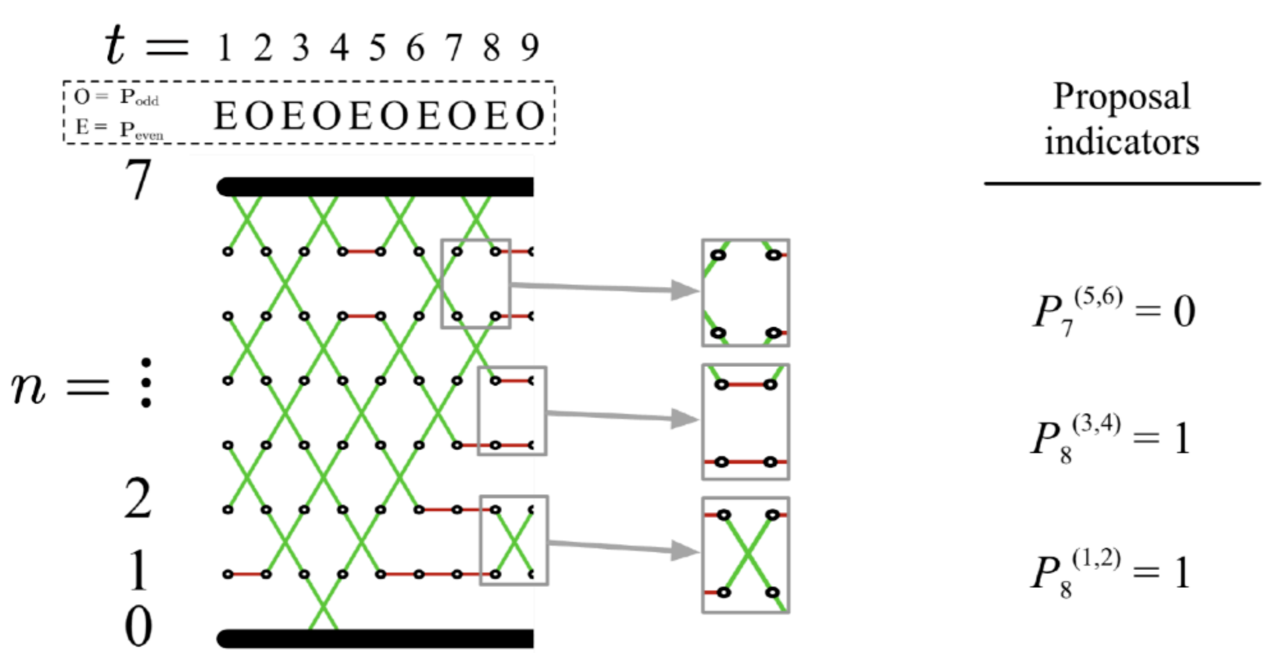

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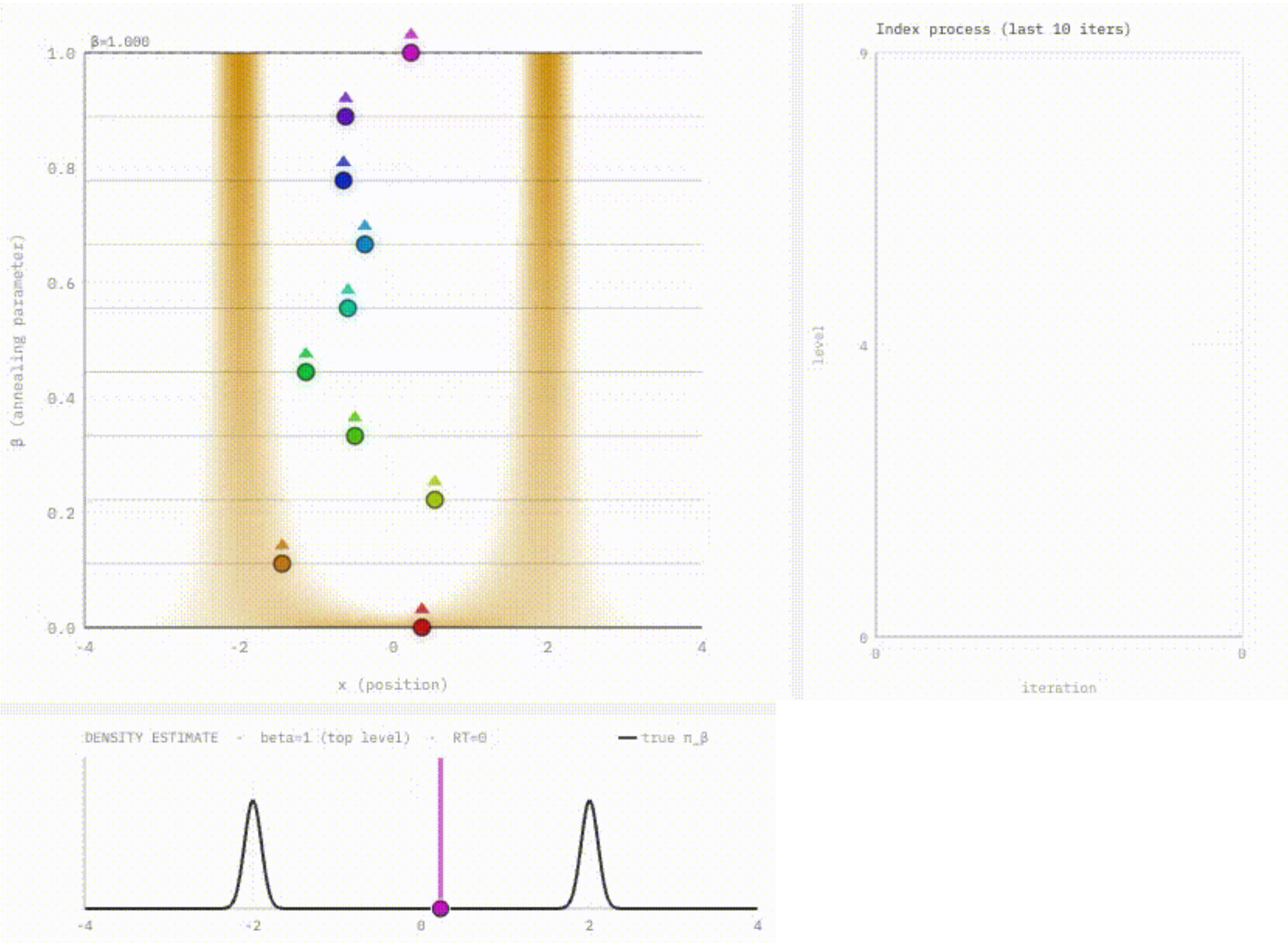

MCMC VS PARALLEL TEMPERING

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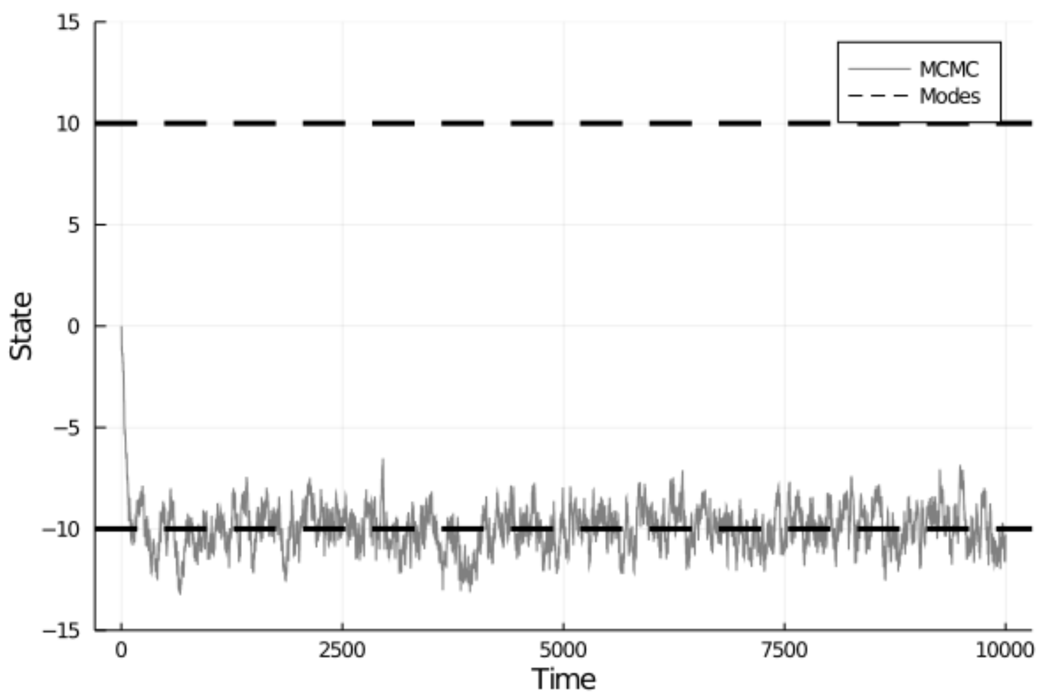
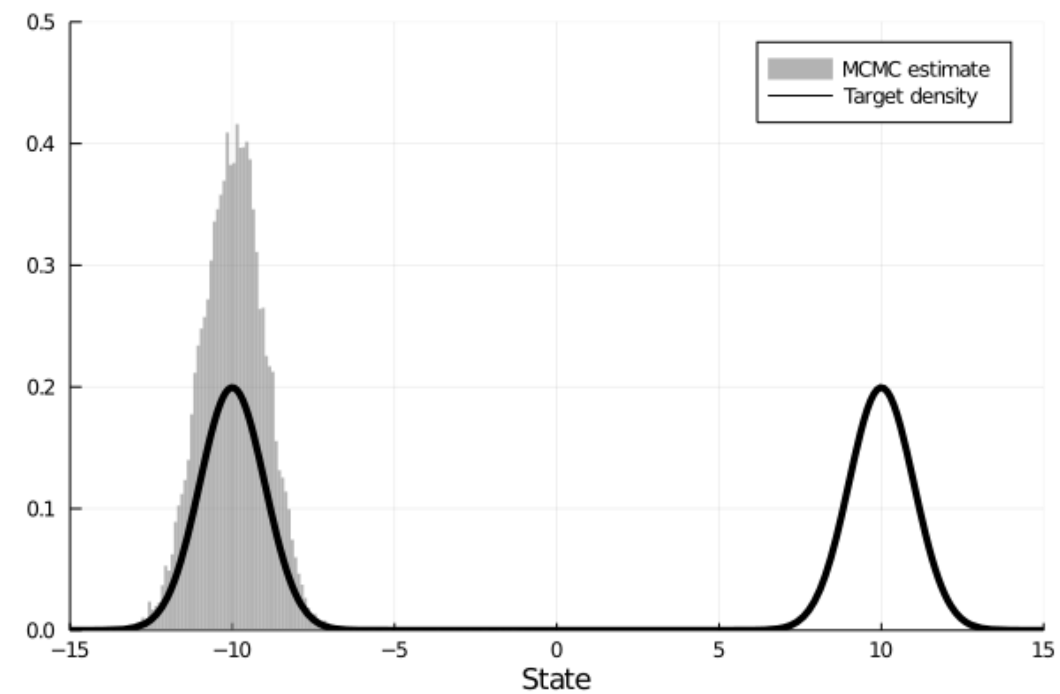
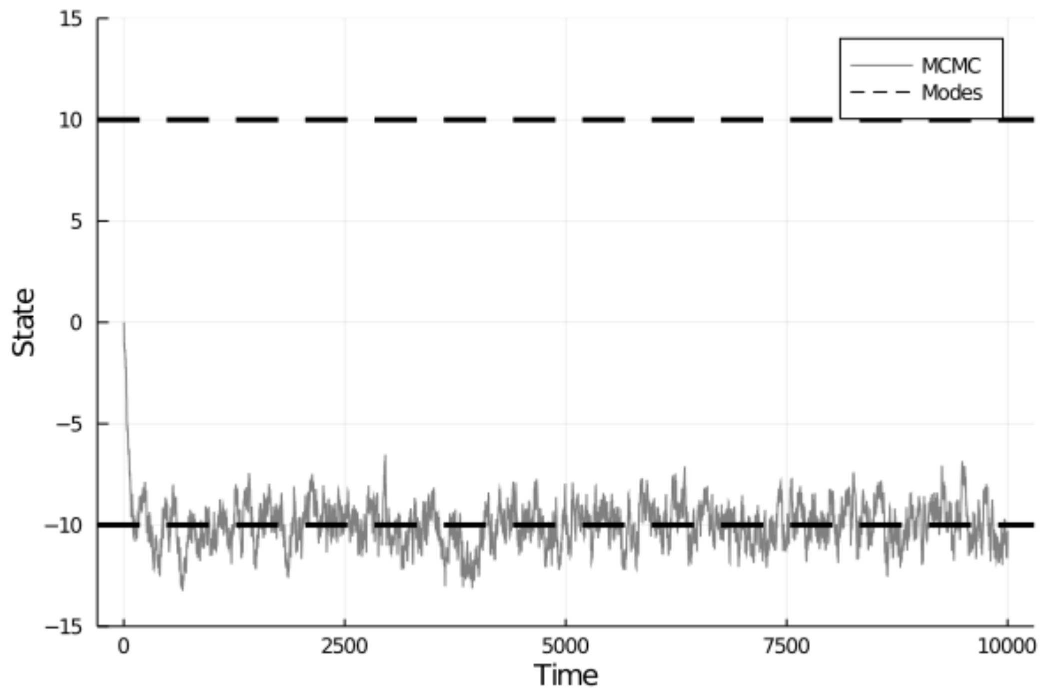
Single Chain MCMC

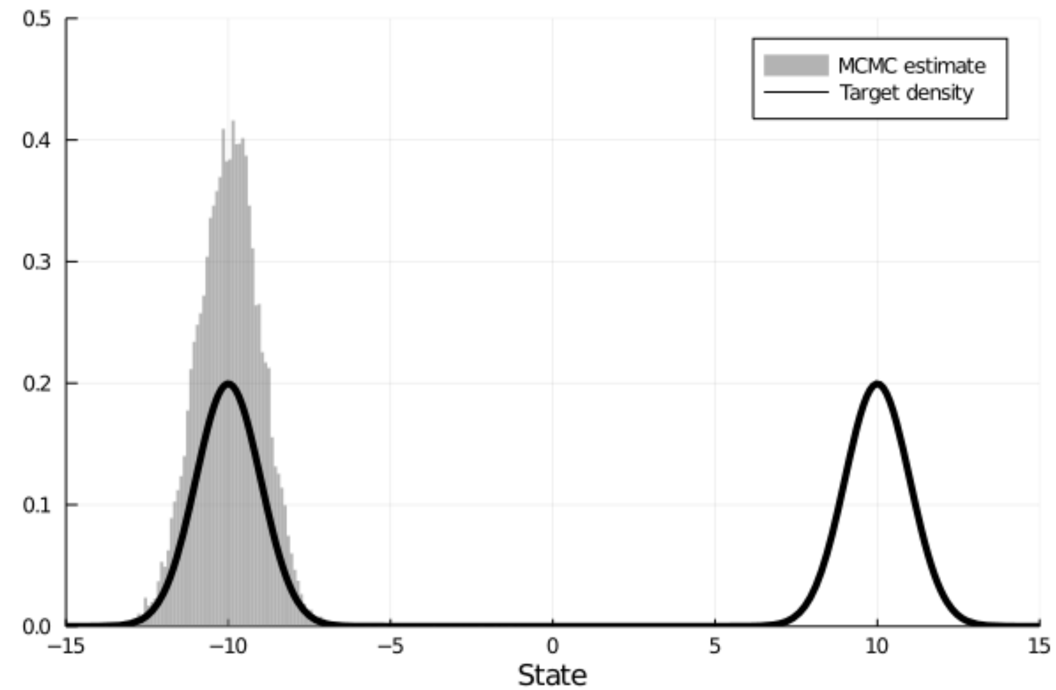




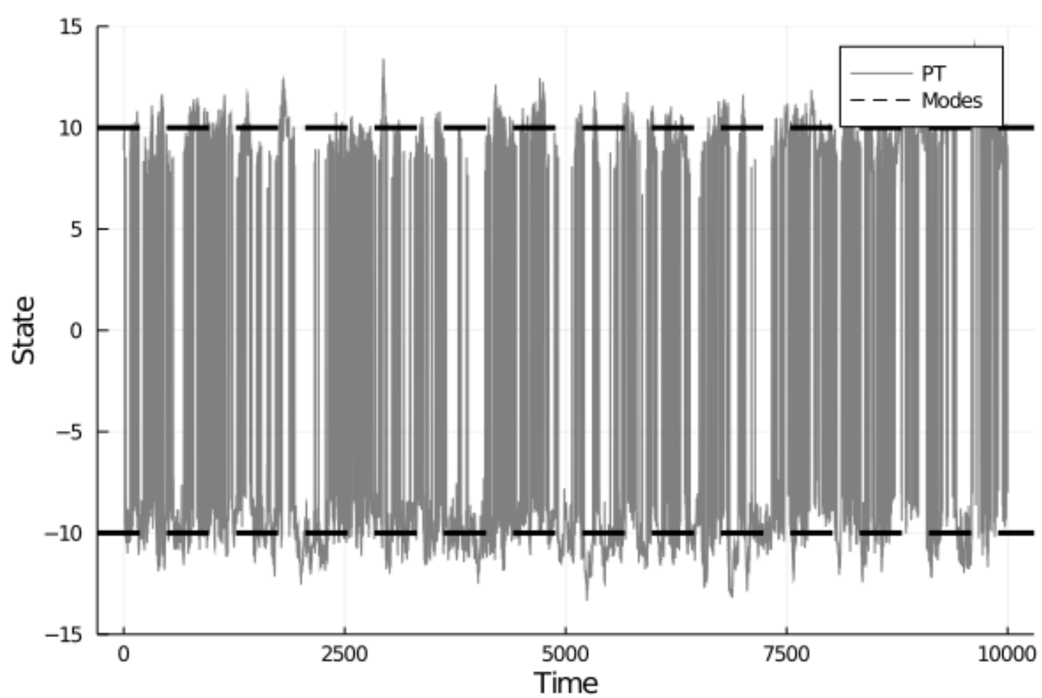
MCMC VS PARALLEL TEMPERING

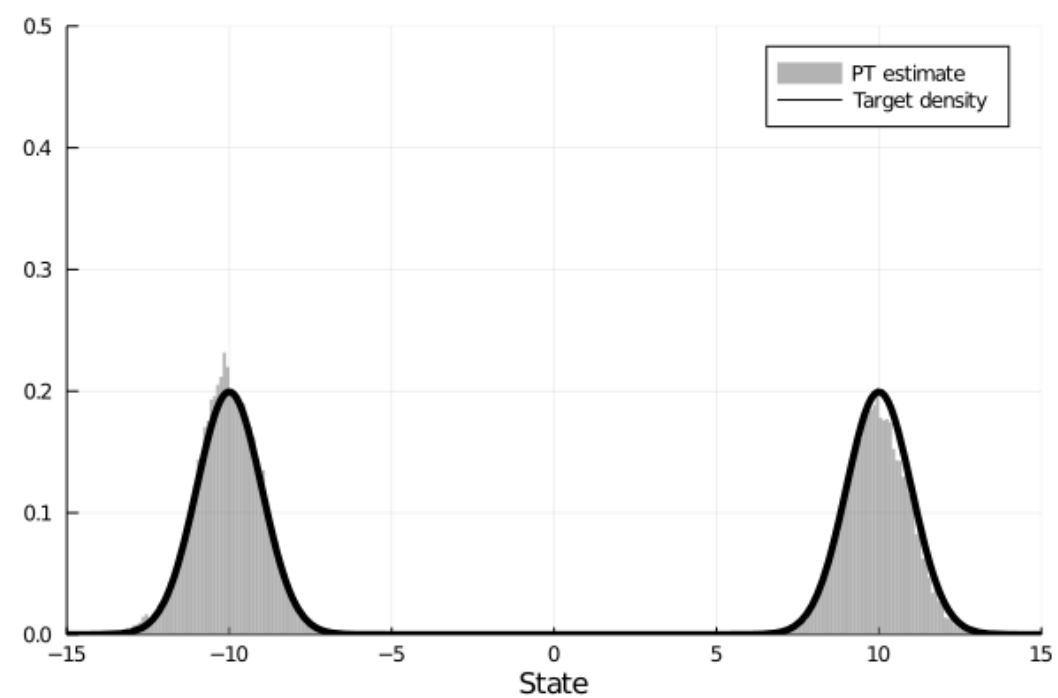
Single Chain MCMC





Parallel Tempering (N = 10)

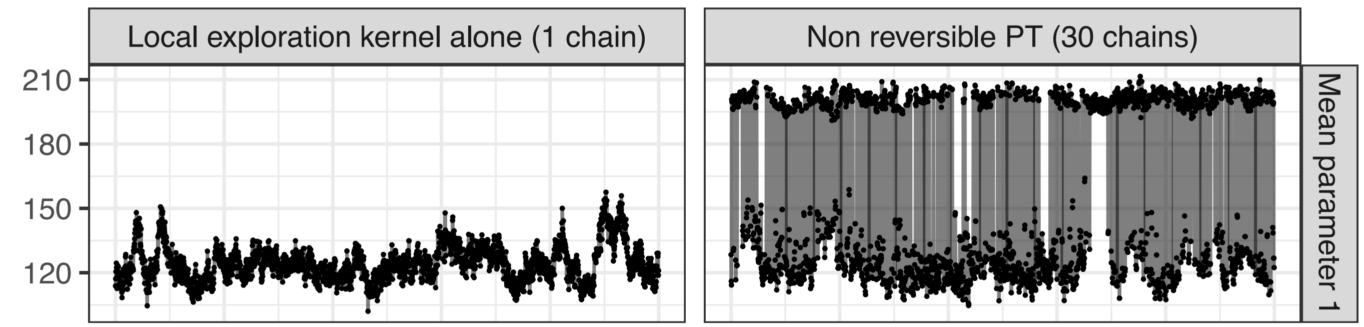




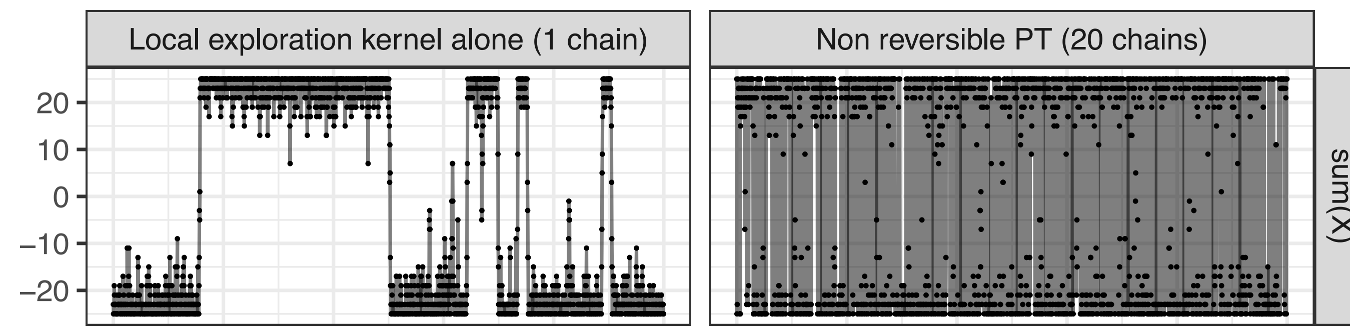
EXAMPLE: MULTIMODAL TARGETS

Single Chain MCMC **Parallel Tempering**

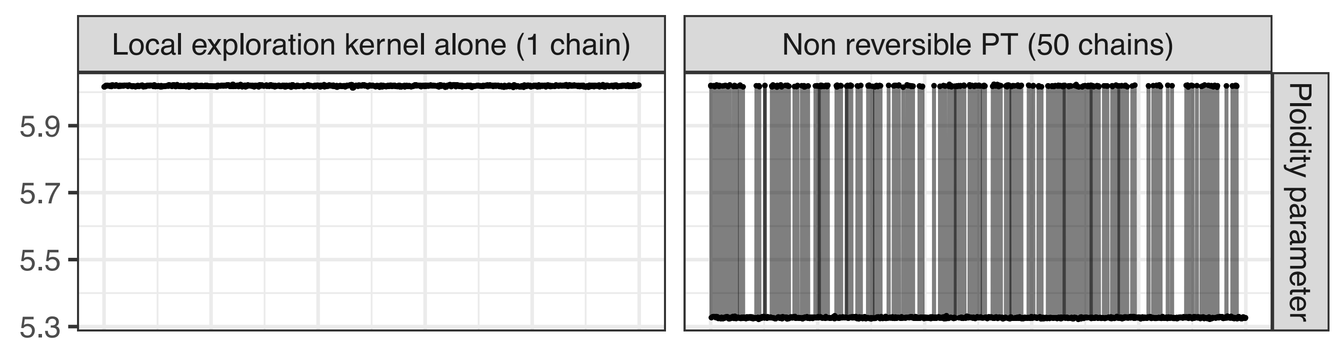
Bayesian Mixture Model
(d = 155)



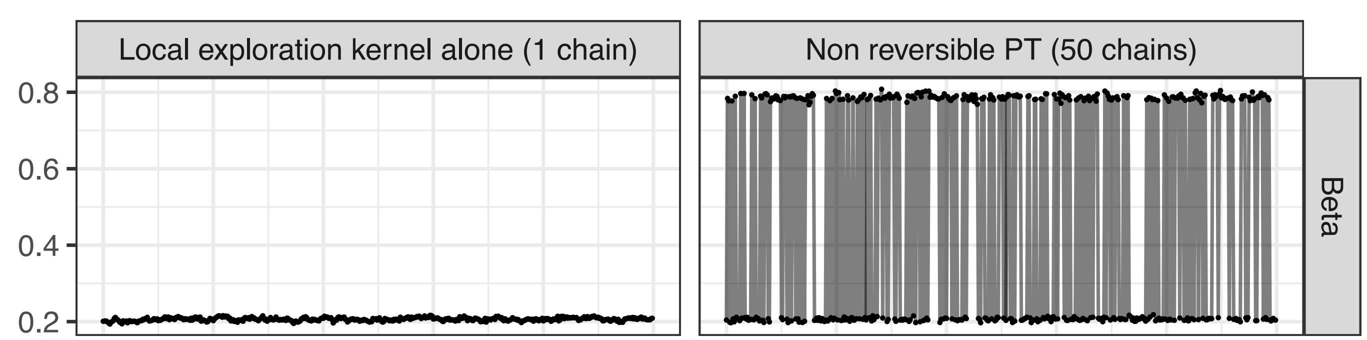
Ising Model
(d = 25)



Copy number inference
w/ whole genome ovarian cancer data
(d = 30)



ODE parameter inference
w/ mRNA data
(d = 9)



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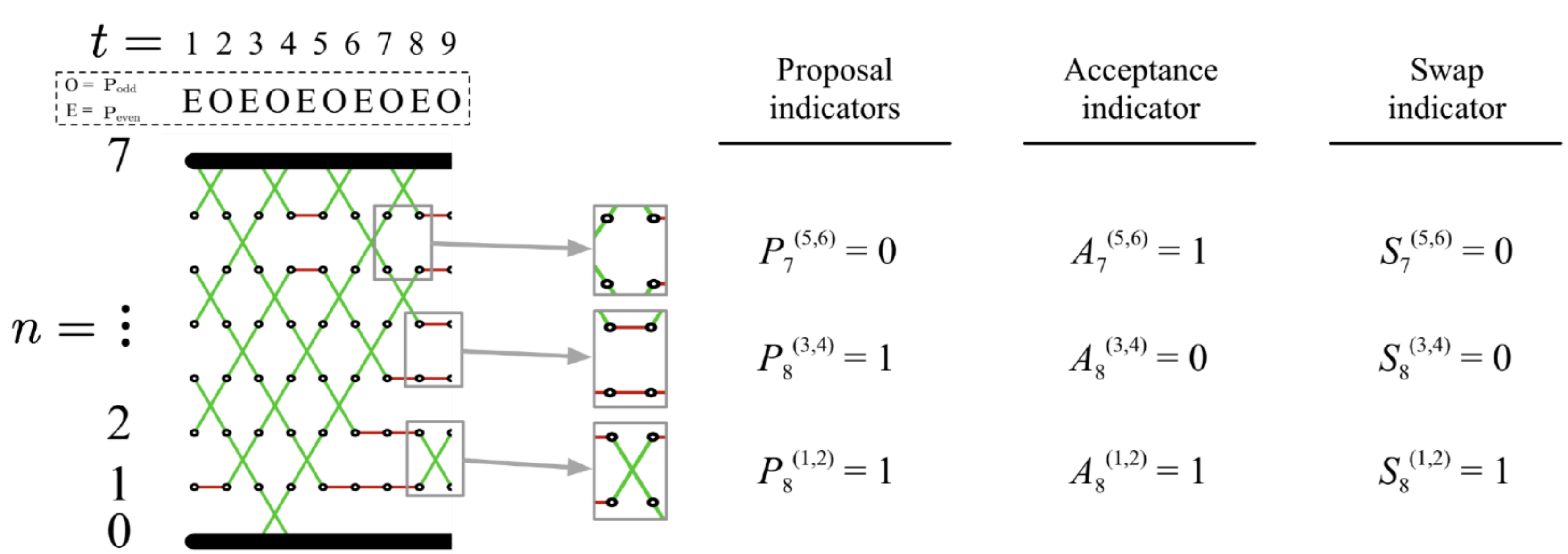

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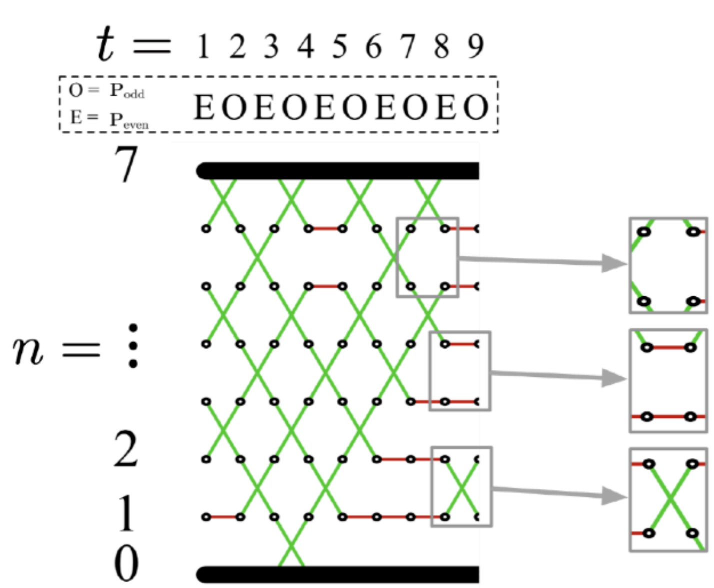

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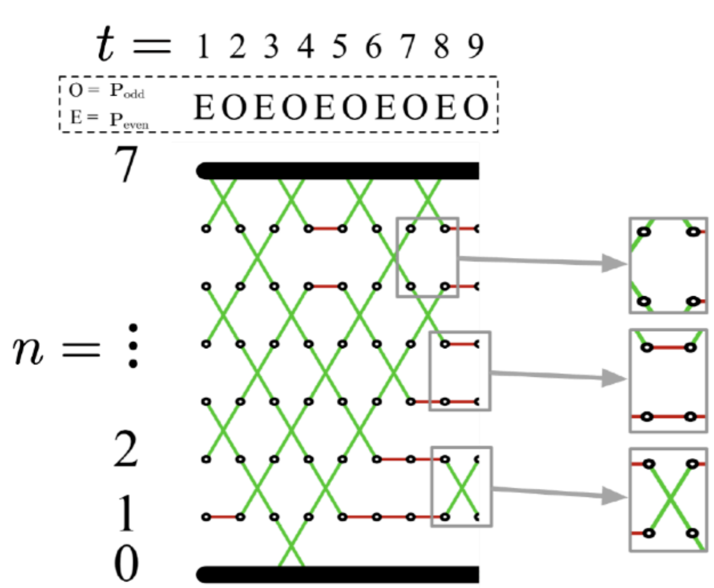

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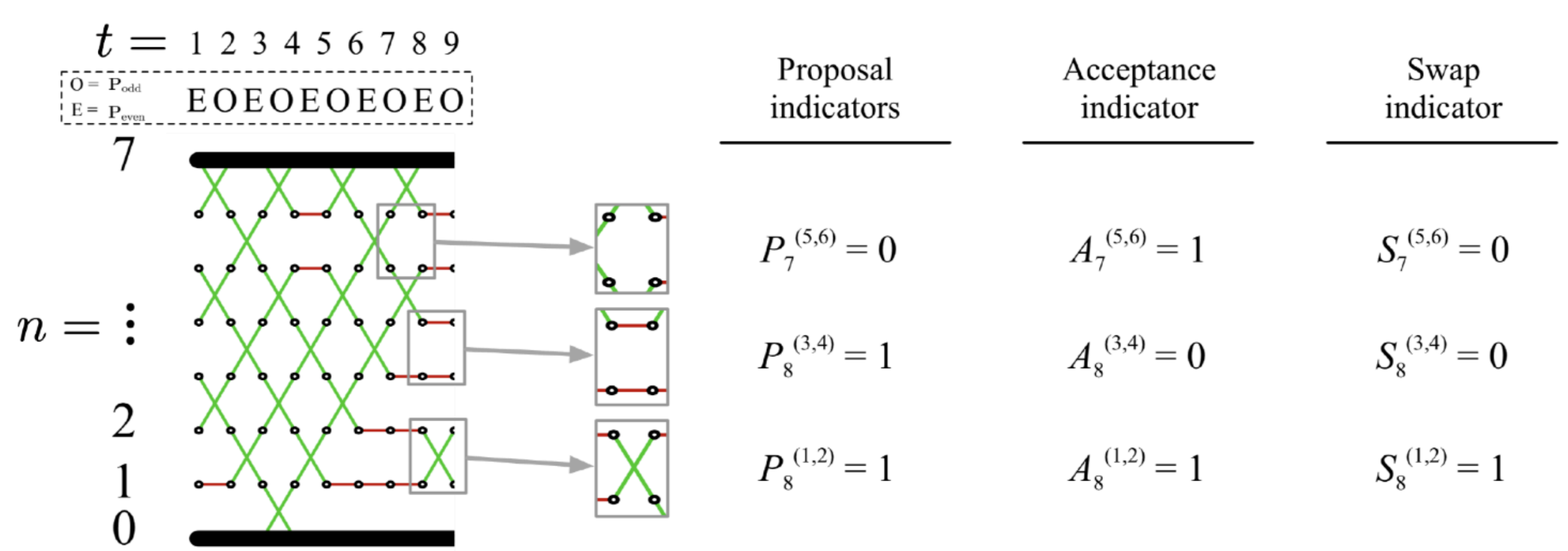

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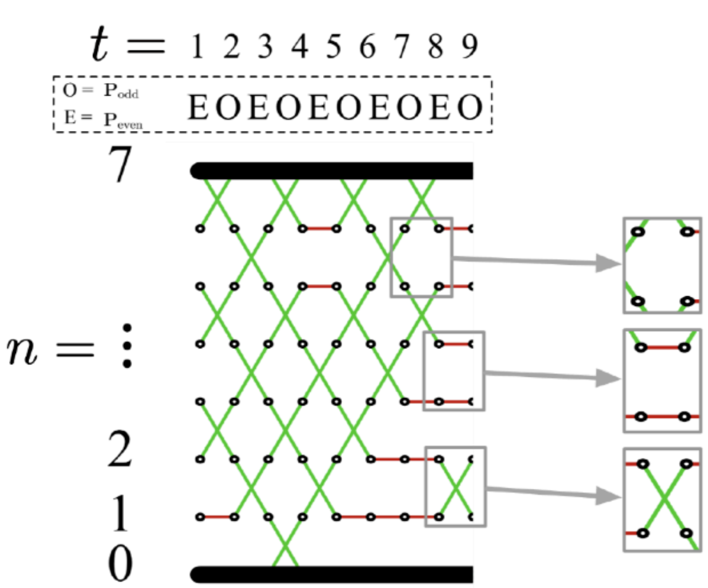

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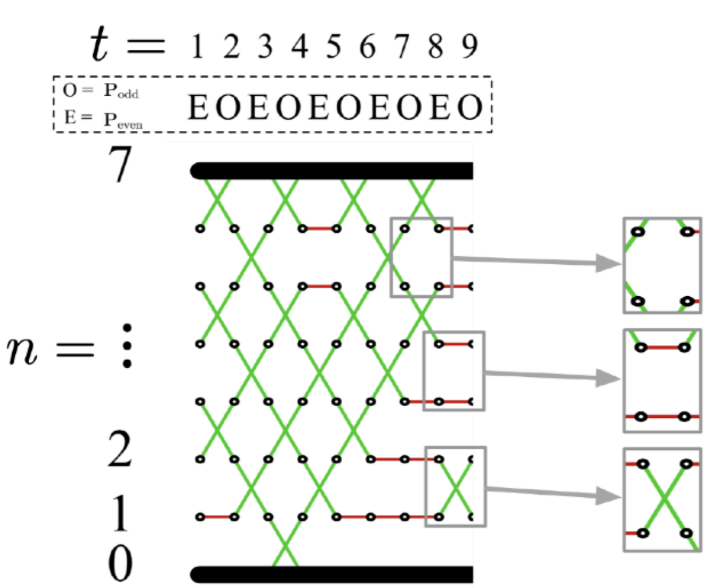

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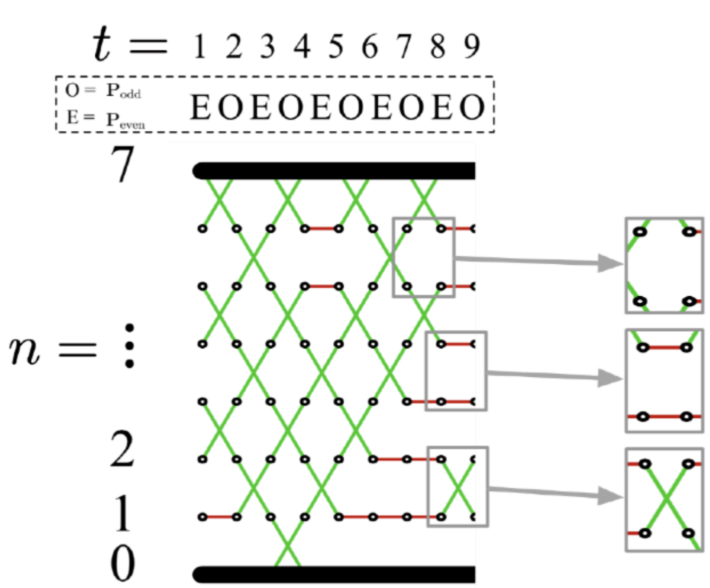

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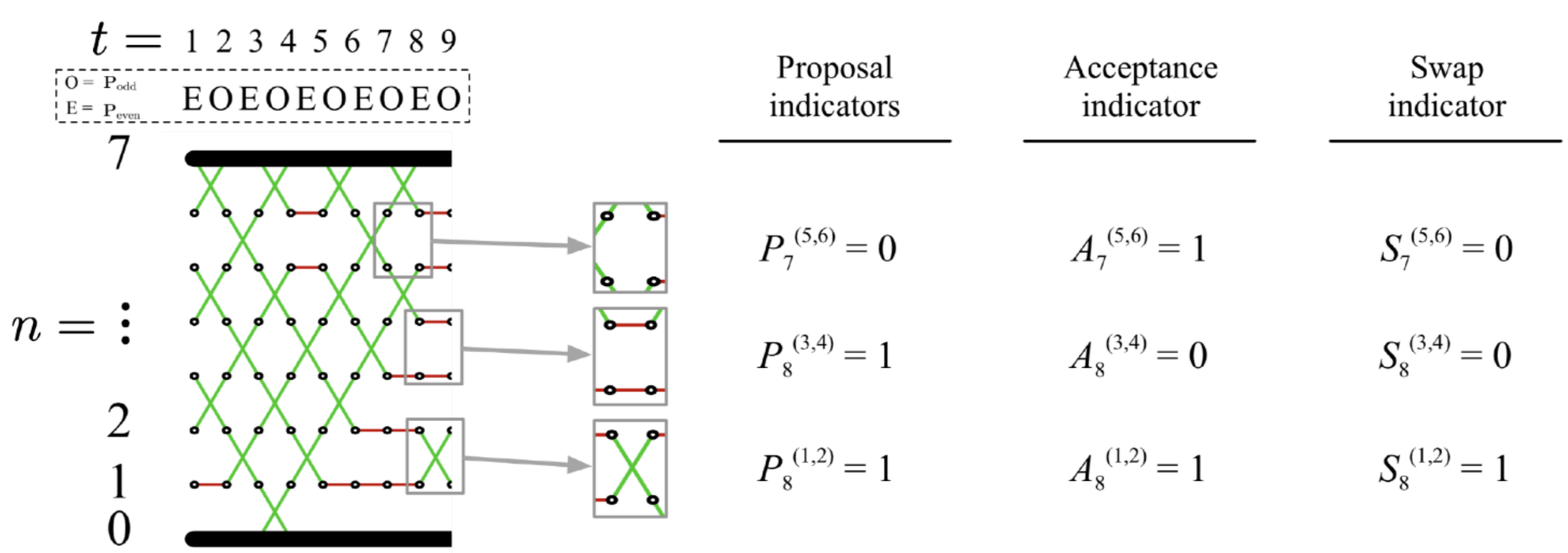

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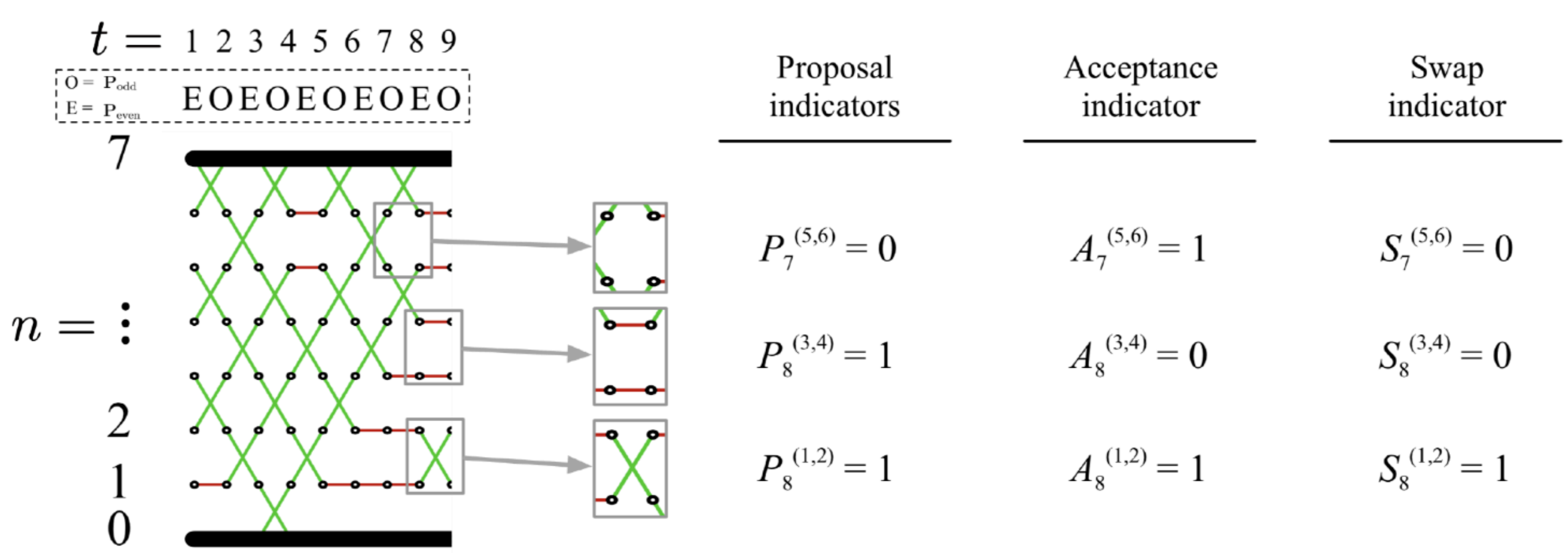

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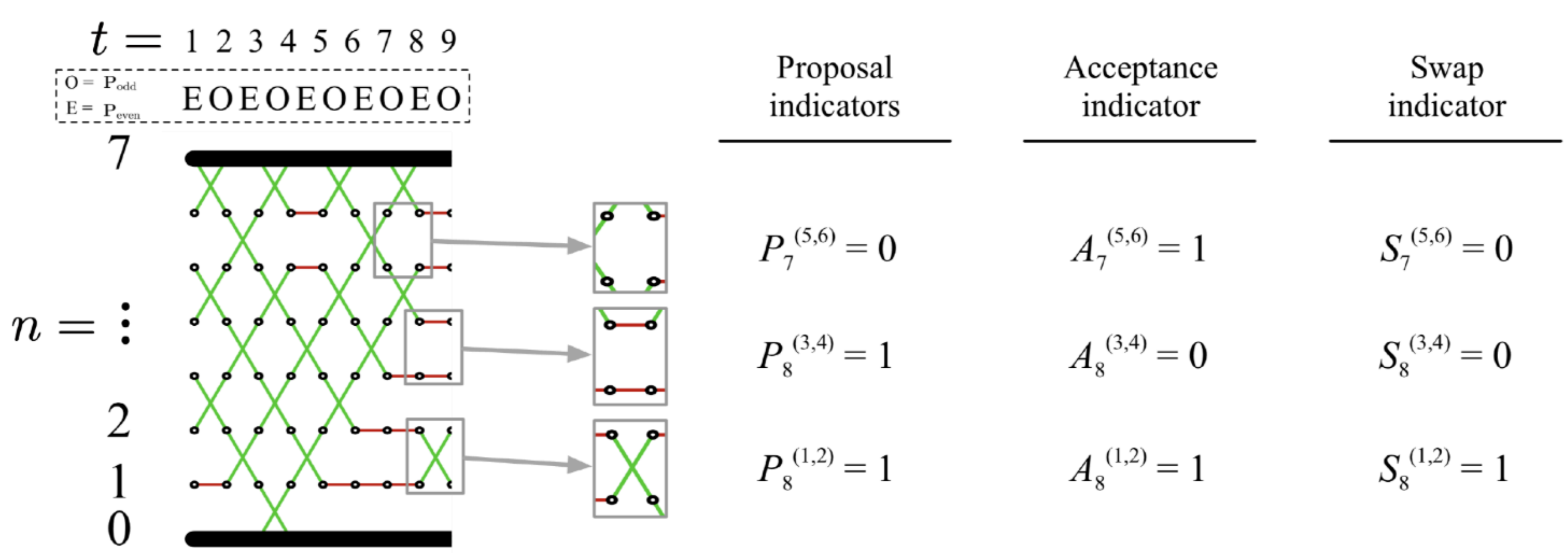

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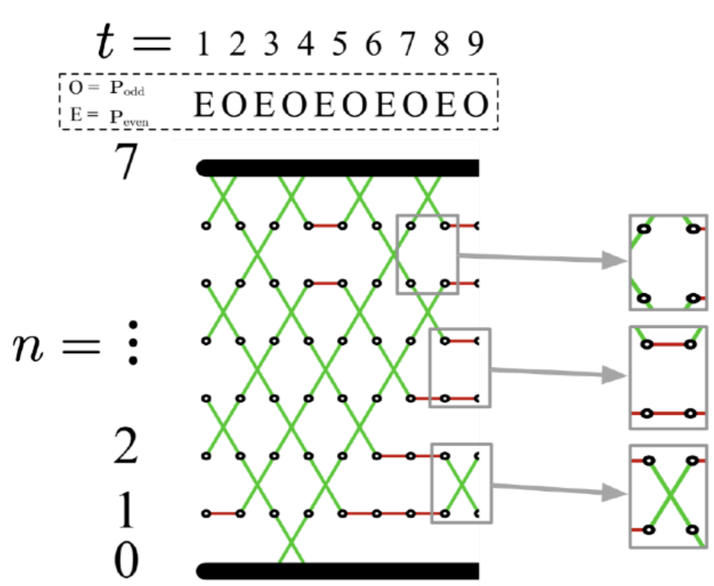

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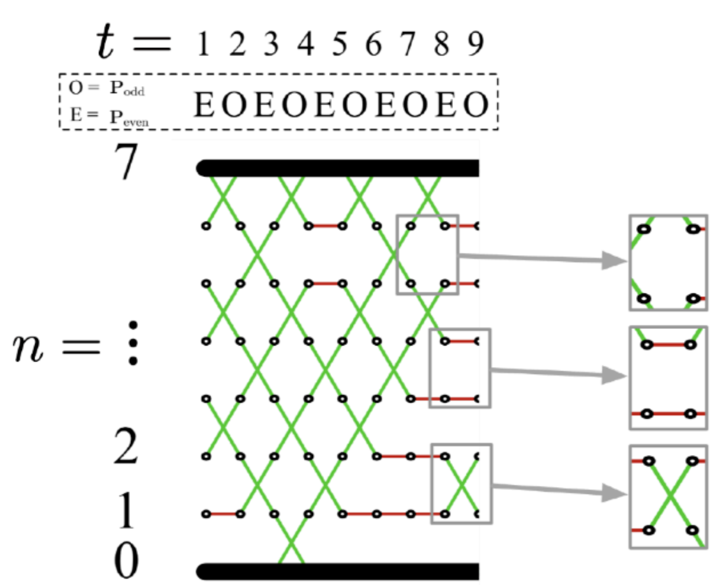

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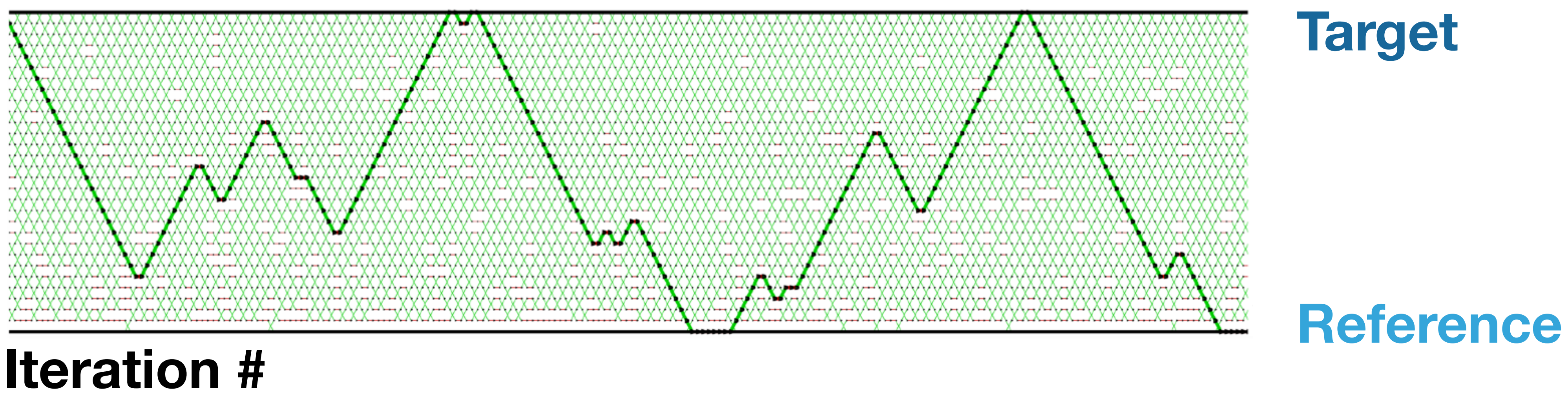

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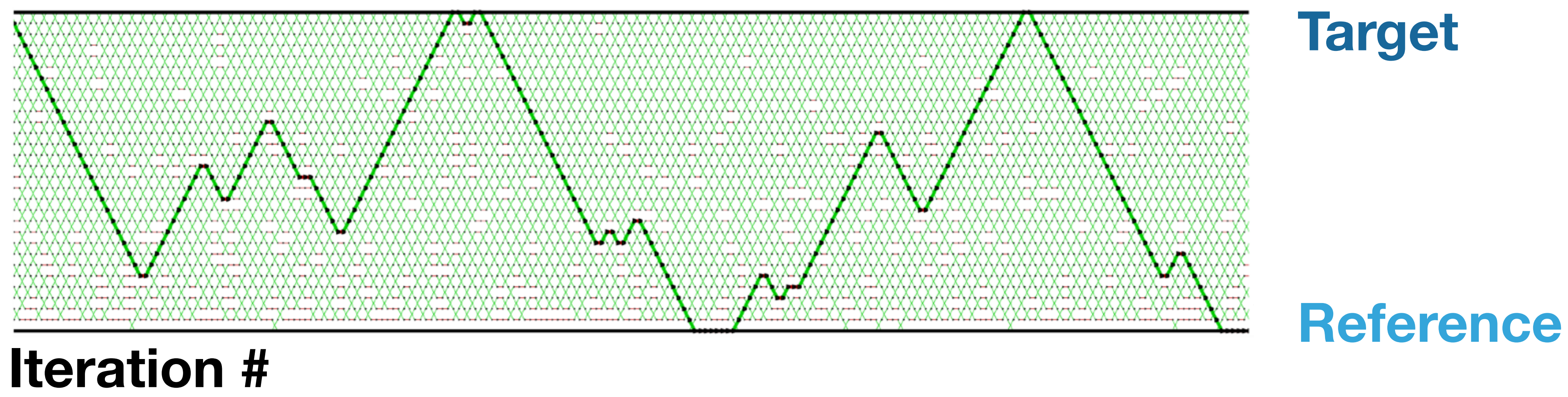

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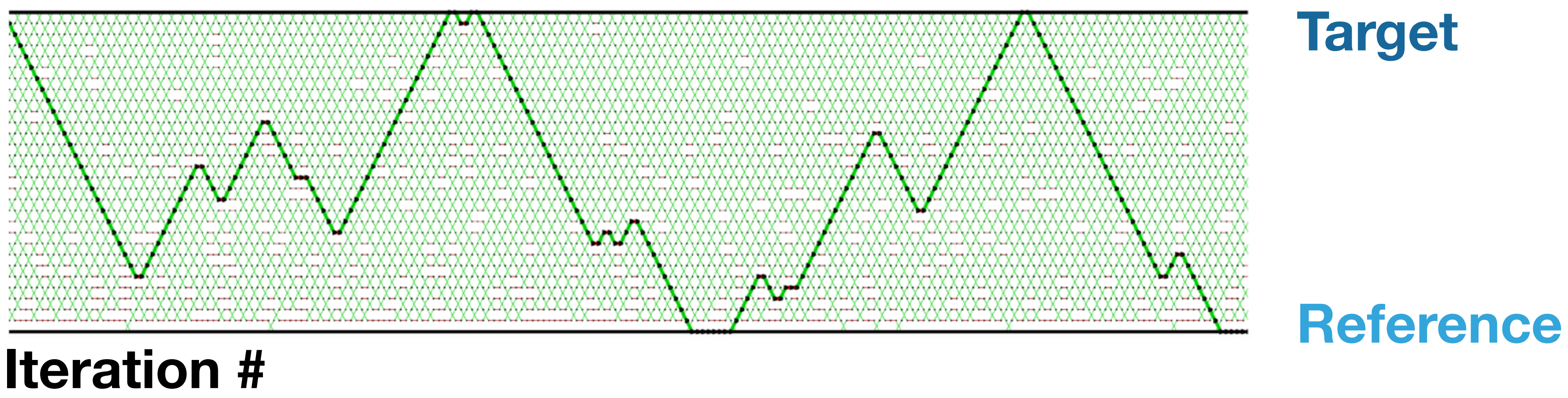

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- **Round trip rate** % of iterations with round trip

$$\tau_N = \lim_{t\to\infty} \frac{\mathbb{E}[\text{total round trips by time t}]}{t}$$

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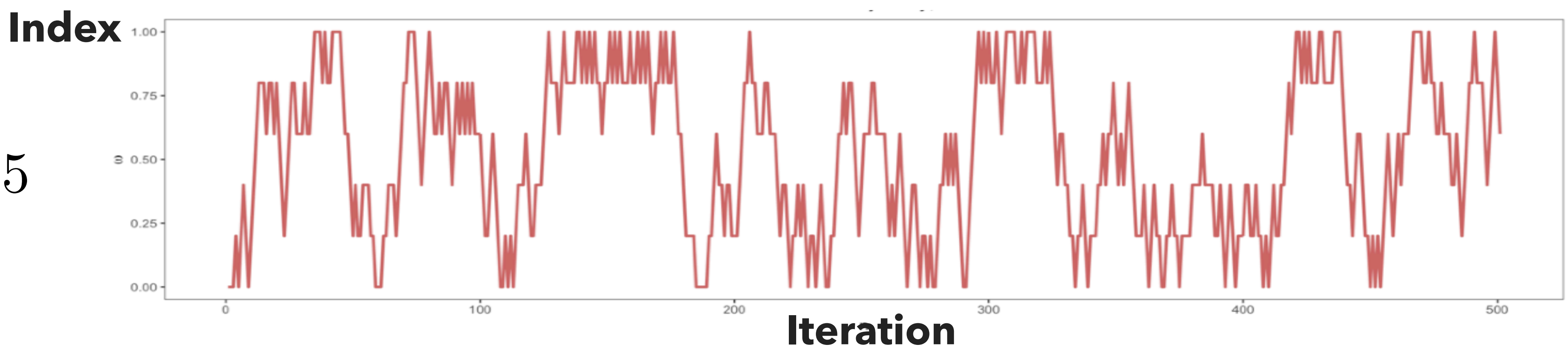

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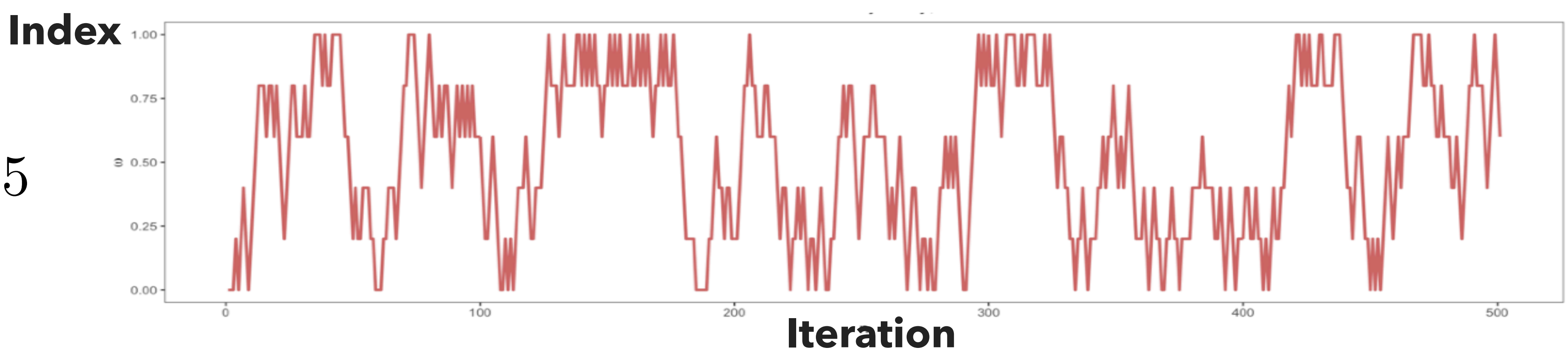

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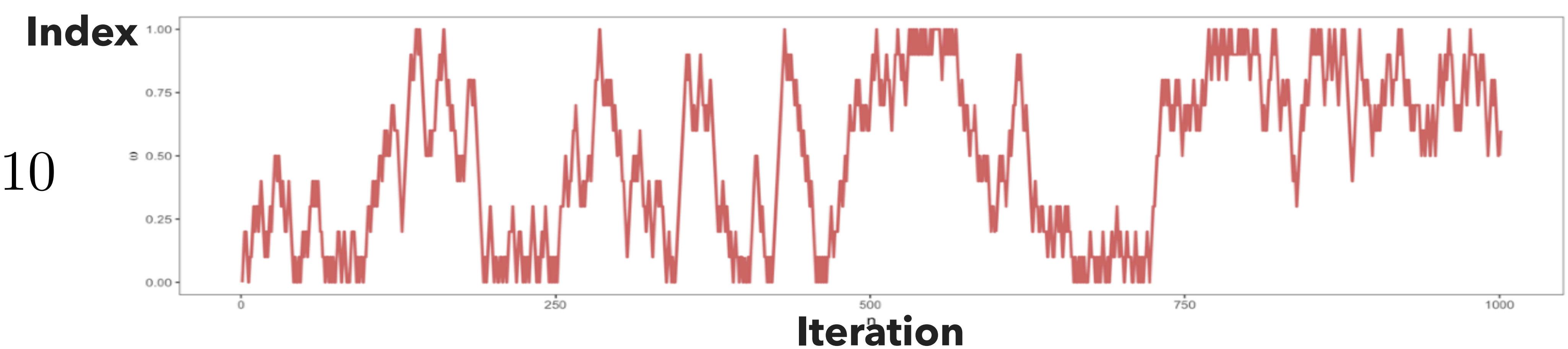

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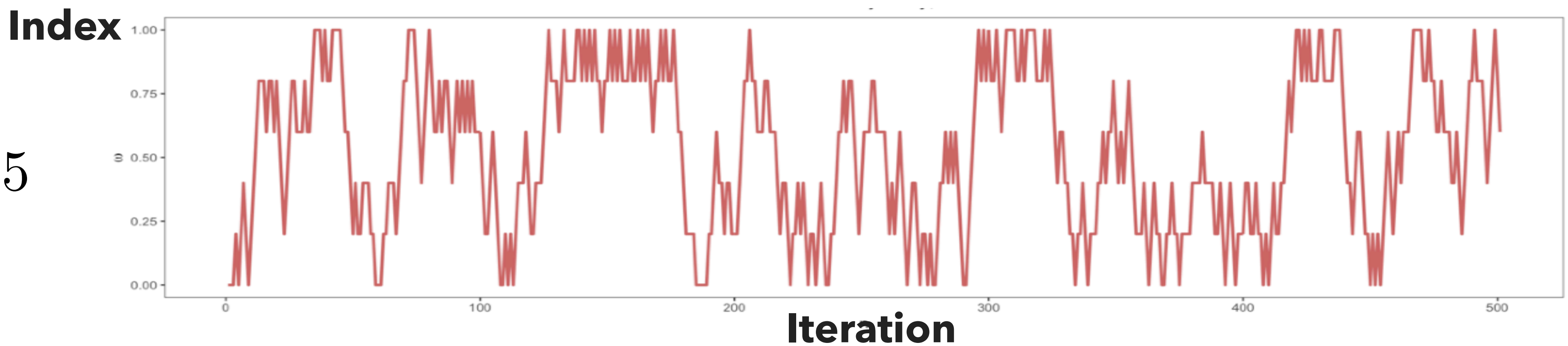

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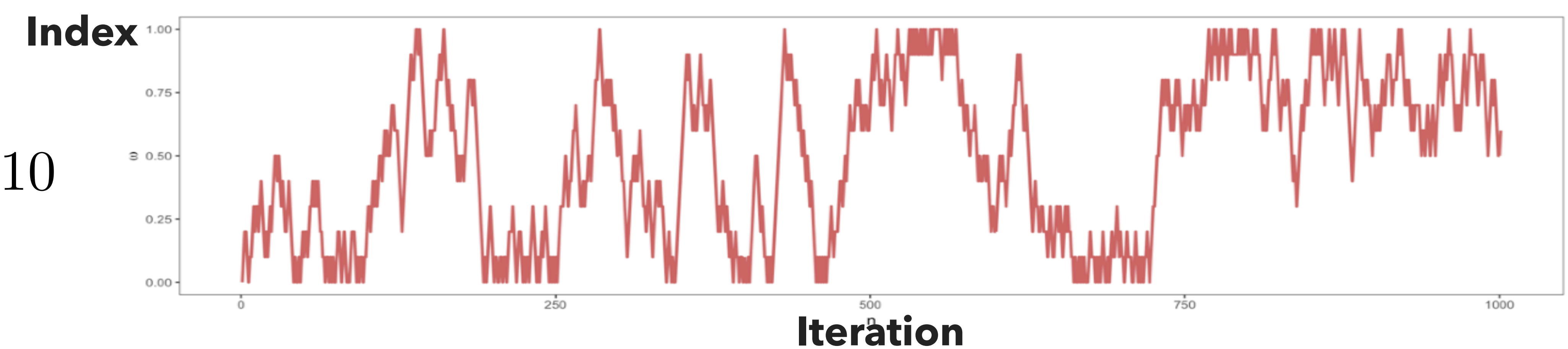

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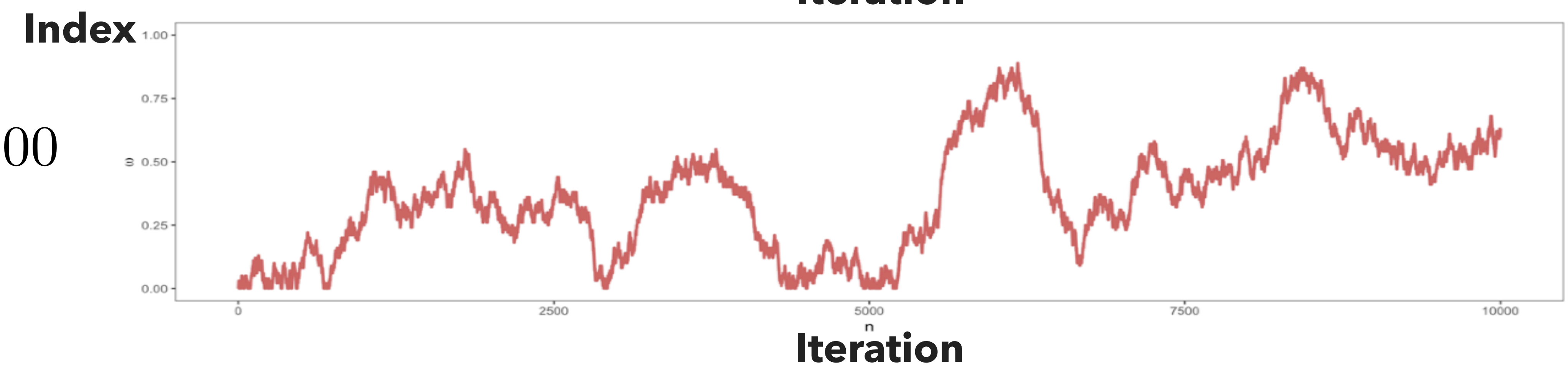

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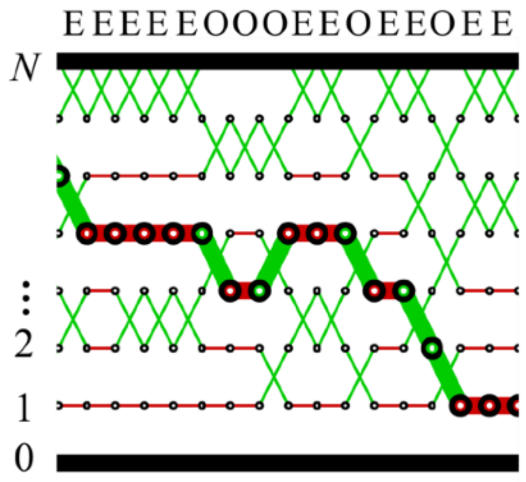

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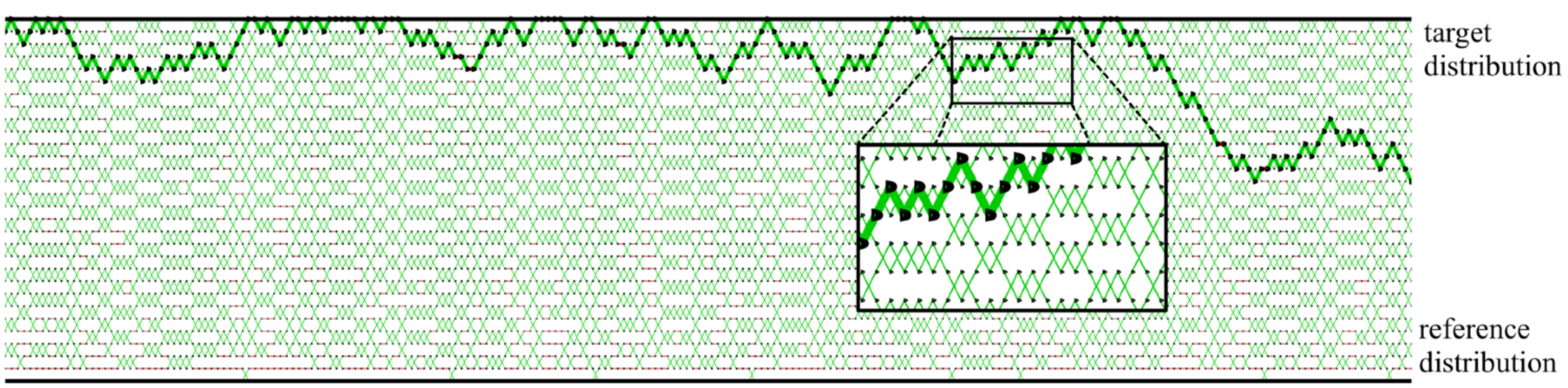

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- The index process behaves like a lazy random walk!

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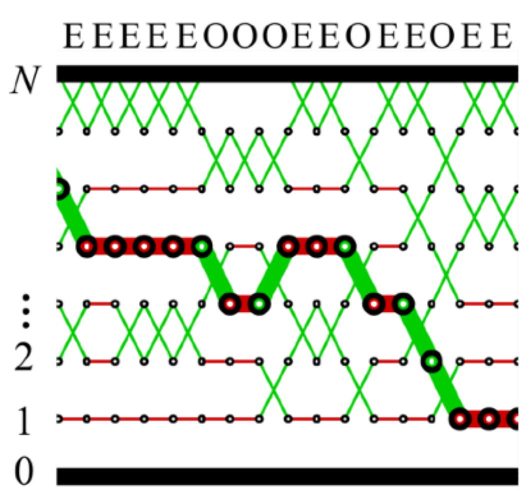

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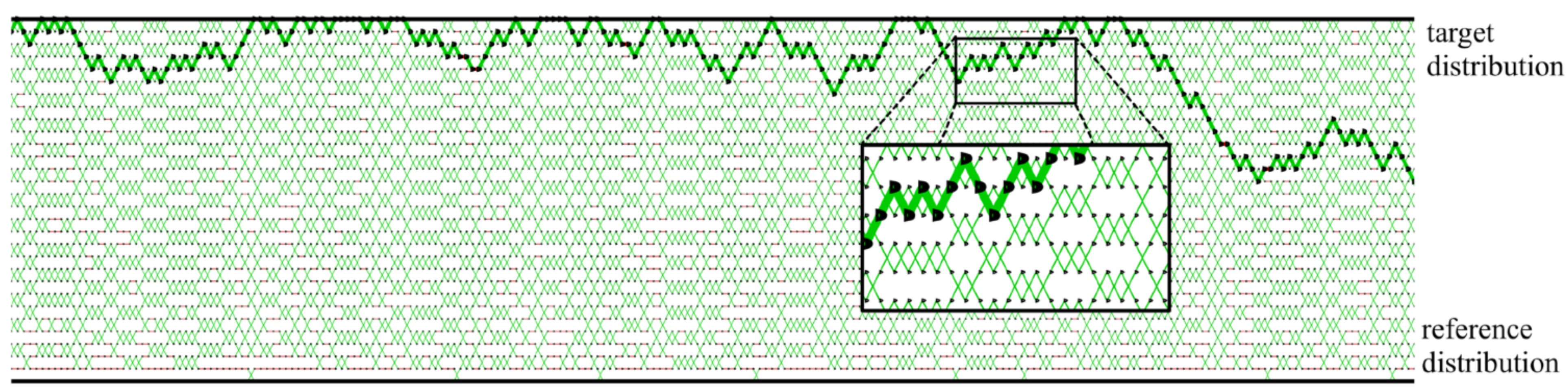
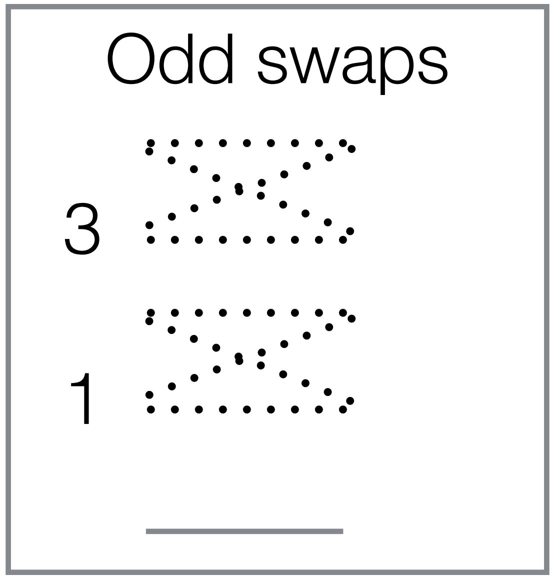
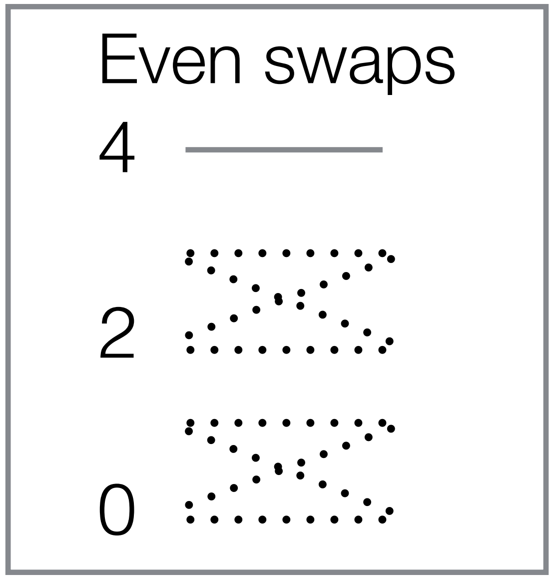
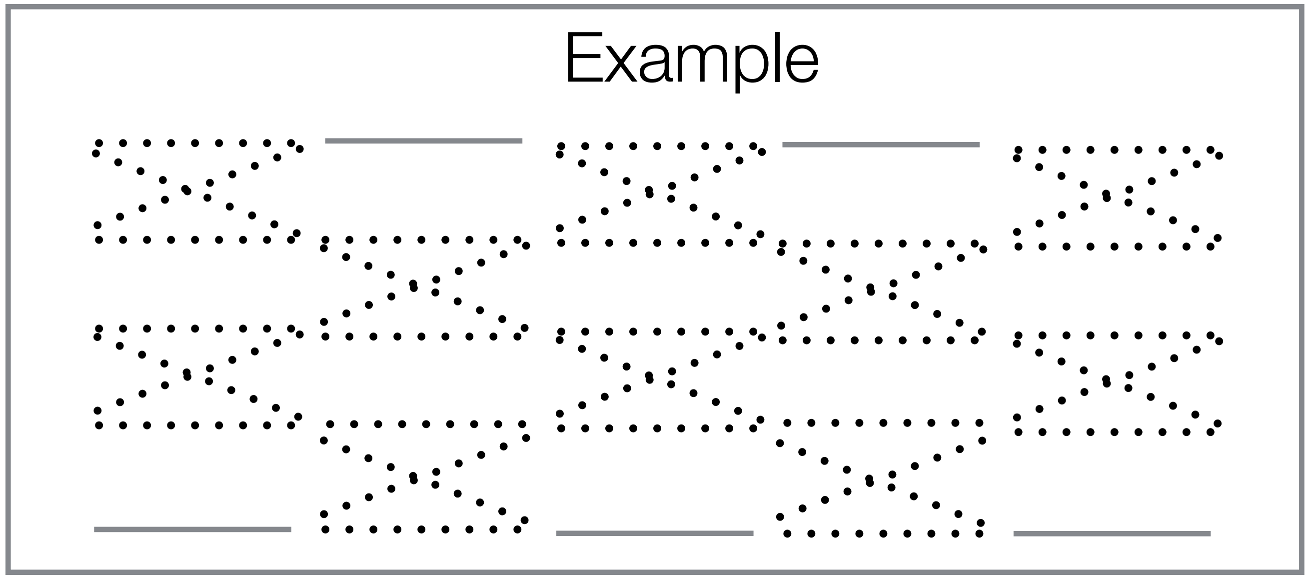


NON-REVERSIBLE PT (OKABE ET AL, 2001)

- Deterministically alternate between **Odd** and **Even** swaps







NON-REVERSIBLE PT (OKABE ET AL, 2001)

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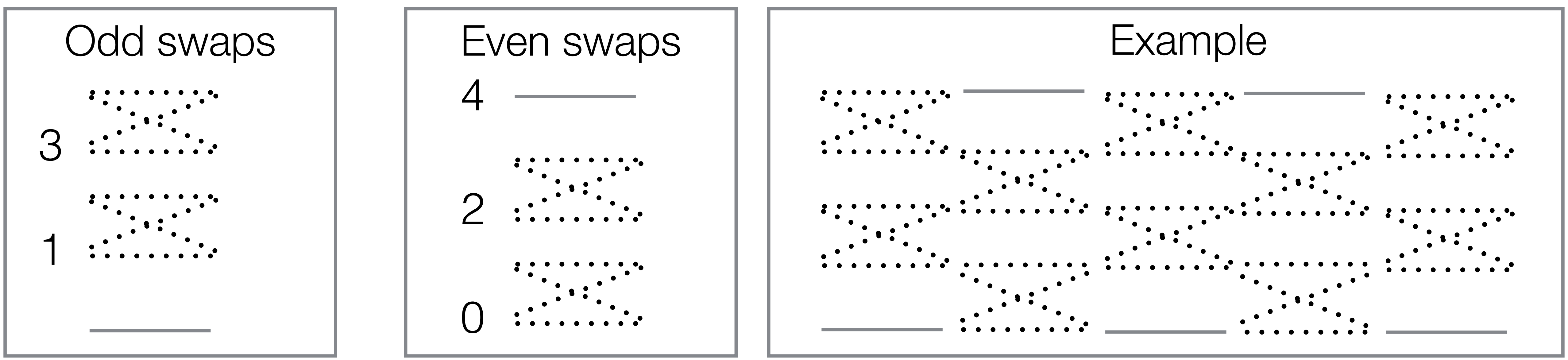

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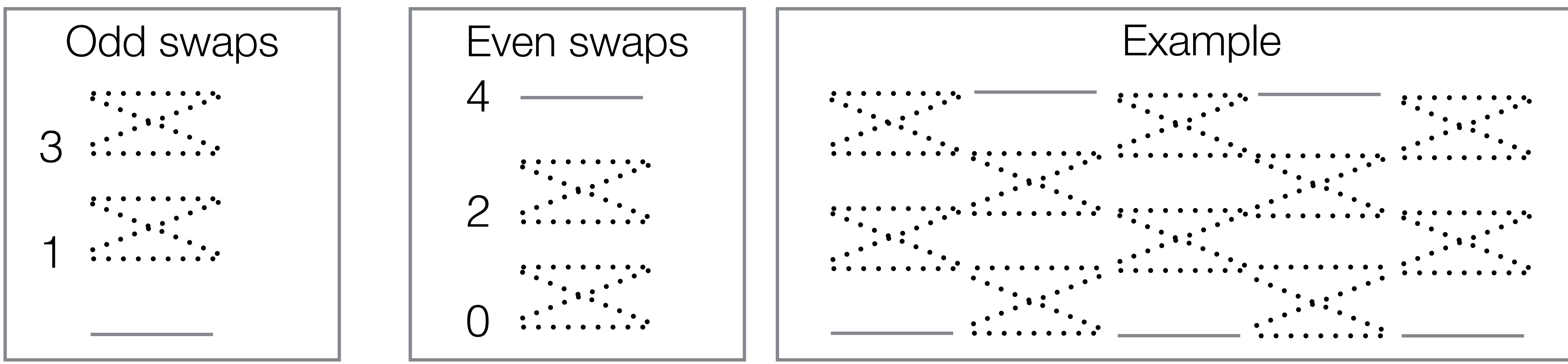

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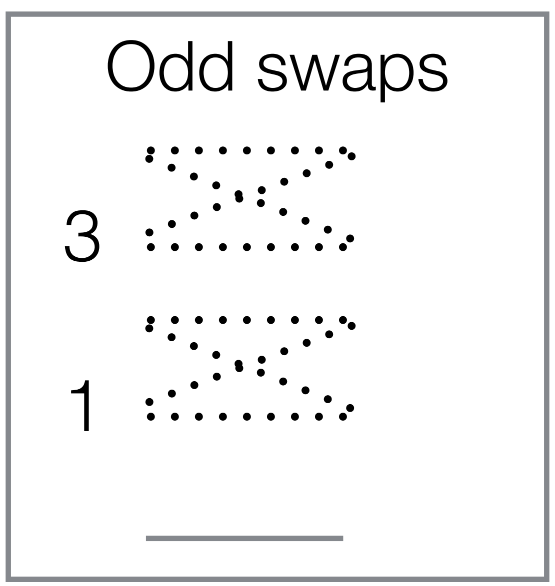

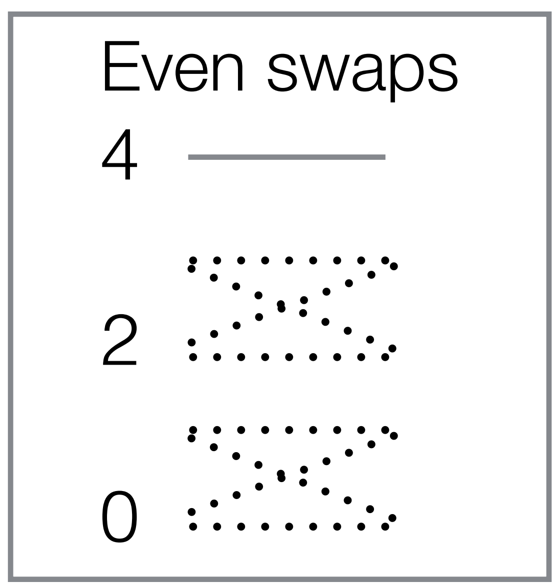

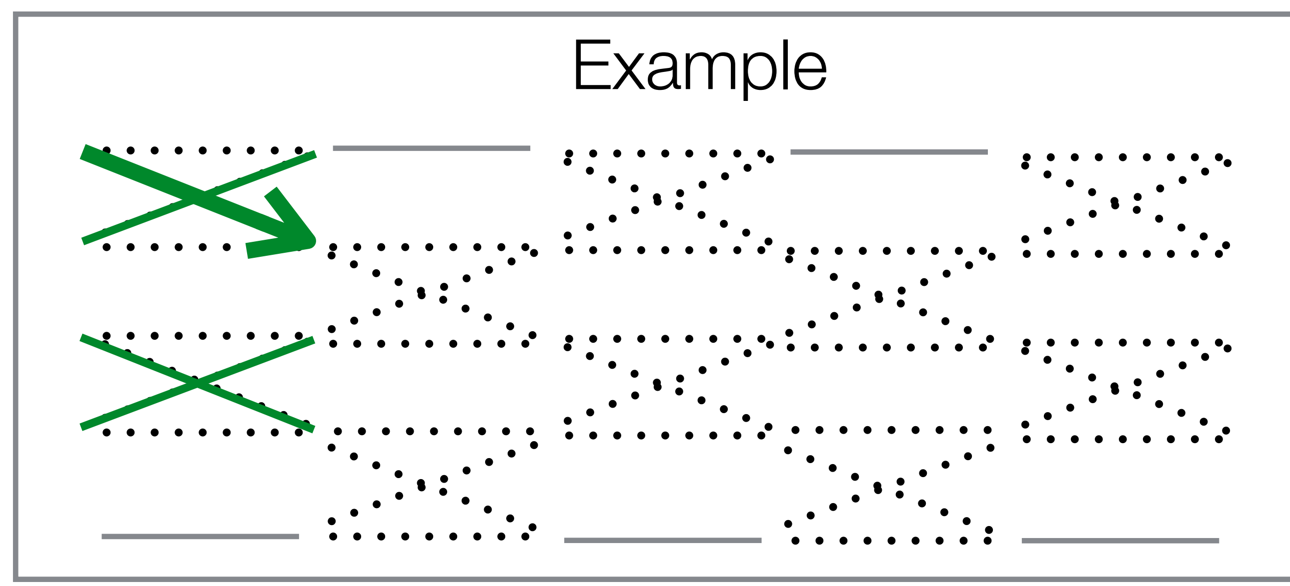

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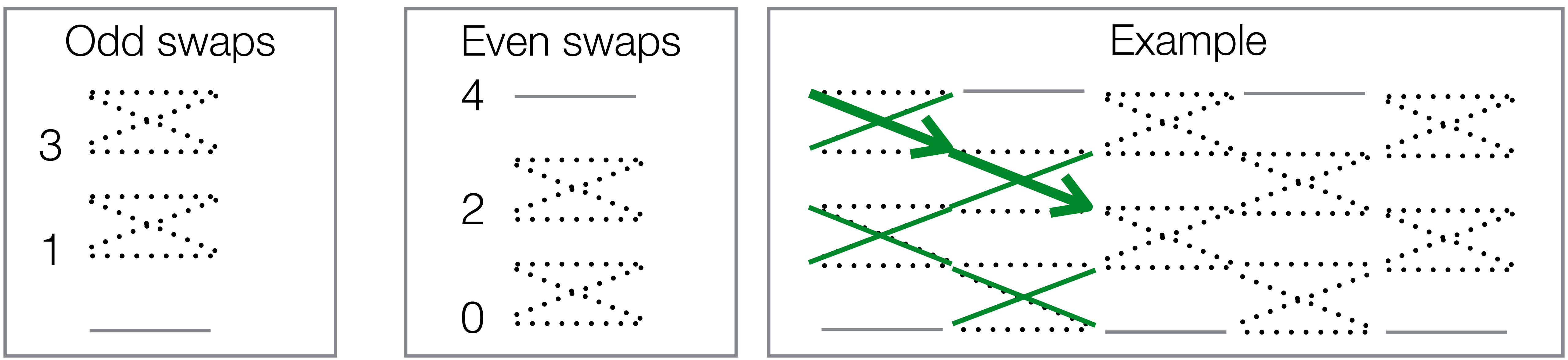

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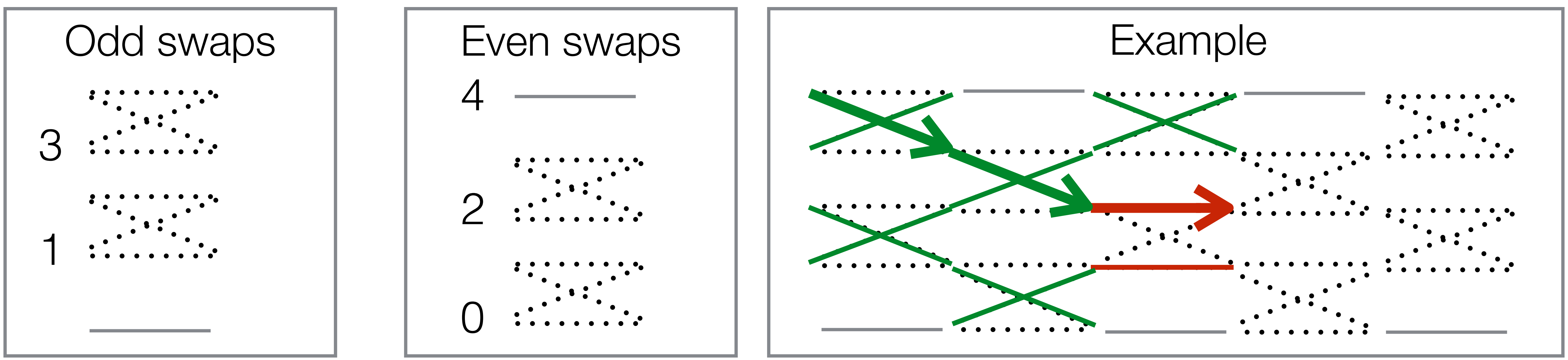

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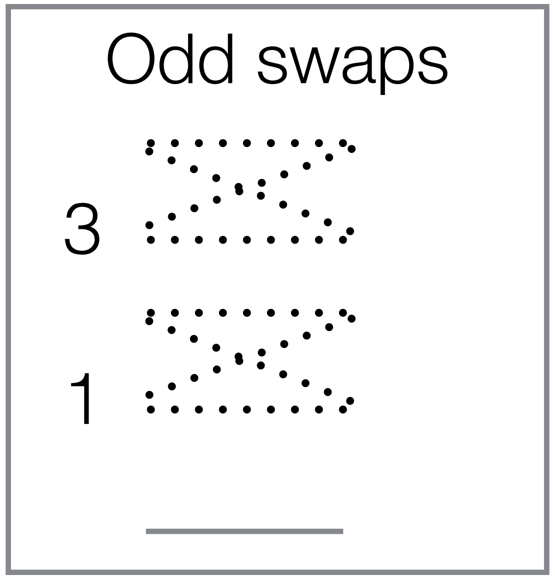

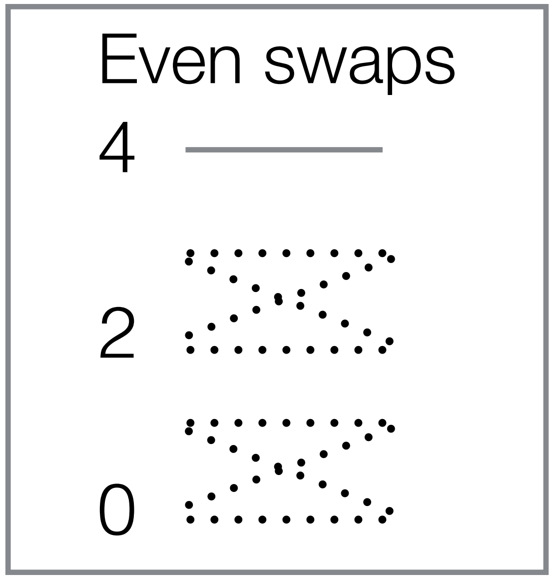

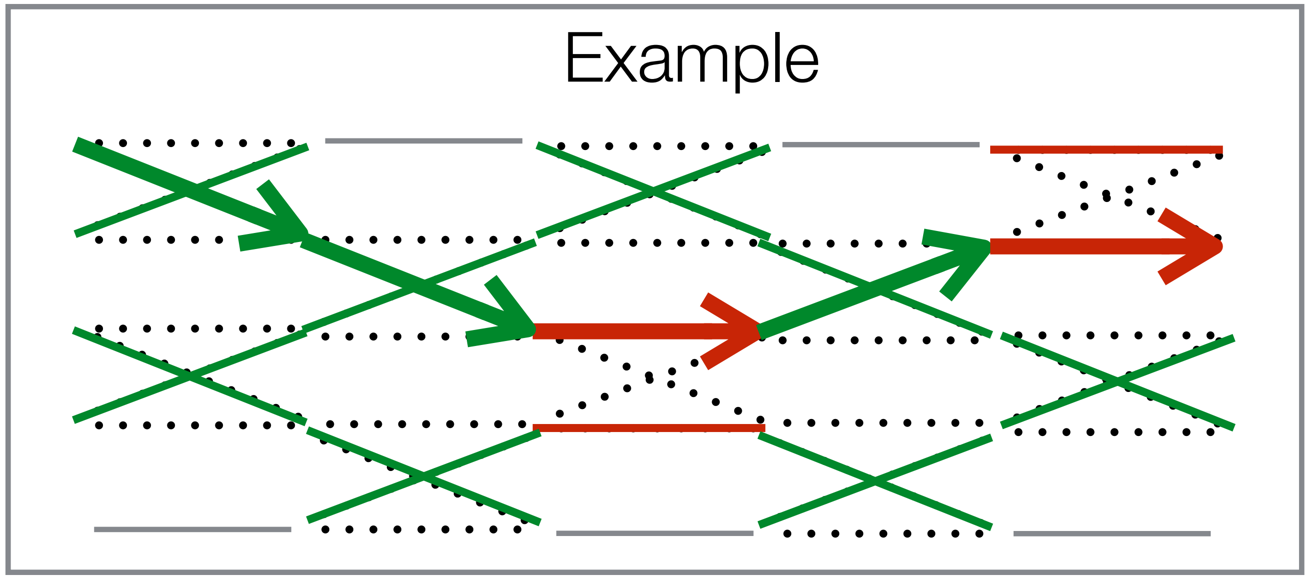

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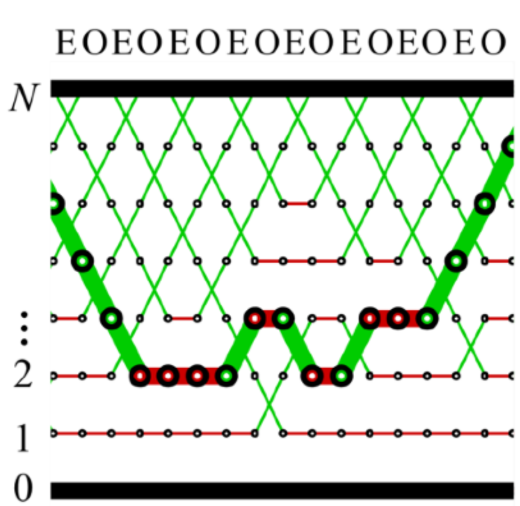

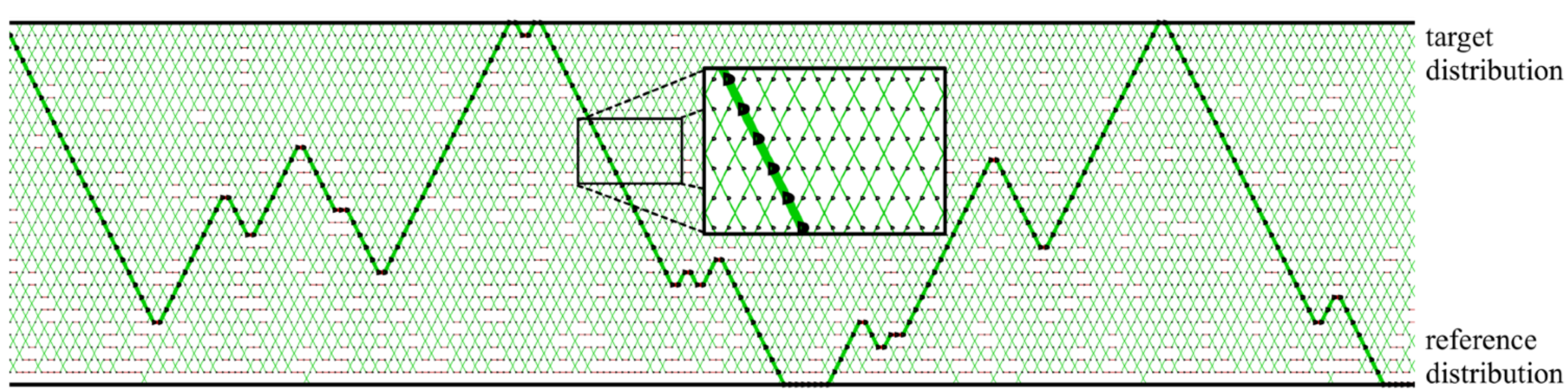

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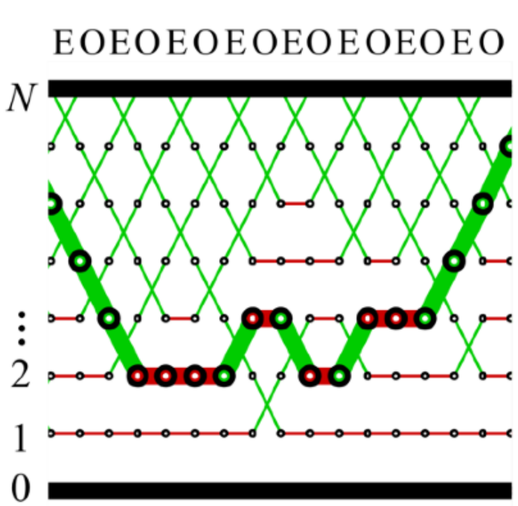

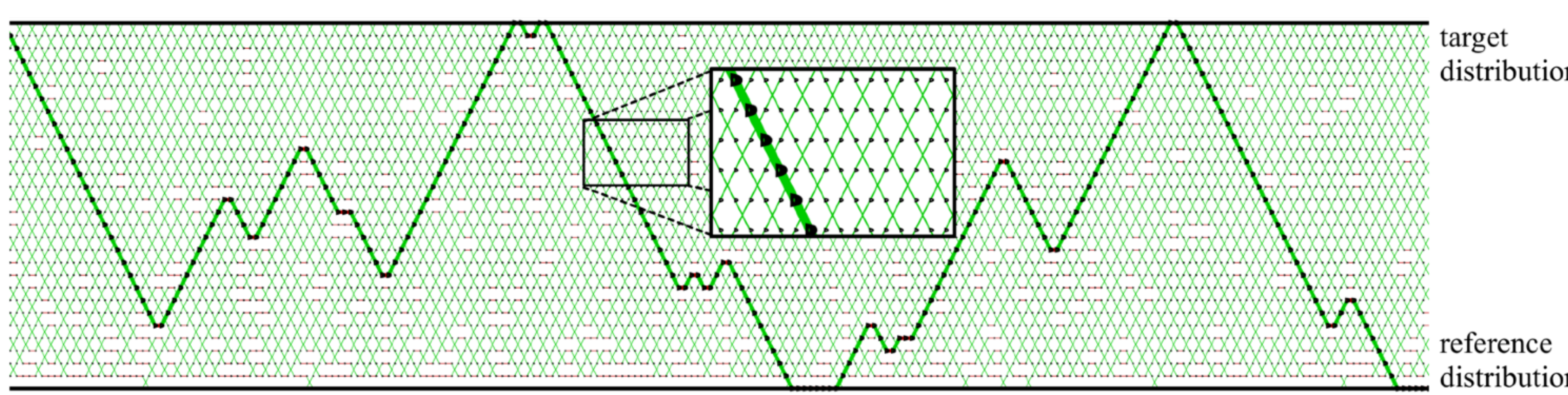

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- Non-reversible PT trajectories have momentum!
    - If swap for machine m is accepted

$$\epsilon^m_{t+1} = \epsilon^m_t$$

    - Otherwise

$$\epsilon^m_{t+1} = -\epsilon^m_t$$